

Detecting cross-firm momentum effects via shared analyst coverage: The role of leaders

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Abstract

Cross-firm momentum effects via shared analyst coverage are well-documented in developed markets, but their robustness remains unclear in emerging markets, where information diffusion is asymmetric and analyst coverage is highly concentrated. Our work revisits this effect in an environment of extreme informational frictions — the Chinese market. We reconstruct the information transmission channel within the analyst coverage network by introducing a novel weighting scheme based on strength centrality (*SC*). This measure identifies influential leader firms that command disproportionate attention from both analysts and the market. Our results demonstrate that *SC*-weighted connected-firm returns robustly predict cross-sectional stock returns, yielding significant and persistent profits even under a rigorous stock filter. This performance cannot be subsumed by strategies based on alternative weighting schemes or by explanations such as intra-industry cross-firm momentum and information discreteness. Further analysis reveals that the superiority of the *SC*-based approach stems from its ability to effectively identify firms with stronger cross-period fundamental linkages. In addition, high-*SC* stocks are characterized by higher investor attention, more efficient information processing, lower arbitrage costs, and greater international exposures. With this evidence, we further confirm a directional spillover: cross-firm momentum effects flow exclusively from these high-*SC* leaders to low-*SC* laggards, and there is no reverse spillover. Our findings suggest that cross-firm momentum may be systematically underestimated in many international markets due to methodological

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limitations rather than economic irrelevance. The *SC*-based framework therefore offers a portable tool for global investors and researchers operating in environments with asymmetric information.

Keywords: Cross-firm momentum, Analyst coverage, Strength centrality, Information diffusion, Emerging markets, International exposure

JEL Codes: G12, G14, G15, C58

1 Introduction

A growing body of literature documents that asset returns exhibit cross-firm predictability when firms are linked through shared information intermediaries, particularly analysts ([Ali and Hirshleifer, 2020](#)). In developed markets, most notably the U.S., strategies based on peer returns constructed via shared analyst coverage generate significant momentum profits, suggesting that information diffuses gradually across fundamentally connected firms. However, this evidence rests on institutional conditions that are far from universal: dense analyst coverage, professional investor dominance, and relatively symmetric information environments. In contrast, the majority of international financial markets, especially emerging economies, are characterized by sparse and highly concentrated analyst attention, retail-dominated trading, and heterogeneous ownership structures. These frictions raise a fundamental question for international finance: does cross-firm momentum via shared analyst coverage persist in such environments, or is it an artifact of data-rich, institutionally mature settings?

The answer hinges on how we model the information transmission channel. Traditional approaches weight peer returns by the number of common analysts or its variants, implicitly assuming that information flows symmetrically between connected firms.¹ However, this assumption is severely violated in many international markets, where market attention is scarce and concentrated, investor bases are heterogeneous, and price adjustment speeds vary systematically across firms. In such environments, information propagates more likely in a directional manner, specifically from a small set of highly visible “hub” firms to a long tail of followers. Under such asymmetry, traditional weighting schemes may dilute or even obscure the true signal, particularly at standard monthly horizons.² Consequently, the apparent

¹A typical approach to studying co-coverage cross-firm momentum uses the number of common analysts or its variants as weights to construct peer average returns and then tests the ability of these returns to predict the future return of a focal firm. According to [Ali and Hirshleifer \(2020\)](#), the number of common analysts measures the fundamental connection between firms and also gauges the cost of information processing channels for rational investors, which corresponds to an increase in the time required for information to be reflected in market outcomes.

²In markets where these resources are highly concentrated, compounded by speculative behavior from dominant retail investors, information tends to be rapidly incorporated into the prices of relevant assets. Therefore, a simple common analyst count fails to discriminate between the information diffusion dynamics

absence of cross-firm momentum in emerging markets may reflect methodological limitations rather than economic irrelevance.

To address this challenge, we propose a novel weighting scheme based on a firm’s strength centrality (SC) within the analyst coverage network. Specifically, SC is calculated as the sum of the weights of all edges connected to a firm, thus providing a measure of its total connectedness (Barrat et al., 2004). By definition, a high- SC firm is not only fundamentally linked to many peers but, more importantly, occupies a pivotal position as an informational hub within the entire ecosystem of analysts and, by extension, the market. Consequently, these high- SC firms become informational *leaders* where information is incorporated rapidly and efficiently. By assigning greater weight to returns from these high- SC stocks, we effectively construct a portfolio that pits “fast-information-incorporating” stocks against “slow-information-incorporating” stocks. This allows us to more precisely capture the gradual diffusion of information (Hong and Stein, 1999) or more generally, the process of sluggish adjustment to information (Peng and Xiong, 2006; Hou, 2007), and in turn the resultant cross-firm return predictability.

We test this framework in an environment that represents a stringent stress test for cross-firm momentum: the Chinese stock market. China exhibits extreme versions of frictions common across emerging economies: analyst coverage is highly concentrated (Xu et al., 2013), retail investors dominate trading volume (Tan et al., 2024), and ownership structures span state-owned, domestic private, and foreign-invested enterprises. Crucially, our goal is not to explain China per se, but to leverage its institutional extremity to isolate whether traditional methods fail under realistic constraints. If the SC -based signal survives where conventional approaches falter, it provides compelling evidence for its applicability to a broader class of international markets with milder but structurally similar frictions.

We begin with a univariate portfolio analysis, where the results align well with our expectations. Under a simple filtering criterion typically employed in prior studies on the Chinese

of a highly influential, well-covered firm and a less-covered firm. Section 2.3 presents a further discussion about this.

market and when considering equal-weighted returns, we find that connected-firm returns (denoted as $CFRet$) based on various weighting schemes significantly predict cross-sectional stock returns. Notably, the long-short portfolio formed by sorting on SC -weighted $CFRet$ (denoted as $CFRet^{SC}$) demonstrates superior performance, yielding an equal-weighted return of 1.042% per month (annualized at 16.8%). Furthermore, all $CFRet$ -sorted long-short portfolios based on different weighting schemes deliver statistically significant alphas across popular risk pricing models, including the single-index model (SIM) (Sharpe, 1963), CH4 model (Liu et al., 2019), q^5 model (Hou et al., 2021), and the six-factor (FF6) model (Fama and French, 2018).

However, when employing a stricter yet more practical filtering criterion that has been frequently adopted in the related literature (Hsu et al., 2018; Liu et al., 2019; Jansen et al., 2021), we find that only the $CFRet^{SC}$ -sorted long-short portfolio continues to deliver statistically significant equal-weighted returns and alphas across various models. In contrast, the cross-firm momentum effects based on other popular schemes weaken, and their portfolios produce significant alphas solely under the q^5 and FF6 models. When further considering value-weighted returns, the results for $CFRet^{SC}$ remain consistent. In contrast, those of the alternatives all become insignificant, highlighting the superior performance of $CFRet^{SC}$ in predicting stock returns. This evidence collectively confirms the presence of more pronounced cross-firm momentum effects because of the SC weighting scheme.

We further examine whether the $CFRet$ -sorted long-short portfolio (denoted as $CFMOM$ for brevity), constructed using a specific weighting scheme, can explain the performance of its counterparts based on other weighting schemes. Specifically, we focus on whether the $CFRet^{SC}$ -sorted long-short portfolio, denoted as $CFMOM^{SC}$, can explain or be explained by other types of $CFMOM$. To address this question, we perform spanning tests to evaluate the alphas of the specified $CFMOM$, which are derived from augmented pricing factor models that incorporate the return of another type of $CFMOM$. Our results show that the return predictability based on $CFRet^{SC}$ cannot be explained by its counterparts, as ev-

identified by the fact that the alphas of $CFMOM^{SC}$ remain significant in various augmented pricing models. In other words, the cross-firm momentum effects based on the SC weighting scheme cannot be subsumed by those based on alternative weighting schemes. Conversely, $CFMOM^{SC}$ can largely account for the performance of the other types of $CFMOM$.

To validate the ability of $CFRet^{SC}$ to predict stock returns under different weighting schemes while controlling for additional firm attributes, we conduct the [Fama and MacBeth \(1973\)](#) (FM) regressions. The results of univariate FM regressions reveal that all individual $CFRet$ measures exhibit superior predictability for cross-sectional stock returns, even after controlling for common firm attributes. In bivariate FM regressions combining $CFRet^{SC}$ with each of the other types of $CFRet$, we find that $CFRet^{SC}$ continues to result in statistically and economically significant profits. For example, the monthly average return spreads between the decile portfolios with the highest and lowest values of $CFRet^{SC}$ range from 8.04% to 12.48% per annum, with t -statistics exceeding 2.7. Even when all $CFRet$ measures are jointly incorporated in a multivariate FM regression, the return spread between the decile portfolios with the highest and lowest values of $CFRet^{SC}$ remains around 8.28% per annum, which is still significant at the 5% level. By contrast, in both the bivariate and multivariate setups the other $CFRet$ measures lose their ability to predict stock returns, either in terms of statistical or economic significance.

It is important to understand the drivers behind these findings. This essentially amounts to exploring the information embedded in SC . First, we examine the fundamental linkages underlying various weighting schemes. Using panel regressions of firms' profitability growth on the average profitability growth of their directly-connected peers, we confirm that all weighting schemes in study capture contemporaneous and cross-period fundamental connections. In comparison, the SC -based scheme excels in identifying cross-period linkages, which are more critical for cross-firm return spillovers. Second, we test whether SC can indeed identify firms that receive extensive attention from analysts and the market. Building on the results of univariate portfolio analysis linking SC to firm-level characteristics, we find

that high- SC stocks associate with larger size, higher online search volume, greater analyst coverage, higher abnormal trading volume, among other attention metrics. Moreover, high- SC stocks have a significantly higher institutional ownership proportion and lower arbitrage costs. These together ensure that relevant information about high- SC stocks is incorporated into prices more quickly.

We also delve into the relationship between SC and a firm’s ownership structure. This represents a key institutional dimension of the Chinese market with direct implications for a firm’s information environment and, consequently, its position within the analyst network. We find that the prevalence of foreign-invested enterprises (FIEs) or firms with high international exposure decreases monotonically from the highest to the lowest SC deciles. In contrast, the proportion of state-owned enterprises (SOEs) increases monotonically as SC declines, while domestic private enterprises (DPEs) exhibit a non-linear, inverted U-shaped relationship with SC . A further scrutiny of the results reveals that although DPEs remain the largest category in the high- SC group, firms with high international exposure tend to concentrate in top network tiers, and centrally connected firms are less likely to be SOEs.

It is reasonable to expect more pronounced cross-firm momentum effects for firm laggards, which can also be identified via the SC measure. We thus conduct a bivariate portfolio analysis to evaluate the performance of portfolios formed through dependent sorting on SC and $CFRet^{SC}$. Our full-sample results indicate that return predictability based on $CFRet^{SC}$ emerges only in the lower SC quantiles, corresponding to those stock laggards. Furthermore, as for alternative weighting schemes, we also observe that the cross-firm momentum effects are more prominent for laggards identified via SC .

It pays to further explore how peer information transmits among firms with different ownership structures. Specifically, we aim to study which type of high- SC stocks transmits peer information and which type of low- SC stocks receives this information, ultimately earning a risk premium for delayed adjustment to information. To address these concerns, we perform a bivariate portfolio analysis on sub-samples partitioned by ownership type:

SOEs, DPEs, and FIEs. For each focal firm in a group, we compute three variants of the cross-firm momentum signal ($CFRet^{SC}$) using peer firms from each ownership category. Within each subsample, we similarly conduct a bivariate dependent sorting.

Our analysis reveals heterogeneous roles across ownership types in cross-firm information diffusion. Low- SC DPEs act as primary recipients of delayed information, with $CFRet^{SC}$ from all peer types significantly predicting their returns. In contrast, low- SC FIEs show no return predictability from any peer group, confirming their role as information leaders due to faster price incorporation and higher transparency. Overall, the cross-firm momentum effect is primarily driven by slow information diffusion to peripheral DPEs, with FIEs exhibiting the faster information assimilation.

In our further tests, we consider several alternative explanations for our SC -based cross-firm momentum via shared analyst coverage. Specifically, our results suggest that our SC weighting scheme captures a unique information transmission channel, one that differs significantly from the intra-industry cross-firm momentum effects (Hou, 2007). We also find that the return predictability of $CFRet^{SC}$ cannot be explained by the theory of information discreteness proposed in (Huang et al., 2022). Moreover, although the $CFRet$ measures using weighting schemes based on different attention metrics generate significant cross-firm momentum effects, but the associated return predictability can be fully subsumed by $CFRet^{SC}$. In addition, we finally propose a “refined” $CFRet$ based on PageRank (PR) centrality ($CFRet^{PR}$), which is expected to capture more comprehensive information regarding market attention. Our results show that $CFRet^{PR}$ exhibits superior return predictability, and this predictability cannot be explained by traditional weighting schemes, thus further validating the logic underlying our construction of the SC weighting scheme.

In summary, our findings carry direct implications for international finance. They suggest that cross-firm momentum would be systematically underestimated in many emerging markets. This pattern arises not because economic linkages are absent, but because standard methodologies fail to account for asymmetric information diffusion. Our SC -based frame-

work thus offers a portable, network-informed tool for international investors to identify early movers and delayed adjusters in real time, even in data-scarce environments. Moreover, the finding that internationally integrated firms (e.g., foreign-invested enterprises) consistently occupy leader positions underscores a key channel through which financial openness enhances price efficiency. This mechanism is likely relevant for other emerging markets undergoing gradual financial liberalization, such as India, Vietnam, or Brazil. For policymakers in emerging markets, this implies that promoting international linkages (through foreign ownership, cross-listing, or capital account liberalization) may accelerate information diffusion and reduce mispricing. For global asset allocators, it suggests that network centrality can serve as a leading indicator of a firm’s role in the international information transmission chain.

1.1 Related literature

A growing body of literature has explored cross-firm or cross-asset return spillovers, with two primary strands of research addressing this phenomenon. The first strand adopts data-driven or model-driven approaches.³ The second focuses on return spillovers driven by specific, economically-meaningful connections between firms. These connections, which serve as conduits for spillovers, can arise from shared industry sectors (Moskowitz and Grinblatt, 1999), supplier-customer relationships (Cohen and Frazzini, 2008; Menzly and Ozbas, 2010), technology adoption (Lee et al., 2019), shared analyst coverage (Ali and Hirshleifer, 2020), common geographical location (Parsons et al., 2020), production complementarity (Lee et al., 2024), and bond links (Feng et al., 2025). Specifically, in the context of the Chinese market, a number of studies have examined most of the above *traditional* cross-firm momentum effects (Duan et al., 2022; Yi and Guo, 2022; Jiang et al., 2023; Xiong et al., 2023; Wu et al., 2024). And more studies explore cross-firm momentum effects based on novel interconnected relationships in this market. These relationships include common investment concepts (Du

³The related methods include Diebold and Yilmaz (2014) framework and its variants (Diebold et al., 2017; Ando et al., 2022), partial correlation (Kenett et al., 2015), lagged correlation (Curme et al., 2015), Granger causality (Billio et al., 2012; Zhang and Broadstock, 2020), Copula methods (Philippas and Siriopoulos, 2013; Wang et al., 2021), among others.

et al., 2022), adjacent locations on the screen of trading software (Chen et al., 2023), and news co-mentions (Ge et al., 2025a).

Our work adds to the literature on the informativeness of shared analyst coverage. Defining peer firms based on shared analyst coverage provides a useful supplement to traditional firm classifications, such as those based on industry (Kaustia and Rantala, 2021). In recent work, Gomes et al. (2023) examine the peer effects of various financial policies among firms, considering shared analyst coverage. Ali and Hirshleifer (2020) confirm the fundamental associations between firms that arise from common analysts, documenting that asset returns spread among peer firms identified by shared analysts, offering an alternative explanation for return spillovers among connected firms. Our work makes a specific contribution by providing updated evidence on the diffusion of return information based on shared analysts. Compared with prior studies, particularly those focused on the Chinese stock markets (Yi and Guo, 2022; Jiang et al., 2023; Xiong et al., 2023), which observe cross-firm momentum effects under a less stringent stock filter, we propose a refined version using centrality-based weighting schemes. Specifically, our cross-firm momentum effects remain robust and significant under a more rigorous and practical filtering criterion that has not been used in prior related studies.⁴

Furthermore, our study contributes to a small but growing literature that explicitly distinguishes between leader and laggard firms in cross-predictability. Among these, an early study by Hou (2007) adopts an industry-level perspective and demonstrates that information primarily flows from large industry leaders to smaller firms within the same industry. Scherbina and Schlusche (2020) employ a statistical methodology to define leaders as stocks that predict the returns of other stocks without being predicted by those other stocks themselves. Their work highlights the role of slow information diffusion across assets. Most recently, Ge et al. (2025b) provide a comprehensive analysis of momentum spillovers in

⁴We note that Yi and Guo (2022) conduct the robustness check using different single-condition stock filters, aiming to support the existence of analyst-coverage-based cross-firm momentum in China. However, these filters do not function in the same way as the multi-condition filter typically used in most of the literature, and they cannot simultaneously avoid the impacts of stocks with “unsatisfactory” characteristics.

China. They decompose news co-mention networks into directed leader-follower (LF) links and undirected peer (PE) links. Their findings indicate that the clearer, more directional LF structure facilitates faster attention spillover and price discovery. Our approach shares the goal of identifying directional spillovers but differs fundamentally from these predecessors. We move beyond the statistical identification of [Scherbina and Schlusche \(2020\)](#), the industry-level analysis of [Hou \(2007\)](#), and the news-based LF/PE decomposition of [Ge et al. \(2025b\)](#). Instead, we utilize the economic structure of the analyst coverage network, specifically employing the strength centrality of firms within this network. This framework allows us to define leaders ex ante based on their positional importance within a meaningful information diffusion channel. Our method provides an economically-grounded mechanism for the leadership effects documented in prior studies, which in turn offers a novel perspective and deeper insight into the cross-predictability literature.

Third, our work relates to the literature that applies network centralities to financial markets. At the micro level, existing research has focused on the role of firms or assets within investment portfolios, typically measured by centrality metrics derived from networks built upon return performance correlations ([Ozsoylev and Walden, 2011](#); [Clemente et al., 2021](#)). At the macro level, [Ozsoylev et al. \(2014\)](#) use eigenvector centrality to study the relationship between an investor’s position in the information network and their trading performance. [El-Khatib et al. \(2015\)](#) employ CEO network centrality measures to investigate how personal connections affect merger and acquisition outcomes. [Rossi et al. \(2018\)](#) show a positive relationship between network centrality and risk-adjusted performance in delegated investment management. [Richmond \(2019\)](#) explores trade network centrality to explain variation in interest rates and currency risk premia across countries, linking a nation’s position in the global trade network to its economic outcomes. [Evgeniou et al. \(2022\)](#) investigate firm centrality in the supplier-customer network and its relationship to managerial market timing ability. In the two studies by [Ali and Hirshleifer \(2020\)](#) and [Yi and Guo \(2022\)](#) most related to our work, the complexity of firm linkages is measured by the degree and eigenvector cen-

tralities of firms within analyst coverage networks. Our study contributes to this literature by linking the importance of firms, identified through strength and PageRank centralities within an analyst coverage network, to their roles as transmitters of return information.

The remainder of the paper is organized as follows. Section 2 outlines the construction of the analyst coverage network and the weighting schemes employed to investigate connected-firm momentum effects. Section 3 presents our baseline results. Section 4 explores the information embedded in *SC*. Section 5 examines the results when firm laggards are identified. Section 6 conducts further tests. Section 7 concludes.

2 Network construction

2.1 Data

For our analysis, we use daily and monthly data for all A-share stocks listed on Chinese stock exchanges, obtained from the China Stock Market & Accounting Research (CSMAR) database. Corresponding quarterly and annual financial statement data are also sourced from the CSMAR.

Our analysis primarily focuses on firm-level characteristics constructed on a monthly basis. To ensure the robustness and reliability of our findings, we adopt a strict yet practical filter (referred to as the “rigorous filter”), which is consistent with prior literature (Hsu et al., 2018; Liu et al., 2019; Jansen et al., 2021). Specifically, stocks are excluded if they meet any of the following criteria: (1) Listed for less than six months. (2) Suspended from trading for more than five days in the past month. (3) Suspended from trading for more than 120 days in the past year. (4) Suspended on portfolio construction or rebalancing days. (5) Classified as “special treatment” (ST) or “particular transfer” (PT). (6) Locked at the daily price limit throughout the rebalancing day.⁵ (7) Among the bottom 30% in terms of market size.

⁵The Chinese A-share market employs a price limit system to curb excessive volatility and speculative trading. Implemented in December 1996, this system restricts daily price fluctuations to $\pm 10\%$ for most stocks on the Shanghai and Shenzhen main boards, with wider limits of $\pm 20\%$ for the ChiNext and STAR

Some of the existing studies rely on relatively simple filters, usually based on a single condition, which may compromise the robustness of their findings. In our analysis, we also compare results derived from the simple filter used in their studies (also referred to as the “lax filter”), which excludes only ST and PT stocks, with those obtained using our rigorous filter that incorporates all seven conditions.⁶

2.2 Analyst coverage network

We now introduce stock networks formed based on shared analyst coverage, referred to as the analyst coverage network. Specifically, at the beginning of each month, a firm or stock is considered covered by an analyst if the analyst has issued at least one earnings forecast or recommendation for that stock within the past 12 months. Two stocks are defined as *connected* if at least one analyst covers both. Using this definition, we construct monthly network structures (Ali and Hirshleifer, 2020; Gomes et al., 2023), where nodes represent firms or stocks, and edges indicate shared analyst coverage. For a focal firm i , its peer or connected firms are all those directly linked to i . From the perspective of network theory, these peers are nodes with a distance of one from the focal firm in the network structure.

Figure 1 illustrates an example of an analyst coverage network. The figure focuses on 44 banking firms as of December 2023. Other firms are linked to these banks if they are included in at least one analyst’s portfolio. It can be observed that the network based on shared analysts, encapsulates information beyond the standard definition of industry sectors, as evidenced by the diverse industries of the connected firms represented by different node colors. Notably, these 44 banks are connected to firms across 34 distinct industries.⁷ Despite

Market, and $\pm 30\%$ for the Beijing Exchange. Filtering stocks based on this criterion ensures our portfolios reflect realistic investment practices.

⁶For example, Jiang et al. (2023) and Xiong et al. (2023) exclude only ST stocks, i.e., condition (5), while Yi and Guo (2022) apply a simpler criterion, removing stocks with closing prices below five Chinese Yuan — akin to condition (7), which filters out small-size stocks. For convenience, we report baseline results using the lax filter (condition (5) alone) but also evaluate results based on each individual condition. Findings are consistent across these comparisons and are available upon request.

⁷In our work, we adopt the 2012 version of the listed company classification method released by the China Securities Regulatory Commission (CSRC). Detailed classifications appear in the China Securities

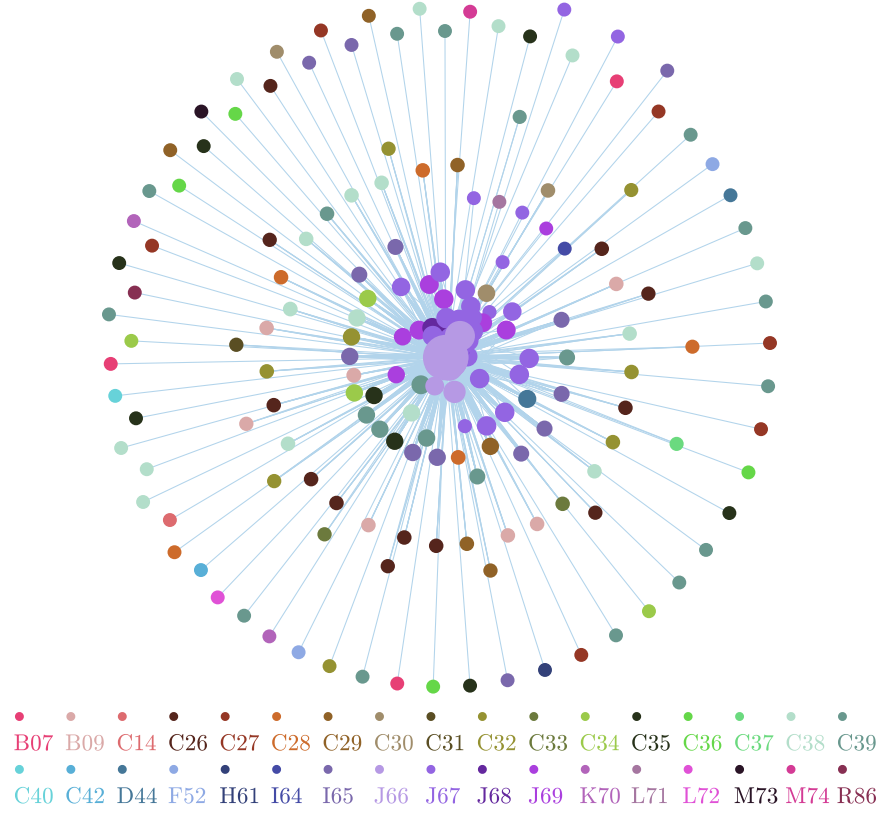


Figure 1 An example of the monthly analyst coverage network as of December 2023

Note: In this network, we concentrate on the focal firms within the banking industry (J66). Other firms are linked to these focal firms if they are covered by at least one common analyst. The size of a firm node is proportional to its node strength, which is defined as the total number of connections with other firms.

these interconnections, firms within the same industry, specifically within the banking sector (industry code J66), tend to exhibit closer intraconnections, as depicted in the figure.

Regulatory Commission (CSRC) Announcement [2012] No. 31, which is accessible via: http://www.csrc.gov.cn/csrc_en/c102034/c1371375/content.shtml. Specifically, these 44 banking firms are also linked to their counterparts from industries with symbol codes prefixed with **B** (mining industry), **C** (manufacturing industry), **D** (industry of electric power, heat, gas and water production and supply), **F** (wholesale and retail industry), **H** (accommodation and catering industry), **I** (industry of information transmission, software and information technology service), **J** (financial industry), **K** (real estate industry), **L** (Leasing and commercial service industry), **M** (scientific research and technical service industry), and **R** (industry of culture, sports and entertainment).

2.3 Weighting schemes and connected-firm returns

To investigate cross-firm momentum effects, we construct CFMOM portfolios by following [Ali and Hirshleifer \(2020\)](#). These portfolios aim to capture whether firms with connected peers exhibiting higher returns also achieve superior performance in the subsequent period. The key input for constructing these portfolios is the connected-firm return ($CFRet$), defined as the weighted average return of a focal firm's connected peers in the analyst coverage network:

$$CFRet_{i,t}^{type} = \sum_{j \in \mathcal{N}_{i,t}} weight_{i,j,t}^{type} \times Ret_{j,t}, \quad (1)$$

where $Ret_{j,t}$ refers to the return of stock j in month t , adjusted for splits and dividends, and $weight_{i,j,t}$ represents the weight assigned to firm j . The weights are determined at the beginning of each month using data from a rolling window spanning from month $t - 1$ to month $t - 12$. The superscript $type$ indicates different weighting schemes, which result in varying CFMOM outcomes. The set $\mathcal{N}_{i,t}$ denotes all connected firms for the focal firm i .

We next review several weighting schemes employed in the literature, starting with the approach introduced by [Ali and Hirshleifer \(2020\)](#) (AH), which is a pioneering work on cross-firm momentum effects driven by shared analyst coverage in the U.S. stock market. Their method emphasizes the number of common analysts between stock pairs, defined as:

$$\omega_{i,j}^{AH} = n_{i,j}, \quad (2)$$

where $n_{i,j}$ represents the number of analysts covering both stocks i and j . If stocks i and j are not connected in the network, $n_{i,j}$ equals zero. The associated weight is scaled such that their summation is equal to one:

$$weight_{i,j}^{AH} = \frac{\omega_{i,j}^{AH}}{\sum_{j \in \mathcal{N}_i} \omega_{i,j}^{AH}} = \frac{n_{i,j}}{\sum_{j \in \mathcal{N}_i} n_{i,j}}. \quad (3)$$

However, the AH weight fails to take into account analyst attentiveness at the individual

level. Several alternative approaches can help address this issue. The first alternative comes from [Israelsen \(2016\)](#) (Isr), and the authors argue that forecasting errors of individual analysts are correlated across different stocks or firms. Therefore, they consider the following proxy for the correlation in consensus forecast errors between a pair of stocks:

$$\omega_{i,j}^{\text{Isr}} = \frac{n_{i,j}}{\sqrt{n_i \times n_j}}, \quad (4)$$

where n_i denotes the number of analysts covering firm i . In a similar way, we then construct $weight_{i,j}^{\text{Isr}}$ as follows:

$$weight_{i,j}^{\text{Isr}} = \frac{\omega_{i,j}^{\text{Isr}}}{\sum_{j \in \mathcal{N}_i} \omega_{i,j}^{\text{Isr}}} = \frac{\frac{n_{i,j}}{\sqrt{n_i \times n_j}}}{\sum_{j \in \mathcal{N}_i} \frac{n_{i,j}}{\sqrt{n_i \times n_j}}}, \quad (5)$$

We notice that [Sorensen \(1948\)](#) (Sor) takes into account a similar measure:

$$\omega_{i,j}^{\text{Sor}} = 2 \times \frac{n_{i,j}}{n_i + n_j}, \quad (6)$$

and its weight can be similarly constructed:

$$weight_{i,j}^{\text{Sor}} = \frac{\omega_{i,j}^{\text{Sor}}}{\sum_{j \in \mathcal{N}_i} \omega_{i,j}^{\text{Sor}}} = \frac{\frac{n_{i,j}}{n_i + n_j}}{\sum_{j \in \mathcal{N}_i} \frac{n_{i,j}}{n_i + n_j}}, \quad (7)$$

Note that $n_i + n_j$ on the right-hand side of [Eq. \(7\)](#) counts pair analysts twice. To avoid double-counting pair analysts, a more precise measure is proposed by [Muslu et al. \(2014\)](#) (MRX). Specifically:

$$\omega_{i,j}^{\text{MRX}} = \frac{n_{i,j}}{n_i + n_j - n_{i,j}}, \quad (8)$$

and we therefore construct $weight_{i,j}^{\text{MRX}}$ as follows:

$$weight_{i,j}^{\text{MRX}} = \frac{\omega_{i,j}^{\text{MRX}}}{\sum_{j \in \mathcal{N}_i} \omega_{i,j}^{\text{MRX}}} = \frac{\frac{n_{i,j}}{n_i + n_j - n_{i,j}}}{\sum_{j \in \mathcal{N}_i} \frac{n_{i,j}}{n_i + n_j - n_{i,j}}}, \quad (9)$$

Evidently, all of the four weighting schemes mentioned above revolve around the number

of common analysts for the given stock pair. According to [Eq. \(1\)](#), a greater number of common analysts between the focal firm i and its connected firm j leads to a higher weight being assigned to firm j . The realized return of firm j thus have a stronger impact. This is also in line with the intuition that more common analysts are associated with a closer fundamental connection between firms ([Ali and Hirshleifer, 2020](#); [Jiang et al., 2023](#)), which are expected to result in more prominent cross-firm momentum effects.

In comparison, the latter three schemes, namely $weight^{Isr}$, $weight^{MRX}$, and $weight^{Sor}$, are variations derived by scaling $weight^{AH}$ with functions involving n_i and n_j . This approach allows for the further consideration of the impact of individual analysts covering either firm i or firm j .⁸ Obviously, a small (large) number of individual (pair) analysts leads to a high weight being assigned to firm j . In this regard, these three types of weights further underscore the importance of common analysts. On the other hand, a potential risk embedded in these three weighting schemes is that they might diminish the influence of connected firms that have a large number of common analysts with respect to the focal firm. This is because in the equations, including [Eq. \(4\)](#), [Eq. \(6\)](#), and [Eq. \(8\)](#), the denominator changes more rapidly than the numerator, although they are positively correlated. Overall, the number of common analysts measures the fundamental connection between firms and also gauges the cost of information processing channels for rational investors ([Ali and Hirshleifer, 2020](#)), which corresponds to an increase in the time required for information to be reflected in market outcomes. Thus, classic weighting schemes that depend on the number of common analysts identify cross-firm momentum effects in mature markets such as the U.S.

However, these schemes may be suboptimal for capturing the full complexity of information diffusion in markets where analyst attention and investor processing of information are highly concentrated, and especially where retail investors dominate. This is because the classic approach implicitly assumes a relatively uniform distribution of analyst attention

⁸Here, we follow the convention in [Muslu et al. \(2014\)](#) to define individual and pair analysts. Specifically, individual analysts are those who cover either the focal firm i or its connected firm j , but not both. Pair analysts are those who cover both the focal firm i and its connected firm j .

and information processing speed across firms. In markets where these resources are highly concentrated, compounded by speculative behavior from dominant retail investors, information tends to be rapidly incorporated into the prices of relevant assets. Therefore, a simple common analyst count fails to discriminate between the information diffusion dynamics of a highly influential, well-covered *leader* firm and a less-covered *follower* firm. The return of the leader firm, by virtue of its centrality in the information network, carries a stronger predictive signal for its peers, yet the classic scheme assigns it the same weight as the return of a peripheral follower if their pairwise common analyst count is identical. This equal weighting of fundamentally unequal information sources dilutes the momentum signal and renders the strategy suboptimal.

To successfully detect cross-firm momentum effects via shared analyst coverage, it is thus crucial to identify the firm leaders within the network, which are defined as stocks that assimilate information into their prices more swiftly. These leaders, characterized by strong fundamental ties to focal firms and robust past performance, act as the primary drivers of cross-firm momentum effects. Identifying these leaders requires assessing both the fundamental linkages between stocks and the speed at which connected firms incorporate information into their prices. The former is captured by the value of $n_{i,j}$, while the latter can be approximated by a firm’s importance within the network. Specifically, a firm’s importance increases with its connections to others beyond the focal firm. The broader a firm’s network, the greater the attention it garners, enabling more timely incorporation of information into its stock price. As a result, these stock leaders should be assigned higher weights when evaluating their influence on the focal firm.

Fortunately, the identification of leaders or prominent nodes within a network structure is relatively straightforward, as network theory provides various well-established methods for this purpose. In particular, we adopt strength centrality (SC) (Barrat et al., 2004), aligning well with our analysis above. SC is defined as the sum of the weights of edges connected to

a given node, or equivalently:

$$SC_j = \sum_{m \in \mathcal{N}_j} n_{j,m}. \quad (10)$$

We therefore readily build our SC -based weights:

$$weight_{i,j}^{SC} = \frac{SC_j}{\sum_{j \in \mathcal{N}_i} SC_j}. \quad (11)$$

By definition, SC_j not only captures firm j 's relationship with the focal firm i but also accounts for j 's connections with all other neighboring firms. Compared to n_j , SC_j offers a distinct advantage in accurately reflecting the attentiveness of market participants, particularly financial analysts.⁹ The SC measure aligns powerfully with the behavioral theory of limited investor attention (Kahneman, 1973; Hirshleifer and Teoh, 2003). Moreover, investors, facing information overload, disproportionately focus on “salient” firms. The high- SC firms that naturally attract this scarce attention, also tend to be large, liquid, held by institutional investors, and associated with lower arbitrage costs.¹⁰ The core innovation lies in using the SC of a peer firm to weight its return when constructing the momentum signal for a focal firm. By doing this, we effectively construct a portfolio that pits “fast-information-incorporating” stocks against “slow-information-incorporating” stocks. As such, the SC scheme can more precisely capture the process of sluggish adjustment to information (Peng and Xiong, 2006; Hou, 2007). The theoretically expected significant cross-firm momentum effect can thus be interpreted as a risk premium compensating for the delayed

⁹It is reasonable to assume that more skilled analysts possess greater endowments of effort, including time and energy, enabling them to efficiently manage their tasks (Harford et al., 2019). Consequently, such analysts are able to cover a larger number of firms in their portfolios. Therefore, the total effort of each analyst can be approximated by the number of firms they cover. From this perspective, SC_j provides a more accurate measure of the cumulative analyst effort allocated to firm j , whereas n_j overlooks this intuitive differentiation and assumes all analysts are identical.

¹⁰Large firms naturally attract more attention due to their size and visibility (Merton, 1987; Bhushan, 1989); institutional investors, driven by research and asset allocation needs, tend to hold significant positions in these firms (O'Brien and Bhushan, 1990); and precisely because these firms have large market capitalizations, high liquidity, and high institutional ownership, the arbitrage costs—including transaction costs and short-selling costs—associated with trading their shares are significantly lower (Amihud and Mendelson, 1986; D'Avolio, 2002), allowing arbitrageurs to more effectively correct any mispricing (Shleifer and Vishny, 1997) and ultimately leading to price behaviors that differ sharply from those of less visible, hard-to-arbitrage small firms (Fang and Peress, 2009).

incorporation of information from these central nodes (Hou and Moskowitz, 2005).

It is also important to note that in Eq. (10), the subscript of the left-hand-side variable is only j , which distinguishes it from the previous four weighting schemes (see Eq. (2), Eq. (4), Eq. (6), and Eq. (8)) for firm pairs, where both i and j appear. This ensures that the importance of a firm, as measured by SC , remains relatively constant for different focal firms. For instance, if firm A has a higher SC value than firm B , then for each focal firm connected to both A and B , the return weight of firm A will always be higher than that of firm B . This *firm importance invariance* matters in detection of cross-firm momentum effects.

We now have five distinct weighting schemes. According to Eq. (1), we can therefore obtain $CFRet^{type}$, where $type \in \{AH, Isr, MRX, Sor, SC\}$. The correlation matrix and associated summary statistics are presented in Table D.1 and Table D.3, respectively, in Section D of the Online Appendix. Notably, based on the results in the correlation matrix, we find that the correlations among the four conventional $CFRet$ measures are relatively high. The average Pearson correlation across these six pairs is 0.922. In contrast, relatively low correlations are observed between $CFRet^{SC}$ and other conventional alternatives. For example, $CFRet^{SC}$ has the highest Pearson correlation of 0.877 with $CFRet^{AH}$, while its correlations with the other three measures drop below 0.76. Moreover, for most pairs of $CFRet$ measures, the Pearson and Spearman correlations are generally very close to each other, suggesting that the relationships between these measures are largely linear. Collectively, these results indicate the uniqueness of the SC -based weighting scheme.

2.4 Network properties

2.4.1 Development of connected components

A deeper understanding of how the network structure evolves, as described above, is critical to our analysis. We also anticipate that these structural changes will align with our expectations.

Specifically, we consider three types of network configurations that may emerge in our samples. In the first configuration, many stocks remain unconnected to any other within the network. In such cases, these focal firms have few peer firms, rendering it impossible to construct connected-firm returns using the previously outlined method. This situation arises when analysts exclusively track individual stocks. The second configuration, which is likely more prevalent, consists of several independent stock groups (or communities) within the network, with no interconnections between these groups. Stocks within each group can “reach” one another through paths (direct or indirect links), forming what network theory identifies as a connected *component*. Notably, in this scenario, the *SC*-based weighting method we propose may lose applicability, as comparing node importance across separate components becomes inconsistent. The third and also relatively ideal configuration includes multiple components within the network, where the majority of firms belong to the largest connected component. Under this setup, we can effectively calculate connected-firm returns for focal firms and reliably compare node importance, as all stocks reside within the same network component.

We therefore present the evolution of connected components in analyst coverage networks from December 2001 to December 2023 in [Figure 2](#). Each bubble represents the number of connected components in a given rolling period, with its size reflecting the number of stocks within the component. As shown in the figure, each of the networks for the period from 2001 to 2007 consists of numerous components. For example, as of July 2003, the updated network features 57 components, with the largest containing only 133 firms. No single component dominates in this period, making it challenging to construct and analyze *SC*-based weights effectively. By contrast, the network structure undergoes a significant shift beginning in 2009. The variation in components becomes more stable. While the number of components continues to fluctuate monthly, a single dominant component consistently emerges, encompassing the majority of firms. And smaller components typically contain only one or two stocks. This creates an ideal configuration for our analysis.

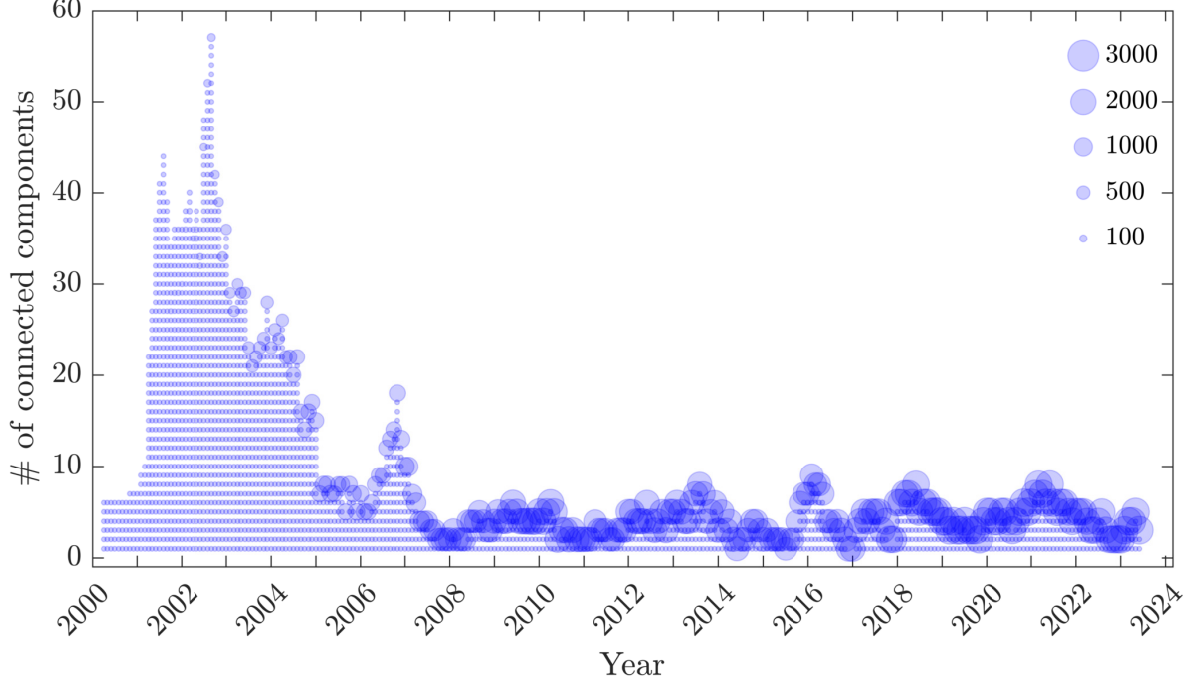


Figure 2 Development of connected components in monthly-updated analyst coverage networks

Note: This figure illustrates the evolution of connected components in the monthly-updated analyst coverage networks from December 2001 to December 2023. Each bubble represents the number of connected components in a given month, with its size corresponding to the number of firms within the component.

As a result, our analysis based on analyst coverage network begins in January 2010. Monthly networks are constructed using 12-month rolling samples starting from January 2009, giving rise to the main results covering the period from January 2010 to December 2023. We finally obtain 363,008 month-firm observations, with an average of 2,161 stocks per month.

2.4.2 Statistics of the components

We further evaluate key structural properties of the largest components in the monthly network structures from 2010 to 2023. Specifically, we examine the number of firms included in each component, their proportion relative to the total number of stocks, the aggregate market capitalization of these firms as a percentage of the total market capitalization, the largest edge weight within the component (i.e., the maximum number of shared analysts

Table 1 Statistics related to the largest components

This table provides statistics for the largest components in analyst coverage networks from January 2010 to December 2023. Key metrics include the number of firms within the largest components (“# of connected firms”), the proportion of these firms relative to the total number of market firms (“% of total number of stocks covered”), their share of the total market capitalization (“% of total market capitalization covered”), the proportion of cross-industry connections (“% cross-industry connections”), and the number of connected industries (“# of connected industries”). Additional metrics include the maximum edge weight within the component (“Max edge weight”) and the component density, defined as the ratio of actual edges to the theoretical maximum number of edges. For each metric, time-series values are summarized by their minimum, median, mean, maximum, and standard deviation.

	Min	Median	Mean	Max	Std
# of connected firms	1366	2216	2216	3103	387
% of total number of stocks covered	52.4	75.7	72.6	92.9	13.5
% of total market capitalization covered	85.7	91.8	91.3	96.8	3.4
% cross-industry connections	0.4	72.8	69.3	100	24.0
# of connected industries	1	23	24	63	13
Max edge weight	38	57	60	83	13
Component density	0.049	0.066	0.077	0.143	0.025

between stocks), and the edge density of the component.¹¹ We focus on the time-series mean, minimum, maximum, median and standard deviation of these properties.

The results in Table 1 reveal that the number of firms included in the largest component ranges from a minimum of 1,366 to a maximum of 3,103, with both the median and mean at 2,216. The proportion of these firms relative to the total number of stocks varies between 52.4% and 92.9%, averaging 72.6%. Notably, the median proportion (75.7%) exceeds the mean, indicating a left-skewed distribution and suggesting that the largest component often contains relatively few firms during many periods. In contrast, the market capitalization of firms within the largest component consistently accounts for a significant share of the total market capitalization. For instance, the lowest observed proportion is 85%, while both the mean and median are 91%, peaking at 96.8%.

We also calculate the percentage of cross-industry connections for each firm and the industry distribution of the associated stocks. Our results show that approximately 70% of connections occur between stocks from different industries. For a focal firm, it connects directly to stocks from an average of 24 industries, with the maximum number of connected

¹¹For a network of size n with m edges, its theoretically maximum edge size reaches $\frac{n \times (n-1)}{2}$. The proportion of the actual edge size to the theoretical value, defined as the *density* of the network, is $\frac{2 \times m}{n \times (n-1)}$.

industries reaching 63. These findings align with the patterns presented in [Figure 1](#), which demonstrate that firm connections based on analyst coverage differ significantly from those based on the traditional industry definition ([Kaustia and Rantala, 2021](#)). In other words, if there exists significant co-coverage-based cross-firm momentum, cross-industry connections should play a more important role in driving these effects.

By examining the maximum edge weight that represents the most frequently co-covered stock pairs, we find that at least 38 analysts simultaneously cover these pairs, with a maximum of 83 analysts and an average of 60 analysts. Despite this, the network density of the largest component remains relatively low, ranging from 4.9% to 14.3%, with an average of 7.7%. These findings, coupled with earlier observations, suggest that the connectivity within the largest component resembles a scale-free network ([Albert and Barabási, 2002](#)): most stock pairs are covered by relatively few analysts, reflecting a structure shaped by market segmentation and analyst specialization, which likely impacts market efficiency and investment strategies.

3 Baseline results

3.1 Cross-firm momentum effects under different weighting schemes

We begin by examining cross-firm momentum effects using the aforementioned weighting schemes. To achieve this, we analyze the performance of CFMOM portfolios through univariate portfolio analysis. At the beginning of each month, individual stocks are sorted into ten decile groups based on their latest connected-firm returns, i.e., $CFRet^{AH}$, $CFRet^{Isr}$, $CFRet^{Sor}$, $CFRet^{MRX}$, and $CFRet^{SC}$. For instance, in the case of $CFRet^{AH}$, decile portfolio 1, labeled High, consists of stocks with the highest average $CFRet^{AH}$, while decile portfolio 10, labeled Low, includes stocks with the lowest values. Each portfolio is held for one month, and its performance is evaluated over the full sample period spanning January 2010 to December 2023.

Table 2 Equal-weighted returns on $CFRet$ -sorted portfolios

This table presents monthly average returns for ten decile portfolios, which are constructed by sorting on $CFRet^{AH}$, $CFRet^{Isr}$, $CFRet^{MRX}$, $CFRet^{Sor}$, and $CFRet^{SC}$, respectively. At the beginning of each month, stocks are sorted into decile portfolios based on their $CFRet$ values, constructed using the procedure in Section 2.3. Portfolio 1 (10), labeled as High (Low), contains the stocks with the average highest (lowest) $CFRet$ value. All portfolios are rebalanced at the beginning of the next month and their one-month-ahead return is obtained. For each decile portfolio, we report the equal-weighted (EW) average monthly excess return. The last rows report return spreads and their alphas between decile 1 (high $CFRet$) and decile 10 (low $CFRet$). SIM alpha is obtained from Sharpe (1963) single-index model (SIM) which consists of the market factor (MKT) only. CH4 alpha is obtained from the CH4 model of Liu et al. (2019). q^5 alpha is obtained from the q^5 -factor model of Hou et al. (2021). FF6 alpha is obtained from the six-factor model of Fama and French (2018). Pricing factors above are specifically constructed for the Chinese market, which are detailed in Section A of the Online Appendix. Panels A and B respectively present the results with lax and rigorous stock filters mentioned in Section 2.1. Newey and West (1987) t -statistics are provided in the parentheses. The results shown in bold signify their statistical significance at the 5% level. The sample period is January 2010 to December 2023.

	Panel A: Lax filter					Panel B: Rigorous filter				
	AH	Isr	MRX	Sor	SC	AH	Isr	MRX	Sor	SC
High	1.787 (2.35)	1.589 (2.07)	1.677 (2.19)	1.729 (2.23)	1.777 (2.36)	1.155 (1.81)	0.976 (1.57)	1.034 (1.68)	1.105 (1.78)	1.223 (1.93)
2	1.098 (1.79)	1.195 (1.94)	1.126 (1.84)	1.123 (1.84)	1.103 (1.85)	0.754 (1.38)	0.821 (1.52)	0.816 (1.49)	0.810 (1.48)	0.755 (1.41)
3	1.004 (1.74)	0.811 (1.48)	0.944 (1.72)	0.925 (1.68)	0.979 (1.65)	0.695 (1.37)	0.589 (1.16)	0.696 (1.35)	0.638 (1.24)	0.695 (1.33)
4	0.956 (1.68)	0.920 (1.65)	0.970 (1.63)	0.952 (1.62)	0.858 (1.57)	0.836 (1.54)	0.689 (1.32)	0.695 (1.30)	0.721 (1.37)	0.678 (1.34)
5	0.946 (1.61)	0.800 (1.39)	0.921 (1.59)	0.886 (1.52)	0.794 (1.41)	0.666 (1.26)	0.620 (1.18)	0.791 (1.46)	0.712 (1.30)	0.589 (1.15)
6	0.736 (1.28)	0.825 (1.39)	0.762 (1.28)	0.773 (1.30)	0.704 (1.22)	0.513 (0.98)	0.623 (1.17)	0.470 (0.88)	0.491 (0.93)	0.557 (1.10)
7	0.542 (0.95)	0.624 (1.08)	0.643 (1.12)	0.606 (1.07)	0.616 (1.07)	0.384 (0.76)	0.437 (0.84)	0.403 (0.79)	0.440 (0.87)	0.317 (0.61)
8	0.497 (0.88)	0.539 (0.93)	0.418 (0.74)	0.399 (0.69)	0.578 (1.02)	0.265 (0.52)	0.324 (0.61)	0.249 (0.49)	0.230 (0.45)	0.468 (0.89)
9	0.438 (0.78)	0.570 (0.99)	0.438 (0.77)	0.508 (0.90)	0.630 (1.12)	0.307 (0.57)	0.478 (0.91)	0.422 (0.80)	0.422 (0.80)	0.464 (0.88)
Low	0.762 (1.32)	0.727 (1.28)	0.700 (1.22)	0.698 (1.22)	0.735 (1.24)	0.576 (1.09)	0.516 (0.99)	0.506 (0.95)	0.529 (1.00)	0.405 (0.75)
High–Low	1.025 (2.22)	0.861 (2.06)	0.977 (2.11)	1.031 (2.20)	1.042 (2.63)	0.579 (1.29)	0.460 (1.16)	0.529 (1.19)	0.576 (1.32)	0.818 (2.21)
SIM alpha	1.013 (2.26)	0.841 (2.12)	0.957 (2.17)	1.012 (2.27)	1.025 (2.69)	0.594 (1.30)	0.475 (1.18)	0.542 (1.21)	0.590 (1.34)	0.819 (2.20)
CH4 alpha	1.153 (2.50)	0.943 (2.38)	1.048 (2.35)	1.085 (2.42)	1.148 (3.05)	0.885 (1.98)	0.708 (1.87)	0.788 (1.80)	0.835 (1.92)	1.087 (3.04)
q^5 alpha	1.313 (2.35)	1.129 (2.28)	1.265 (2.34)	1.296 (2.43)	1.314 (2.84)	1.188 (2.13)	1.050 (2.11)	1.147 (2.11)	1.202 (2.22)	1.326 (2.94)
FF6 alpha	1.311 (2.49)	1.153 (2.60)	1.250 (2.49)	1.291 (2.56)	1.300 (3.00)	1.091 (2.15)	0.945 (2.22)	1.016 (2.07)	1.076 (2.20)	1.259 (3.10)

Table 2 presents the equal-weighted returns of CFMOM portfolios. Panels A and B display the results under lax and rigorous filters, respectively. The findings in Panel A suggest

that under the lax filter (i.e., excluding only ST and PT stocks), CFMOM portfolios exhibit robust performance. It is evident that all High portfolios, constructed using the five weighting schemes, show statistical significance at the 5% level. The lowest return, 1.589% per month (approximately 19% annualized, calculated as $1.589\% \times 12$), is achieved by the portfolio sorted on $CFRet^{Isr}$, whereas the highest return, 1.787% per month (approximately 21.4% annualized), is delivered by the portfolio based on $CFRet^{AH}$. Notably, the High portfolio constructed using the SC weighting scheme also performs exceptionally well, delivering the second-highest return and achieving the highest t -statistic of 2.36.¹²

Additionally, from decile 1 to decile 9, portfolios based on the various weighting schemes exhibit a clear monotonic decline in raw returns, while decile 10 (Low portfolio) shows a slight increase. These trends collectively yield statistically significant return spreads for all five long-short portfolios constructed by longing the High portfolio and shorting the Low. Among them, the long-short portfolio based on $CFRet^{SC}$ demonstrates the strongest performance, with a monthly average return of 1.042% (annualized at 16.8%). The last four rows of the table present risk-adjusted returns for the long-short portfolios using various factor models: the single-index model (SIM) from [Sharpe \(1963\)](#), the CH4 model from [Liu et al. \(2019\)](#), the q^5 model from [Hou et al. \(2021\)](#), and the six-factor model (FF6) from [Fama and French \(2018\)](#). The pricing factors are constructed specifically for the Chinese market, as detailed in Section A of the Online Appendix. These results reveal that the risk-adjusted returns for all five long-short portfolios remain statistically significant at the 5% level. Furthermore, the $CFRet^{SC}$ -sorted long-short portfolio consistently achieves the highest t -statistics under different pricing models, ranging from 2.20 to 3.10, highlighting its superior return predictability.

As shown in Panel B of [Table 2](#), applying the stricter stock filter (i.e., rigorous filter) leads to a decline in the profitability of all CFMOM portfolios. Notably, none of the High portfolios achieve statistical significance at the 5% level. Despite this, the High portfolio

¹²Unless otherwise stated, the t -statistics reported in our analysis refer to those calculated using [Newey and West \(1987\)](#) methodology.

Table 3 Value-weighted returns on $CFRet$ -sorted portfolios

This table presents monthly average returns for ten decile portfolios, which are constructed by sorting on $CFRet^{AH}$, $CFRet^{Isr}$, $CFRet^{MRX}$, $CFRet^{Sor}$, and $CFRet^{SC}$, respectively. At the beginning of each month, stocks are sorted into decile portfolios based on their $CFRet$ values, constructed using the procedure in Section 2.3. Portfolio 1 (10), labeled as High (Low), contains the stocks with the average highest (lowest) $CFRet$ value. All portfolios are rebalanced at the beginning of the next month and their one-month-ahead return is obtained. For each decile portfolio, we report the value-weighted (VW) average monthly excess return. The last rows report return spreads and their alphas between decile 1 (high $CFRet$) and decile 10 (low $CFRet$). SIM alpha is obtained from (Sharpe, 1963) single-index model (SIM) which consists of the market factor (MKT) only. CH4 alpha is obtained from the CH4 model of Liu et al. (2019). q^5 alpha is obtained from the q^5 -factor model of Hou et al. (2021). FF6 alpha is obtained from the six-factor model of Fama and French (2018). Pricing factors above are specifically constructed for the Chinese market, which are detailed in Section A of the Online Appendix. Panels A and B respectively present the results with lax and rigorous stock filters mentioned in Section 2.1. Newey and West (1987) t -statistics are provided in the parentheses. The results shown in bold signify their statistical significance at the 5% level. The sample period is January 2010 to December 2023.

	Panel A: Lax filter					Panel B: Rigorous filter				
	AH	Isr	MRX	Sor	SC	AH	Isr	MRX	Sor	SC
High	0.894 (1.42)	0.656 (1.06)	0.782 (1.26)	0.795 (1.28)	0.994 (1.44)	0.766 (1.26)	0.460 (0.77)	0.692 (1.17)	0.732 (1.24)	0.888 (1.36)
2	0.256 (0.46)	0.689 (1.26)	0.413 (0.72)	0.345 (0.61)	0.456 (0.89)	0.057 (0.11)	0.533 (1.04)	0.287 (0.51)	0.178 (0.33)	0.354 (0.72)
3	0.292 (0.57)	0.141 (0.28)	0.232 (0.49)	0.366 (0.74)	0.710 (1.27)	0.324 (0.64)	0.066 (0.13)	0.213 (0.44)	0.240 (0.48)	0.582 (1.12)
4	0.432 (0.79)	0.378 (0.75)	0.508 (0.88)	0.408 (0.74)	0.403 (0.84)	0.433 (0.80)	0.432 (0.85)	0.393 (0.72)	0.467 (0.88)	0.438 (0.93)
5	0.683 (1.24)	0.467 (0.84)	0.740 (1.34)	0.604 (1.11)	0.434 (0.92)	0.694 (1.30)	0.267 (0.52)	0.675 (1.26)	0.581 (1.08)	0.285 (0.60)
6	0.167 (0.32)	0.244 (0.47)	0.099 (0.18)	0.085 (0.16)	0.037 (0.08)	-0.035 (-0.07)	0.278 (0.53)	0.111 (0.22)	0.078 (0.16)	0.017 (0.04)
7	0.206 (0.41)	0.298 (0.56)	0.062 (0.12)	0.116 (0.23)	0.037 (0.07)	0.100 (0.20)	0.214 (0.42)	-0.104 (-0.21)	-0.038 (-0.08)	0.004 (0.01)
8	0.052 (0.11)	0.112 (0.22)	-0.036 (-0.07)	0.098 (0.19)	0.221 (0.42)	-0.010 (-0.02)	0.159 (0.31)	0.002 (0.00)	0.092 (0.19)	0.157 (0.30)
9	-0.015 (-0.03)	0.066 (0.14)	0.090 (0.17)	-0.003 (-0.01)	0.053 (0.11)	0.018 (0.03)	0.120 (0.26)	0.207 (0.40)	0.146 (0.29)	0.175 (0.36)
Low	0.334 (0.71)	0.369 (0.78)	0.276 (0.59)	0.329 (0.70)	0.174 (0.34)	0.335 (0.73)	0.333 (0.73)	0.279 (0.61)	0.264 (0.58)	-0.002 (-0.00)
High-Low	0.560 (1.12)	0.287 (0.64)	0.506 (1.01)	0.467 (0.94)	0.820 (1.84)	0.430 (0.86)	0.127 (0.28)	0.413 (0.82)	0.467 (0.97)	0.889 (2.01)
SIM alpha	0.572 (1.13)	0.298 (0.66)	0.515 (1.03)	0.477 (0.96)	0.815 (1.84)	0.450 (0.89)	0.149 (0.33)	0.432 (0.86)	0.488 (1.00)	0.891 (2.01)
CH4 alpha	0.805 (1.49)	0.476 (1.02)	0.709 (1.35)	0.644 (1.22)	1.077 (2.34)	0.714 (1.35)	0.328 (0.71)	0.646 (1.23)	0.702 (1.39)	1.198 (2.80)
q^5 alpha	0.995 (1.60)	0.580 (1.03)	0.852 (1.39)	0.801 (1.31)	1.180 (2.33)	0.838 (1.36)	0.426 (0.77)	0.830 (1.34)	0.893 (1.50)	1.312 (2.64)
FF6 alpha	0.871 (1.45)	0.527 (1.06)	0.792 (1.37)	0.722 (1.24)	1.145 (2.25)	0.782 (1.32)	0.410 (0.83)	0.728 (1.26)	0.781 (1.41)	1.297 (2.72)

based on $CFRet^{SC}$ remains the best performer, yielding the highest return of 1.223% per month, with a t -statistic of 1.93, close to the conventional significance threshold of two. In

contrast, the returns of the other four long-short portfolios, constructed using conventional weighting schemes, decline by over 40% compared to those in Panel A, falling below 0.58% per month and losing statistical significance. The $CFRet^{SC}$ -sorted long-short portfolio experiences a smaller decline, with returns decreasing by only 20%, from 1.042% to 0.818% per month (annualized at 9.82%), while maintaining statistical significance at the 5% level. After adjusting for common risk factors, only the long-short portfolio based on $CFRet^{SC}$ continues to exhibit statistically significant alphas across all pricing models. In contrast, the other long-short portfolios generally show significant alphas only under the q^5 and FF6 models. These findings confirm that $CFRet^{SC}$ outperforms other weighting schemes in predicting cross-sectional returns. The substantial differences in results for other versions of $CFRet$ between Panels A and B further suggest that their return predictability may stem primarily from stocks with abnormal performance.

When analyzing value-weighted returns, the superiority of the SC weighting scheme becomes even more apparent. As shown in Table 3, under both lax and rigorous filters in Panels A and B, only the long-short portfolios constructed using the SC weighting scheme demonstrate statistically significant raw or risk-adjusted returns at the 5% level. Under the rigorous filter, the long-short portfolio based on $CFRet^{SC}$ even outperforms its counterpart in Panel B of Table 2, which relies on equal-weighted returns. Specifically, the $CFRet^{SC}$ -sorted long-short portfolio achieves an equal-weighted return of 0.818% per month, compared to a value-weighted return of 0.889%. Moreover, after controlling for common risk factors, the $CFRet^{SC}$ -based long-short portfolio generates higher alphas in value-weighted results relative to equal-weighted results across all models except the q^5 . These findings indicate that cross-firm return predictability captured using the SC weighting scheme are less influenced by small-cap stocks. Together with the results from Table 2, it is evident that employing SC to identify leaders within the system and assigning them greater weights effectively uncovers more pronounced cross-firm momentum effects, aligning well with our expectations. We proceed with the rigorous filter for subsequent analyses.

Based on the above analysis, we also derive a natural conjecture: the four weighting schemes above that do not capture the monthly-frequency spillover of peer information, are more likely to capture faster and more timely information incorporation into prices (though the process may involve overreactions driven by sentiment or herding).¹³ To verify this conjecture, we conduct the univariate portfolio analysis on a weekly basis, which is detailed in Section B of the Online Appendix. Consistent with our expectations, the results in Table B.1 show that *CFRet*-sorted long-short portfolios based on the five weighting schemes all yield significantly positive profits. Moreover, these portfolios also generate significant alphas after adjustment for different sets of risk factors. By comparison, the traditional four schemes lead to higher returns with stronger statistical significance and lower sensitivity to market capitalization, while *CFRet*^{SC} exhibits weaker performance. This evidence aligns with our expectations. To determine whether these schemes contain unique pricing information, however, additional spanning tests are necessary.

3.2 Spanning tests

We further explore whether the CFMOM long-short portfolio (denoted as *CFMOM* for brevity), constructed using a specific weighting scheme, can account for its counterparts based on alternative weighting schemes. This analysis aims to determine whether cross-firm momentum effects derived from one weighting scheme can effectively explain those from other schemes. Our particular attention is given to the *CFMOM* based on the *SC* weighting scheme, referred to as *CFMOM*^{SC}, to evaluate whether it can explain or be explained by other versions of *CFMOM*. We therefore conduct spanning tests to evaluate the alphas of *CFMOM* obtained from augmented factor models that include each of the other four *CFMOM* returns.

Panel A of Table 4 reports the alphas of *CFMOM*^{SC}. These alphas are derived from time-series regressions using an augmented model that integrates factors from the corre-

¹³We are grateful to an anonymous referee for highlighting this point, which motivates us to conduct the tests using weekly data.

Table 4 Spanning tests for the CFMOM long-short portfolios

This table presents the alphas of various CFMOM long-short portfolios (denoted as $CFMOM$). Panel A highlights the alpha of $CFMOM^{SC}$, derived from time-series regressions based on an augmented model that incorporates factors from the specified model in the corresponding column and the $CFMOM$ listed in the corresponding row. Panel B displays the alphas of four alternative versions of $CFMOM$, namely $CFMOM^{AH}$, $CFMOM^{Isr}$, $CFMOM^{MRX}$, and $CFMOM^{Sor}$, as indicated in the first column. These results are obtained from time-series regressions using an augmented model that integrates factors from the corresponding column's model and $CFMOM^{SC}$. The models employed are tailored specifically for the Chinese market and include the single-index model (SIM) from [Sharpe \(1963\)](#), the CH4 model from [Liu et al. \(2019\)](#), the q^5 model by [Hou et al. \(2021\)](#), and the six-factor model (FF6) from [Fama and French \(2018\)](#). Details on the construction of these pricing factors are provided in Section A of the Online Appendix. The left and right panels correspond to results based on equal-weighted (EW) and value-weighted (VW) portfolios, respectively. [Newey and West \(1987\)](#) t -statistics are presented in the parentheses. Statistically significant results at the 5% level are highlighted in bold. The sample period spans from January 2010 to December 2023.

	EW				VW			
	SIM	CH4	q^5	FF6	SIM	CH4	q^5	FF6
Panel A: $CFMOM^{SC}$'s alpha								
$CFMOM^{AH}$	0.325 (3.50)	0.354 (3.64)	0.343 (3.08)	0.357 (3.31)	0.534 (2.98)	0.636 (3.47)	0.649 (3.01)	0.684 (3.41)
$CFMOM^{Isr}$	0.412 (3.14)	0.481 (3.61)	0.429 (2.77)	0.452 (3.20)	0.770 (3.13)	0.934 (3.72)	0.967 (3.95)	0.968 (3.37)
$CFMOM^{MRX}$	0.369 (2.89)	0.434 (3.25)	0.376 (2.72)	0.419 (2.93)	0.549 (2.52)	0.692 (2.84)	0.658 (2.73)	0.729 (2.80)
$CFMOM^{Sor}$	0.326 (2.82)	0.390 (3.18)	0.325 (2.44)	0.362 (2.78)	0.491 (2.32)	0.629 (2.70)	0.583 (2.53)	0.666 (2.68)
Panel B: Other $CFMOM$'s alpha								
$CFMOM^{AH}$	-0.327 (-2.87)	-0.337 (-2.80)	-0.307 (-2.24)	-0.327 (-2.42)	-0.414 (-1.96)	-0.449 (-1.84)	-0.437 (-1.52)	-0.486 (-1.77)
$CFMOM^{Isr}$	-0.356 (-2.62)	-0.390 (-2.94)	-0.297 (-1.83)	-0.329 (-2.37)	-0.616 (-2.56)	-0.705 (-2.83)	-0.707 (-2.76)	-0.714 (-2.52)
$CFMOM^{MRX}$	-0.339 (-2.26)	-0.379 (-2.43)	-0.284 (-1.74)	-0.341 (-2.03)	-0.385 (-1.56)	-0.457 (-1.57)	-0.377 (-1.22)	-0.475 (-1.47)
$CFMOM^{Sor}$	-0.298 (-2.26)	-0.342 (-2.42)	-0.242 (-1.57)	-0.291 (-1.93)	-0.326 (-1.44)	-0.399 (-1.50)	-0.310 (-1.12)	-0.419 (-1.44)

sponding column's specified model and the $CFMOM$ indicated in the row. The results show that all $CFMOM^{SC}$ alphas remain statistically significant at the 5% level, with t -statistics exceeding 2.3. Notably, in the right panel, which displays value-weighted results, $CFMOM^{SC}$ demonstrates superior performance, as reflected in its higher alpha magnitudes compared to the equal-weighted results in the left panel. Overall, the return predictability of $CFMOM^{SC}$ cannot be attributed to its counterparts, indicating that cross-firm return predictability based on the SC weighting scheme are distinct and not subsumed by those

derived from other weighting schemes.

Panel B presents the results for the other four versions of *CFMOM*: $CFMOM^{AH}$, $CFMOM^{Isr}$, $CFMOM^{MRX}$, and $CFMOM^{Sor}$. Their alphas are derived from time-series regressions using an augmented model that incorporates factors from the corresponding column’s specified model and $CFMOM^{SC}$. Interestingly, all alphas are negative, with some achieving statistical significance at the 5% level. It is worth noting that their raw and risk-adjusted returns are positive, as reported in Panels B of [Table 2](#) and [Table 3](#).

We also conduct similar tests using weekly data, as shown in [Table B.2](#) of the Online Appendix. Recall that in the univariate portfolio analysis using weekly data, we observe that *CFRet* metrics based on the four traditional weighting schemes exhibit more significant return predictability. In the spanning tests, however, none of the weighting schemes delivers distinct return predictability, as all can be explained by other alternatives and common factors. This finding implies that at the weekly frequency, the *SC* weighting scheme contains pricing information similar to that of the other four traditional weighting schemes, even though it performs not good in the univariate portfolio analysis. In summary, the results from monthly and weekly data together suggest that the return of $CFMOM^{SC}$ can largely explain those of the *CFMOM* long-short portfolios based on the four classic weighting schemes.

3.3 Fama-Macbeth regressions

To further validate the predictive ability of *CFRet* under different weighting schemes while controlling for more firm characteristics, we employ [Fama and MacBeth \(1973\)](#) (FM, hereafter) regressions. In these regressions, the left-hand-side (LHS) variable is the one-month-ahead excess return of individual stocks, while the right-hand-side (RHS) variables include either univariate *CFRet* (i.e., $CFRet^{AH}$, $CFRet^{Isr}$, $CFRet^{MRX}$, $CFRet^{Sor}$, and $CFRet^{SC}$) or combinations of these measures. To ensure robustness, all RHS variables are winsorized at the 1st and 99th percentiles.

Consistent with prior literature, we incorporate several control variables closely related to

Table 5 The results of Fama-Macbeth regressions

This table presents results from the cross-sectional regressions of future equity returns on different $CFRet$ measures and various control variables between 2010 and 2023. Panel A presents results estimated for one-month-ahead returns using the ordinary least squares (OLS) methodology. Panel B presents results estimated for one-month-ahead returns using the weighted least squares (WLS) methodology of [Asparouhova et al. \(2013\)](#), where each observed return is weighted by one plus the observed prior return on the stock. All right-hand-side variables are winsorized at the first and 99th percentiles. The reported coefficients represent time-series averages obtained from monthly [Fama and MacBeth \(1973\)](#) regressions, and the corresponding [Newey and West \(1987\)](#) t -statistics are presented in parentheses. The results shown in bold signify their statistical significance at the 5% level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: OLS										
$CFRet^{SC}$	0.119 (3.95)					0.060 (2.72)	0.093 (3.70)	0.087 (4.06)	0.080 (3.78)	0.061 (2.41)
$CFRet^{AH}$		0.109 (3.60)				0.062 (1.88)				-0.023 (-0.33)
$CFRet^{Isr}$			0.093 (3.45)				0.035 (1.64)			-0.048 (-0.92)
$CFRet^{MRX}$				0.096 (3.46)				0.041 (1.76)		-0.196 (-1.22)
$CFRet^{Sor}$					0.103 (3.53)				0.048 (1.88)	0.332 (1.76)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj- R^2	0.100	0.102	0.101	0.102	0.102	0.104	0.104	0.104	0.104	0.109
Panel B: WLS										
$CFRet^{SC}$	0.123 (3.96)					0.061 (2.73)	0.094 (3.74)	0.088 (4.10)	0.081 (3.83)	0.062 (2.44)
$CFRet^{AH}$		0.112 (3.61)				0.065 (1.93)				-0.019 (-0.27)
$CFRet^{Isr}$			0.094 (3.43)				0.037 (1.68)			-0.049 (-0.91)
$CFRet^{MRX}$				0.099 (3.48)				0.043 (1.84)		-0.184 (-1.10)
$CFRet^{Sor}$					0.106 (3.55)				0.050 (1.95)	0.319 (1.63)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj- R^2	0.099	0.102	0.101	0.102	0.102	0.104	0.104	0.104	0.104	0.109

cross-firm momentum effects, denoted as *Controls*. These include: (1) Investment-to-assets (*IA*), (2) Return-on-equity (*ROE*), (3) Logarithm of market equity (*LnSize*), (4) Earnings-to-price ratio (*EP*), (5) Abnormal turnover rate (*Turn*), (6) Cross-sectional momentum (*CSMOM*), and (7) [Amihud \(2002\)](#) illiquidity factor (*Illiq*). Definitions and descriptive statistics for these variables are detailed in Section D of the Online Appendix.

Panel A of [Table 5](#) presents the results estimated using the ordinary least squares (OLS).

In model (1), $CFRet^{SC}$ is the sole RHS variable. The slope coefficient is 0.119, with a t -statistic of 3.95, demonstrating its robust explanatory power for cross-sectional stock returns. Using the results from [Table D.2](#) of the Online Appendix, this corresponds to a monthly average return spread between the decile portfolios with the highest and lowest values of $CFRet^{SC}$ of 1.34% ($= 0.119 \times 11.23$), resulting in an annual return of 16.08%.

Models (2) to (5) focus on $CFRet$ versions based on alternative weighting schemes. It can be observed that all associated slopes hover around 0.10, and exhibit statistically significant predictive ability for cross-sectional stock returns, as evidenced by their high t -statistics exceeding three. Specifically, the slope of $CFRet^{AH}$ is 0.109. This implies that a monthly average return spread between the decile portfolios with the highest and lowest values of $CFRet^{AH}$ of 1.40% ($= 0.109 \times 12.80$), resulting in an annual return of 16.74%. This also aligns well with prior evidence, which detect superior performance of $CFRet^{AH}$ using FM regressions in China ([Jiang et al., 2023](#); [Xiong et al., 2023](#); [Yi and Guo, 2022](#)). Comparable results can be observed for the other three weighting schemes. Specifically, the monthly return spreads between the highest and lowest decile portfolios are 1.22% ($= 0.093 \times 13.17$) for $CFRet^{Isr}$ (annualized at 14.70%), 1.44% ($= 0.096 \times 14.97$) for $CFRet^{MRX}$ (annualized at 17.25%), and 1.33% ($= 0.103 \times 12.93$) for $CFRet^{Sor}$ (annualized at 15.98%). These findings confirm the superior predictability of all $CFRet$ measures in explaining cross-sectional returns, even after controlling for common firm-level characteristics.

In models (6) to (9), we examine the FM regressions incorporating $CFRet^{SC}$ and one alternative $CFRet$ based on conventional weighting schemes as their RHS variables. Evidently, after additionally considering the influence of $CFRet^{SC}$, the return predictability of other four $CFRet$ versions diminishes significantly, as indicated by their decreased slopes and associated t -statistics falling below 1.88. By contrast, although the slope coefficients of $CFRet^{SC}$ in all four models exhibit varying degrees of reduction compared to that in model (1), they remain statistically significant at the 5% level (at least), with t -statistics exceeding 2.7. According to [Table D.2](#), the monthly average return spreads between the decile portfolios

with the highest and lowest values of $CFRet^{SC}$ are approximately 0.67% ($= 0.060 \times 11.23$), 1.04% ($= 0.093 \times 11.23$), 0.98% ($= 0.087 \times 11.23$), and 0.90% ($= 0.080 \times 11.23$), respectively. Correspondingly, the long-short portfolios yield annual returns of 8.04%, 12.48%, 11.76%, and 10.80%, respectively. Overall, $CFRet^{SC}$ still leads to considerable profits.

Model (10) includes all five $CFRet$ versions. The slope for $CFRet^{SC}$ remains statistically significant at the 5% level, evidenced by a t -statistic of 2.41. The return spread between the decile portfolios with the highest and lowest values of $CFRet^{SC}$ is around 0.69% ($= 0.061 \times 11.23$) per month, equivalent to 8.28% per annum. Meanwhile, we observe that the coefficients for other $CFRet$ versions are insignificant at the 5% level. Those for $CFRet^{AH}$, $CFRet^{Isr}$, and $CFRet^{MRX}$ even turn negative. These results together suggest the superior ability of return forecasting based on $CFRet^{SC}$, which can not be explained by the other four $CFRet$ versions as well as related firm-level characteristics. Conversely, $CFRet^{SC}$ subsumes the predictability of the other four $CFRet$ measures, which consolidates our findings from the spanning tests.

In light of [Asparouhova et al. \(2013\)](#), temporary deviations of prices from fundamental values can bias estimates of individual stocks' mean returns, differences in mean returns across portfolios, and parameters in return regressions. To address this, the authors propose a correction using weighted least squares (WLS) estimation, where the prior-period excess return serves as the weighting variable. This method is argued to provide consistent estimates of the true mean return for individual stocks or stock portfolios. Accordingly, we apply the FM regressions using the WLS methodology from [Asparouhova et al. \(2013\)](#), with the weighting variable defined as one plus the observed prior return of each stock. The results are presented in Panel B of [Table 5](#).

Overall, we observe similar or even better performance for $CFRet^{SC}$. Specifically, compared to the results in Panel A, the slope coefficient of $CFRet^{SC}$ increases to 0.123, with a comparable t -statistic of 3.96. Additionally, the coefficients for the other four weighting schemes also improve while remaining statistically significant at the 5% level. In the mul-

tivariate FM regressions, as presented in models (6) to (10), the slopes and t -statistics of $CFRet^{SC}$ show slight increases relative to those in Panel A. Meanwhile, the coefficients for the other four $CFRet$ measures become insignificant at the 5% level. These results further support our earlier finding that $CFRet^{SC}$ demonstrates superior performance in predicting cross-sectional stock returns.

4 The information embedded in the SC weighting scheme

In the previous sections, we demonstrate through various tests that cross-firm momentum effects are more pronounced under the SC weighting scheme. This section seeks to further investigate the drivers behind these findings.

4.1 Fundamental linkages

We first examine whether fundamental linkages exist for stock pairs under various weighting schemes. We hypothesize that these linkages play a significant role in cross-firm momentum effects, with stronger linkages expected to result in more pronounced return spillovers or improved return predictability based on the given $CFRet$.

To test this hypothesis, we regress firms' profitability growth measures on the average growth measures of their connected firms.¹⁴ The measure of profitability growth, PG , for each stock is defined as:

$$PG_{i,t} = 2 \times \frac{Profit_t - Profit_{t-1}}{Assets_t + Assets_{t-1}}, \quad (12)$$

where $Profit$ and $Assets$ denote the operating income before depreciation and total assets, respectively. Similarly, we can construct the average PG performance for connected firms,

¹⁴We also consider another widely-used variable related to firm fundamentals, namely sales growth, which is defined as $(sales-per-share_t / sales-per-share_{t-1}) - 1$. However, our unreported results indicate no evidence of fundamental linkages between focal firms and their peers under any of weighting schemes in our study, suggesting that sales growth is unsuitable for our analysis, despite its effectiveness in studies of the U.S. market (Ali and Hirshleifer, 2020).

denoted as $CFPG$:

$$CFPG_{i,t}^{type} = \sum_{j \in \mathcal{N}_{i,t}} weight_{i,j,t}^{type} \times PG_{j,t}, \quad type = \{AH, Isr, Sor, MRX, SC\}. \quad (13)$$

In the Chinese market, all listed firms have a fiscal year ending in December, and annual reports must be released within the subsequent four months. By April 30 of each year, we collect financial data from annual reports to calculate firms' profitability growth for the previous fiscal year. Consequently, the PG observations remain unchanged from May of one year to April of the following year. To ensure that the growth variables of connected firms ($CFPG$) align within the same time frame, we consider only the monthly network structure as of December of each calendar year.

We thus build the following panel regression models to evaluate the relationship between $CFPG_t$ and PG_t , as well as between $CFPG_t$ and PG_{t+1} :

$$PG_{i,t} = \beta_1 \times CFPG_{i,t}^{type} + Controls + YearEffect + FirmEffect + \epsilon_{i,t}, \quad (14)$$

$$PG_{i,t+1} = \beta_1 \times CFPG_{i,t}^{type} + Controls + YearEffect + FirmEffect + \epsilon_{i,t}. \quad (15)$$

Both regressions include year and firm fixed effects ($YearEffect$ and $FirmEffect$) and the logarithm of market equity for each firm ($LnSize$) in the control variables ($Controls$). For brevity, the coefficients on $LnSize$ are not reported. All dependent and independent variables are cross-sectionally standardized to have a mean of zero and a standard deviation of one.

The results are presented in [Table 6](#). As shown in models (1) to (5) in Panel A, a contemporaneously close fundamental relationship between PG and $CFPG$ is observed across different weighting schemes, as indicated by their large t -statistics exceeding eight. In contrast, relatively weaker fundamental linkages are evident in the SC -based results. Specifically, the slope of $CFPG^{SC}$ is 0.052, whereas the slopes for the other four counterparts exceed 0.08.

Table 6 Fundamental linkage tests for various weighting schemes

This table presents the results of the panel regressions. Profit growth (PG) is calculated as defined in Eq. (12). Connected-firm profit growth ($CFPG$) is computed as the weighted average profit growth of connected firms using various weighting schemes. The results shown in bold signify their statistical significance at the 5% level. The sample period for the analysis spans from January 2010 to December 2023.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Dependent variable being PG_t									
$CFPG^{SC}$	0.052 (8.15)					-0.065 (-6.55)	0.009 (1.14)	-0.011 (-1.36)	-0.020 (-2.45)
$CFPG^{AH}$		0.101 (13.93)				0.152 (13.36)			
$CFPG^{Isr}$			0.080 (10.95)				0.076 (9.16)		
$CFPG^{MRX}$				0.104 (13.69)				0.110 (12.51)	
$CFPG^{Sor}$					0.101 (13.63)				0.113 (12.55)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# obs.	24277	24277	22627	22627	22627	24276	22626	22626	22626
Panel B: Dependent variable being PG_{t+1}									
$CFPG^{SC}$	0.048 (7.02)					0.045 (4.90)	0.040 (5.42)	0.036 (4.86)	0.037 (4.79)
$CFPG^{AH}$		0.039 (5.70)				0.004 (0.40)			
$CFPG^{Isr}$			0.037 (5.93)				0.019 (2.78)		
$CFPG^{MRX}$				0.043 (6.67)				0.024 (3.38)	
$CFPG^{Sor}$					0.042 (6.49)				0.021 (2.92)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# obs.	23008	23008	22835	22835	22835	23007	22834	22834	22834

Further evidence from models (6) to (9) reinforces this finding. After controlling individually for the four counterparts in the panel regression, the contemporaneous relationship between $CFPG^{SC}$ and PG becomes insignificant and even negative in some cases, whereas other versions of $CFPG$ remain statistically significant. These findings collectively suggest that the SC -based weighting scheme is not well-suited for accurately identifying contemporaneous fundamental linkages.

Turning to the cross-period results in Panel B, which are of greater interest, we observe a different pattern. As shown in models (1) to (5), there are strong cross-period fundamental

linkages between PG and each $CFPG$. Unlike the findings in Panel A, the cross-period relationship between PG and $CFPG^{SC}$ emerges as the strongest among the weighting schemes. This is evidenced by a large slope coefficient of 0.048 for $CFPG^{SC}$, accompanied by a high t -statistic of 7.02. Furthermore, the relationship between PG and $CFPG^{SC}$ remains robust after controlling for other alternatives. Specifically, as shown in models (6) to (9), the slopes for $CFPG^{SC}$ exceed 0.036, with associated t -statistics surpassing 4.7. In contrast, the slopes for $CFPG^{Isr}$, $CFPG^{MRX}$, and $CFPG^{Sor}$ drop below 0.025, though they remain statistically significant at the 5% level. Additionally, the slope for $CFPG^{AH}$ even becomes insignificant.

In summary, these findings demonstrate that within the analyst coverage network, connected firms exhibit both contemporaneous and cross-period fundamental linkages under various weighting schemes. However, the SC -based weighting scheme proves more effective in identifying cross-period fundamental connections, which are evidently more critical for the cross-firm momentum effects identified in the previous sections.

4.2 Average firm attributes

Another key reason (or more importantly, the critical factor) that the SC -based weighting scheme leads to more significant cross-firm momentum lies in the fact that SC captures differences in the level of investor attention. This section therefore examines the relationship between SC and relevant firm-level characteristics.

In light of prior studies (Da et al., 2011; Koester et al., 2016; Durand et al., 2019; Driskill et al., 2020), we first consider the following traditional attention metrics, including: the logarithm of firm size ($LnSize$); the number of analysts covering the firm (NOA); the search volume index with stock codes as keywords ($SVIC$); the search volume index with stock codes, company abbreviations, and full company names as keywords ($SVIA$); abnormal trading volume in Chinese Yuan ($AbnVolume$); cumulative returns over the past 12 months ($CSMOM$).¹⁵ We also construct an attention metric that relies on real market reactions,

¹⁵The data on the Search Volume Index (SVI) for Chinese listed firms are derived from the Chinese

which describes stock performance during a firm’s earnings announcement window. Assets widely followed by the market tend to exhibit more pronounced abnormal returns, so we include the absolute value of cumulative abnormal returns during the earnings announcement window (*AbsCAR*).

Aside from these attention metrics, we consider proxy variables for other channels related to the incorporation of information into prices. Specifically, a large body of literature consistently shows that institutional investors outperform retail investors significantly in terms of information processing efficiency and speed (Henderson and Pearson, 2011; Kelley and Tetlock, 2013; Hendershott et al., 2015). We thus use the proportion of institutional ownership (*InstProp*) as a proxy variable for information dissemination efficiency. Arbitrage costs (which include transaction costs, financing constraints, and short-selling restrictions) are the core channel that causes information to be slowly incorporated into prices. We thus follow Gu et al. (2018) to construct a limit-to-arbitrage (*LoA*) index.

Beyond these conventional firm attributes, we delve into a firm’s ownership structure. This represents a key institutional dimension of the Chinese market with direct implications for a firm’s information environment and, consequently, its position within the analyst network. We examine three distinct types of ownership that capture a firm’s integration into global capital markets and its domestic institutional role: (1) High international-exposure firms. We consider two proxy variables including QFII ownership proportion (*QFIIProp*) and the dummy variable of foreign-invested enterprise (FIE), i.e., *DFIE*. We posit that firms with greater international exposure are subject to heightened scrutiny from a broader set of sophisticated global investors and analysts. This elevated monitoring demand and their role as bridges between domestic and global information flows likely cement their positions as central hubs in the network. A positive association between these proxies and *SC* is therefore

Research Data Services Platform (CNRDS). SVI is a composite search index calculated using Baidu platform data that incorporates various types of internet search data and news sentiment information. It aggregates the number of search queries related to each listed firm and reflects the intensity of online searches for that firm. Specifically, we employ two SVI-related metrics: the search volume with stock codes as keywords (*SVIC*) and the sum of search volumes with stock codes, company abbreviations, and full company names as keywords (*SVIA*).

expected. (2) State-owned enterprises (SOEs). SOEs that are characterized by significant government ownership, represent a pivotal component of China’s institutional landscape. Their prominence is often driven by policy mandates and systemic importance rather than pure market dynamics. We consider the dummy variable *DSOE* to identify whether a firm is an SOE. It remains an empirical question whether this form of state-affiliated prominence translates into a central role within the market-driven information network proxied by *SC*. (3) Domestic private enterprises (DPEs). These smaller, more agile private firms rely heavily on market mechanisms to attract capital and attention. For them, gaining analyst coverage and becoming informationally central is crucial for reducing financing frictions and signaling their quality. We consider the dummy variable *DDPE* to identify whether a firm is a DPE. We hypothesize that DPE status may be associated with a stronger need to achieve high *SC* to overcome liabilities of newness and smaller size.

Detailed definitions and associated statistics for the variables above are provided in Section D of the Online Appendix. Rather than constructing a linear regression model, we turn to a univariate portfolio analysis, as it can capture both linear and nonlinear relationships between each firm-level variable and *SC*. Specifically, each month, we sort the stocks based on their latest *SC* values and divide them into ten decile groups. For each decile group, we record the average values of each variable. Finally, we calculate the time-series average values of these cross-sectional variables over the entire sample period. The results are shown in Table 7.

Consistent with our prior predictions, the results confirm that high-*SC* stocks correspond to a higher level of market attention. As *SC* decreases, all attention metrics exhibit a downward trend, and the values of different characteristics for the high-*SC* portfolio are significantly higher than those for the low-*SC* portfolio. Specifically, high-*SC* (low-*SC*) stocks associate with larger (smaller) size, higher (lower) online search volume, greater (fewer) analyst coverage, higher (lower) abnormal trading volume, and more pronounced (less pronounced) return reactions during earnings announcement windows.

Table 7 Average firm-level characteristics conditional on strength centrality (SC)

This table presents the time-series averages of the monthly cross-sectional means of strength centrality (SC) and the related firm-level characteristics for each of ten decile portfolios constructed by sorting on SC . Each month, the stocks are sorted based on their latest SC values and are divided into ten decile groups. For each decile group, we record the average values of each of firm-level characteristics, including: (1) strength centrality (SC); (2) the logarithm of firm size ($LnSize$); (3) the number of analysts covering the firm (NOA); (4) cumulative returns over the past 12 months ($CSMOM$); (5) abnormal trading volume in Chinese Yuan ($AbnVolume$); (6) the absolute of cumulative abnormal returns during the earnings announcement window ($AbsCAR$); (7) the search volume index with stock codes as keywords ($SVIC$); (8) the search volume index with stock codes, company abbreviations, and full company names as keywords ($SVIA$); (9) the proportion of institutional ownership ($InstProp$); (10) the limit-to-arbitrage index (LoA); (11) the proportion of QFII ownership ($QFIIProp$); (12) the dummy variable of foreign-invested firms ($DFIE$); (13) the dummy variable of state-owned enterprises ($DSOE$); (14) the dummy variable of domestic private enterprises ($DDPE$). Their definitions and summary statistics are detailed in Section D of the Online Appendix. Finally, we calculate the time-series average values of these cross-sectional variables over the entire sample period. The sample period for the test spans from January 2010 to December 2023.

	High SC	2	3	4	5	6	7	8	9	Low SC
SC	1239.237	854.894	662.111	517.126	399.812	301.393	218.925	147.102	85.672	32.118
$LnSize$	17.098	16.726	16.509	16.314	16.146	16.021	15.921	15.845	15.767	15.711
NOA	41.838	29.598	22.923	17.794	13.306	9.971	7.191	4.966	3.117	1.671
$CSMOM$	23.370	19.620	18.870	19.100	19.400	17.880	17.910	17.290	14.670	17.520
$AbnVolume$	1.794	1.211	1.000	0.827	0.756	0.700	0.671	0.666	0.630	0.756
$AbsCAR$	4.360	4.200	4.080	3.990	3.870	3.880	3.930	3.800	3.750	3.810
$SVIC$	11759	10800	10480	10002	9783	9731	9560	9776	9952	10329
$SVIA$	48305	42424	38768	34493	30579	28417	27257	26888	26503	27723
$InstProp$	49.253	47.619	46.837	46.203	44.552	43.883	43.406	43.404	43.043	42.350
LoA	5.204	5.510	5.724	5.972	6.233	6.478	6.702	6.862	6.950	7.020
$QFIIProp$	0.767	0.650	0.615	0.581	0.500	0.470	0.422	0.416	0.383	0.363
$DFIE$	10.9	10.4	8.8	8.2	8.5	8.1	8.3	7.5	6.9	6.6
$DSOE$	39.8	40.0	40.1	39.7	40.0	41.3	42.9	45.3	46.9	48.3
$DDPE$	48.5	48.8	50.4	51.3	51.0	50.2	48.4	46.7	45.6	44.1

We also find that high- SC stocks have a significantly higher institutional ownership proportion ($InstProp$), which decreases monotonically as SC declines. A similar monotonic pattern emerges in the relationship between LoA and SC . High- SC stocks correspond to smaller LoA values, meaning lower arbitrage costs, while low- SC stocks have significantly higher arbitrage costs. These results indicate that the stock leaders captured by SC not only attract widespread attention but also involve more efficient information processing by investors and lower costs in actual trading. These together ensure that relevant information about high- SC stocks is incorporated into prices more quickly.

We next examine the relationship between SC and a firm's ownership structure. As shown

in the last four rows of the table, patterns in ownership structure across SC reveal significant differences in how firms of various types embed within the information network. Specifically, the prevalence of FIEs exhibits a monotonically decreasing trend from the highest to the lowest SC deciles. We observe that the $QFIIProp$ follows a similar trend. In stark contrast, the proportion of SOEs exhibits a monotonically increasing trend as we move from the highest- SC to the lowest- SC deciles. The proportion of DPEs displays a non-linear, inverted U-shaped relationship with SC .

We further compare the results of specific proportions with the benchmarks. For the high- SC portfolio, we observe that the proportions of FIEs, SOEs, and DPEs are 10.9%, 39.8%, and 48.5%, respectively. Relative to the average levels (see the last three rows of [Table D.3](#), which are 7.8%, 44.2%, and 47.5%, respectively), the proportion of FIEs is “unusually” high, while that of SOEs is lower. Although DPEs still account for the highest proportion, comparisons reveal that high international-exposure firms are more likely to concentrate in the top tiers of the network, and most centrally connected firms in the network are less likely to be SOEs. Overall, the clear stratification of ownership types across the SC spectrum strongly supports our hypothesis that the analyst network is shaped by a firm’s integration into global capital markets and its domestic institutional role.

5 Identification of laggards

5.1 Full-sample analysis

Recall that our SC weighting scheme is designed to capture the channel through which information transmits from fast-information-incorporating stocks to slow-information-incorporating stocks. Given that our results above show low- SC firms attract relatively less attention, exhibit lower information processing efficiency, and face higher arbitrage costs, it is reasonable to expect more pronounced cross-firm momentum effects when laggards are identified by SC . In other words, we aim to test whether cross-firm momentum effects primarily occur when

Table 8 The performance of portfolios formed by dependent sorting on SC and $CFRet^{SC}$

This table presents the results of bivariate portfolios formed by dependent double sorting of one control variable — SC depicted in Eq. (10) and $CFRet^{SC}$ depicted in Eq. (1) during January 2010 and December 2023. Each month, the individual stocks are first divided into five quintile groups according to their respective SC values. This process is further followed by an additional set of quintile portfolios constructed by sorting on $CFRet^{SC}$. One-month-ahead excess returns of the portfolios are presented. “High” (“Low”) designates portfolios with the highest (lowest) SC or $CFRet^{SC}$ values. The last rows report the $CFRet^{SC}$ -sorted long-short portfolio’s return spread (“High–Low”) and associated alphas within each SC quantile. Specifically, SIM alpha is obtained from Sharpe (1963) single-index model (SIM) which consists of the market factor (MKT) only. CH4 alpha is obtained from the CH4 model of Liu et al. (2019). q^5 alpha is obtained from the q^5 -factor model of Hou et al. (2021). FF6 alpha is obtained from the six-factor model of Fama and French (2018). Pricing factors above are specifically constructed for the Chinese market, which are detailed in Section A of the Online Appendix. Left and right panels present the equal-weighted (EW) and value-weighted (VW) results, respectively. Newey and West (1987) t -statistics are provided in the parentheses. The results shown in bold signify their statistical significance at the 5% level.

	EW					VW				
	High SC	2	3	4	Low SC	High SC	2	3	4	Low SC
High $CFRet^{SC}$	0.828 (1.45)	0.716 (1.27)	1.177 (2.00)	1.135 (1.88)	1.115 (1.80)	0.731 (1.29)	0.561 (0.93)	0.882 (1.50)	0.390 (0.64)	0.561 (0.91)
2	0.475 (0.92)	0.605 (1.21)	0.741 (1.38)	0.808 (1.41)	0.779 (1.35)	0.377 (0.74)	0.308 (0.68)	0.610 (1.05)	0.650 (1.17)	0.289 (0.48)
3	0.507 (0.94)	0.548 (1.06)	0.676 (1.24)	0.698 (1.34)	0.464 (0.86)	0.287 (0.54)	0.295 (0.62)	0.050 (0.10)	0.223 (0.43)	0.102 (0.18)
4	0.422 (0.80)	0.153 (0.29)	0.420 (0.76)	0.548 (1.02)	0.548 (1.06)	0.182 (0.35)	−0.068 (−0.14)	0.166 (0.30)	0.108 (0.21)	0.295 (0.54)
Low $CFRet^{SC}$	0.448 (0.81)	0.349 (0.65)	0.425 (0.76)	0.417 (0.80)	0.392 (0.69)	0.270 (0.55)	0.058 (0.12)	0.076 (0.14)	−0.019 (−0.03)	−0.032 (−0.06)
High–Low	0.380 (0.89)	0.367 (0.86)	0.752 (1.93)	0.718 (2.26)	0.724 (2.54)	0.461 (1.09)	0.502 (1.13)	0.806 (1.91)	0.409 (1.17)	0.593 (2.12)
SIM alpha	0.397 (0.93)	0.388 (0.91)	0.764 (1.97)	0.715 (2.25)	0.726 (2.51)	0.474 (1.11)	0.509 (1.15)	0.821 (1.96)	0.406 (1.15)	0.592 (2.09)
CH4 alpha	0.482 (1.01)	0.573 (1.29)	0.904 (2.29)	0.891 (2.79)	0.937 (3.38)	0.594 (1.25)	0.635 (1.31)	0.889 (2.02)	0.602 (1.64)	0.823 (2.91)
q^5 alpha	0.666 (1.30)	0.868 (1.77)	1.300 (3.14)	1.079 (2.92)	1.121 (2.99)	0.687 (1.28)	0.935 (1.78)	1.389 (3.32)	0.729 (1.68)	0.949 (2.49)
FF6 alpha	0.564 (1.09)	0.749 (1.53)	1.103 (2.59)	1.015 (2.91)	1.101 (3.47)	0.607 (1.15)	0.766 (1.51)	1.043 (2.19)	0.722 (1.80)	0.984 (3.04)

information flows from leaders to laggards; in contrast, such effects are expected to weaken or diminish in the reverse direction, that is, from laggards to leaders.

To test this conjecture, we perform a bivariate portfolio analysis. Specifically, we first divide stocks into five quintile groups based on their SC values. Then, within each SC quintile, we further sort stocks into five quintile groups based on $CFRet^{SC}$. This allows

us to examine the differences in cross-firm return predictability across firm groups with varying degrees of importance, as proxied by SC . We expect to observe significant return predictability based on $CFRet^{SC}$ only in the low SC quantiles, which would also indicate the presence of cross-firm momentum effects. The results are presented in [Table 8](#).

First, in the left panel presenting the equal-weighted returns of portfolios, we observe that the $CFRet^{SC}$ -sorted long-short portfolios yield statistically significant returns only in SC group 3 and lower SC groups, at the 5% level. Furthermore, significant return spreads and notable alphas are observed solely in SC group 4 and the Low group. From the perspective of return magnitude, the long-short portfolio returns exhibit a strong non-linear relationship across different SC groups. Notably, we find that the return spreads and alphas for the long-short portfolios in the High SC group and SC group 2 are considerably lower. For instance, in terms of the raw return spread and SIM alpha, the monthly average returns for these two groups are both below 0.4%, whereas the results for SC groups 3, 4, and 5 exceed a monthly average return of 0.7%. Similar findings emerge from the risk-adjusted results using CH4, q^5 , and FF6. It can be observed that the significant returns primarily stem from improved performance in the long positions. Specifically, the returns generated by High $CFRet^{SC}$ portfolios in SC groups 1 and 2 range from 0.7% to 0.83% per month. In contrast, in SC groups 3 to 5, these returns sharply increase to over 1.11%, while the returns on the short side, represented by Low $CFRet^{SC}$ portfolios, do not exhibit significant changes across the SC quantiles. These findings align with our expectations that identifying laggards allows for more pronounced cross-firm momentum effects.

In the results based on value-weighted (VW) returns, as shown in the right panel, our primary findings remain consistent. Specifically, we observe significant results only in SC groups 3 and 5. Notably, only in SC group 5 does the $CFRet^{SC}$ -sorted long-short portfolio generate a significant return spread, along with notable alphas across all pricing models considered. Given that the value-weighted returns primarily reflect the monotonic trend of market equity, the return magnitude also shows greater non-linearity.

While the other four weighting schemes do not assign proper weights to the focal firm’s connected leaders and laggards (otherwise, we should expect to observe more significant cross-firm momentum effects), they still provide some information related to this distinction. This is also indicated by the results from the univariate portfolio analysis and the FM regressions. We therefore expect to observe more pronounced cross-firm momentum effects for the laggards when applying the other four weighting schemes. These results are presented in Section C of the Online Appendix. Overall, our findings are consistent: in the low SC quantile groups, corresponding to laggards, the $CFRet$ -sorted long-short portfolios under different weighting schemes exhibit significant return performance. In other words, cross-firm momentum effects become more pronounced when laggards are identified.¹⁶

5.2 Sub-sample analysis

In the results of the previous section, we observe distinct distribution patterns of firms with different ownership structures across ten SC deciles. In particular, high- SC stocks contain a relatively higher proportion of firms with international exposure and a relatively lower proportion of SOEs. Along with the finding that cross-firm momentum effects primarily occur from leaders to laggards, an intriguing question arises: how does peer information transmit among firms with different ownership structures? Specifically, which type of high- SC stocks transmits peer information, and which type of low- SC stocks receives such information and thereby earn a risk premium for delayed adjustment to information?

To address the above concerns, we conduct a detailed bivariate portfolio analysis on

¹⁶It is also worth noting that our result above does not contradict the findings in [Ali and Hirshleifer \(2020\)](#). In fact, these two results are not comparable. [Ali and Hirshleifer \(2020\)](#) employ the degree and eigenvector centralities of firms to characterize the complexity of firm linkages. Through the interaction terms of the FM regression, the authors find that when the above-mentioned two measures are high, attention and cognitive processing costs are higher, and thus the cross-firm momentum effect is more significant. We emphasize that there are significant differences between the degree centrality and strength centrality. The strength centrality we adopt considers the weights of firm connections, i.e., the number of common analysts, while the degree centrality only counts the number of distinct firms connected to the focal firm. And our unreported findings indicate that throughout the entire sample period, the groups consisting of 86 stocks (corresponding to the average number of stocks following quintile-quintile grouping) with the highest degree and strength centralities have only an average of 51.67% identical samples in common.

partitioned subsamples. We first segment the entire universe of stocks into three distinct groups based on their ultimate ownership: state-owned enterprises (SOEs), domestic private enterprises (DPEs), and foreign-invested enterprises (FIEs). This classification allows us to treat each group as a separate set of focal firms for which we subsequently calculate future returns. For each focal firm in each ownership group, we construct three versions of the cross-firm momentum signal, $CFRet^{SC}$. Each version is computed using a different set of peer firms: one consisting exclusively of SOE peers, another of DPE peers, and a third of FIE peers. This 3×3 research design yields nine independent analyses, enabling us to disentangle whether information spillovers are more potent within the same ownership class or across different classes. Within each of the nine sub-samples, we perform a bivariate dependent sorting. For a given subsample (e.g., DPE focal firms using SOE peers), we first sort all individual stocks into two portfolios on the basis of SC using the median as the breakpoint. Then, within each SC portfolio, we perform a second sort into five quintile portfolios based on their $CFRet^{SC}$ values. Similarly, we record the one-month-ahead excess equal-weighted return and the corresponding alphas for different $CFRet^{SC}$ -sorted long-short portfolios. The results are presented in [Table 9](#).

Our sub-sample analysis based on ownership structure unveils the heterogeneous roles played by different types of firms in the cross-firm information diffusion process, clarifying the source of the aggregate momentum effect. The results indicate that low- SC DPEs act as the primary recipients of delayed information. This is evidenced by the significant predictive power of $CFRet^{SC}$ constructed from all three types of peer firms (SOE, DPE, and FIE) for their future returns. Their position on the periphery of the information network and likely higher informational frictions make them the slowest to incorporate relevant news, thus constituting the main driver of the observable momentum.

In contrast, low- SC SOEs exhibit a moderate delay, primarily in response to information originating from SOE and FIE peers (as shown by significant alphas in some models). This partial integration suggests that while SOEs are not entirely immune to information

Table 9 The results of the bivariate portfolio analysis using sub-samples

This table presents the results of a bivariate portfolio analysis conducted on sub-samples partitioned by firm ownership type: state-owned enterprises (SOEs), domestic private enterprises (DPEs), and foreign-invested enterprises (FIEs). For each focal firm type, we construct three distinct cross-firm momentum signals ($CFRet^{SC}$) using peers from each ownership group (SOE, DPE, and FIE peers), resulting in a 3×3 design. Within each of the nine sub-samples, stocks are first divided into two groups according to the median of their SC values. Within the high- SC and low- SC groups, the stocks are further sorted into five quintile portfolios based on their $CFRet^{SC}$ values. The table reports the one-month-ahead excess equal-weighted returns ("High-Low") and the corresponding alphas for the $CFRet^{SC}$ -sorted long-short portfolio within each SC group. The pricing models here are consistent with those in Table 8. Newey and West (1987) t -statistics are shown in parentheses. Estimates significant at the 5% level are highlighted in bold. The sample period spans from January 2010 to December 2023.

	SOEs		DPEs		FIEs	
	High SC	Low SC	High SC	Low SC	High SC	Low SC
Panel A: SOE peers						
High-Low	0.495 (1.54)	0.419 (1.55)	0.452 (1.34)	0.603 (2.30)	0.405 (0.84)	0.323 (0.92)
SIM alpha	0.485 (1.54)	0.396 (1.51)	0.464 (1.37)	0.595 (2.28)	0.452 (0.93)	0.335 (0.95)
CH4 alpha	0.896 (2.55)	0.658 (2.90)	0.827 (2.51)	0.762 (2.69)	0.730 (1.34)	0.529 (1.50)
q^5 alpha	0.478 (1.73)	0.406 (1.22)	0.438 (1.29)	0.481 (1.47)	0.704 (1.36)	0.640 (1.50)
FF6 alpha	0.655 (1.99)	0.538 (2.10)	0.723 (2.08)	0.768 (2.65)	0.705 (1.32)	0.661 (1.81)
Panel B: DPE peers						
High-Low	0.447 (1.42)	0.304 (1.18)	0.329 (0.84)	0.797 (2.61)	0.093 (0.19)	0.408 (0.90)
SIM alpha	0.432 (1.40)	0.287 (1.16)	0.342 (0.87)	0.786 (2.59)	0.096 (0.20)	0.378 (0.83)
CH4 alpha	0.708 (2.00)	0.378 (1.51)	0.585 (1.49)	0.802 (2.59)	0.297 (0.58)	0.730 (1.60)
q^5 alpha	0.218 (0.82)	0.232 (0.83)	0.284 (0.68)	0.650 (2.04)	0.277 (0.57)	0.820 (1.73)
FF6 alpha	0.446 (1.44)	0.392 (1.67)	0.627 (1.59)	0.933 (2.98)	0.327 (0.69)	0.874 (1.82)
Panel C: FIE peers						
High-Low	0.292 (1.06)	0.315 (1.24)	-0.019 (-0.06)	0.584 (2.30)	0.277 (0.62)	0.356 (0.88)
SIM alpha	0.301 (1.10)	0.314 (1.24)	-0.003 (-0.01)	0.589 (2.31)	0.267 (0.59)	0.358 (0.87)
CH4 alpha	0.523 (1.72)	0.566 (2.34)	0.235 (0.70)	0.719 (2.80)	0.547 (1.07)	0.439 (1.03)
q^5 alpha	0.152 (0.46)	0.366 (1.34)	0.079 (0.21)	0.377 (1.35)	0.465 (0.86)	0.350 (0.73)
FF6 alpha	0.457 (1.40)	0.582 (2.09)	0.307 (0.91)	0.805 (2.94)	0.581 (1.19)	0.656 (1.55)

spillovers, their distinct institutional setting and policy-driven nature modulate the diffusion process.

Most importantly, we find no statistically significant monthly return predictability for low-*SC* FIEs using any peer group. This null result is not a failure of the measure but a crucial confirmation of their role as information leaders. It suggests that information relevant to FIEs is incorporated into their prices with relative speed, leaving little delayed reaction to be captured at the monthly frequency. This finding aligns with the view that FIEs, characterized by higher transparency and scrutiny from sophisticated global investors, reside in the most informationally efficient segment of the market.

In summary, the aggregate cross-firm momentum phenomenon is predominantly fueled by the slow information diffusion toward the peripheral DPEs. The market exhibits a multi-speed information incorporation process, where the pace is fastest for internationally exposed FIEs and slowest for lesser-known private firms, with SOEs occupying an intermediate space.

6 Further tests

In this section, we further consider several alternative explanations regarding our *SC*-based cross-firm momentum via shared analyst coverage.¹⁷ We also examine the results based on an alternative network centrality. For the following tests, we rely primarily on the Fama-Macbeth (FM) framework, where the left-hand-side (LHS) variable is always a stock's monthly excess return and all right-hand-side (RHS) variables are winsorized at the first and 99th percentiles. To facilitate interpretation, we scale the coefficients so that they represent the change in a stock's monthly average excess return associated with a one-standard-deviation increase in each RHS variable.

¹⁷We thank the anonymous reviewers for motivating us to consider these issues.

6.1 Intra-industry cross-firm momentum

Hou (2007) proposes an industry-based channel of information diffusion. The author documents that information diffuses from large firms to small ones within the same industry. A potential concern is that our SC weighting scheme that aims to identify stock leaders, may capture a mechanism similar to the role of industry leaders as proposed by Hou (2007), or intra-industry cross-firm momentum effects. Although the evidence presented in Figure 1 and Table 1 suggests that industry leadership is less likely to drive our results, we further design a series of tests to disentangle these effects.

First, we construct industry-based peer returns. Specifically, we adopt the CSRC’s 2012 version of the listed company classification method to identify peer firms for each focal firm, and compute the value-weighted average return, denoted as $CFRet^{Ind}$. We aim to examine whether $CFRet^{Ind}$ explains the return predictability of $CFRet^{SC}$. In a further design, to fully eliminate the influence of industry leaders, we reconstruct $CFRet^{SC}$ by excluding firms in the same industry that are directly connected to the focal firm and test whether this reconstructed measure retains its return predictability. Correspondingly, we also construct an alternative $CFRet^{Ind}$ that removes common analyst connections and retains only within-industry connections.

The results appear in the first six columns of Table 10. Results in column (1) show that a one-standard-deviation increase in $CFRet^{Ind}$ generates a monthly average return of 0.275% ($t = 3.22$), indicating the presence of significant intra-industry cross-firm momentum in the market. When we include $CFRet^{Ind}$ as an additional control variable, as in column (2), we observe that the return predictive power of $CFRet^{SC}$ remains significant, with a one-standard-deviation increase in this variable leading to an average return of 0.302% per month ($t = 4.29$). Meanwhile, the slope of $CFRet^{Ind}$ also retains statistically significant. These findings align with prior evidence (Duan et al., 2022; Yi and Guo, 2022; Jiang et al., 2023; Xiong et al., 2023; Ge et al., 2025a).

In column (3), we test the performance of $CFRet^{SC}$ reconstructed after excluding peers

Table 10 The results of alternative explanations to $CFRet^{SC}$ -based return predictability

This table presents results from the cross-sectional regressions of future equity returns on the $CFRet$ -sorted long-short portfolios and various control variables between 2010 and 2023. All right-hand-side variables are winsorized at the first and 99th percentiles. The related control variables are introduced in Section 4. The coefficients reported represent time-series averages obtained from monthly Fama and MacBeth (1973) regressions, and the corresponding Newey and West (1987) t -statistics are presented in parentheses. For comparison, these coefficients are scaled to be interpreted as the change in a stock's monthly average excess return resulting from a one-standard-deviation increase in each right-hand-side variable. The results shown in bold signify their statistical significance at the 5% level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$CFRet^{SC}$		0.302								0.333	0.287	0.341
		(4.29)								(2.04)	(2.18)	(2.82)
$CFRet^{Ind}$	0.275	0.172		0.239								
	(3.22)	(2.62)		(3.25)								
$CFRet^{SC}$ (w.o. co-industry links)			0.315	0.261		0.291						
			(4.86)	(4.95)		(4.99)						
$CFRet^{Ind}$ (w.o. co-coverage links)					0.143	0.109						
					(1.94)	(1.71)						
$CFRet^{ID3}$						0.366				0.045		
						(4.05)				(0.27)		
$CFRet^{ID6}$								0.365			0.086	
								(3.91)			(0.57)	
$CFRet^{ID12}$									0.355			0.029
									(3.77)			(0.19)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj- R^2	0.08	0.10	0.09	0.10	0.08	0.09	0.09	0.09	0.09	0.10	0.10	0.10

within the same industry, and we find that $CFRet^{SC}$ retains its significant return predictive power in this case. Even when including $CFRet^{Ind}$ as an additional control, as shown in column (4), a one-standard-deviation increase in $CFRet^{SC}$ still generates a monthly average return of 0.261% ($t = 4.95$). In contrast, if we exclude peers based on shared analyst coverage, the performance of reconstructed $CFRet^{Ind}$ deteriorates substantially. Column (5) shows that its slope declines to 0.143, a value equal to half of its original magnitude reported in column (1), and it no longer achieves statistical significance at the 5% level. When we further include $CFRet^{SC}$ (reconstructed by excluding same-industry peers) in this specification, $CFRet^{SC}$ still maintains significant return predictive power, with a one-standard-deviation increase leading to a monthly average return of 0.291%.

The findings above firmly indicate that our SC weighting scheme captures a unique

information transmission channel that differs significantly from the traditional industry-based channel.

6.2 Information discreteness

Another alternative explanation is related to cognitive trigger of investors. In light of [Huang et al. \(2022\)](#), in lead-lag relationships, when the return information of the leading firm is discrete (such as that associated with typical news events), it triggers the attention of investors in lagging firms, leading to rapid information absorption and a weak lag effect. In contrast, when the return information of the leading firm is continuous (unlike that associated with typical news events), investors remain inattentive, resulting in insufficient information response and a strong lag effect.

To examine how the theory of information discreteness affects the robustness of our results, we follow [Da et al. \(2014\)](#) and [Huang et al. \(2022\)](#) to define information discreteness ID as follows:

$$ID_{i,t} = \text{sgn}(CRET) \times [\%neg_{i,t} - \%pos_{i,t}], \quad (16)$$

where the cumulative return of $CFRet^{SC}$ during the past J -month ($J = 3, 6, 12$) period is denoted $CRET$. The sign of $CRET$ is denoted $\text{sgn}(CRET)$ and equals: $+1$ when $CRET > 0$ and -1 when $CRET < 0$. The percentage of days during the past J months with positive and negative returns are denoted $\%pos$ and $\%neg$, respectively. And a high ID value indicates a high degree of information discreteness, while a low ID indicates a high degree of information continuity.

We first conduct a bivariate portfolio analysis following [Huang et al. \(2022\)](#). Specifically, we sort individual stocks into high and low groups on the basis of ID values using the median as the breakpoint. Within each group, we further sort the stocks into five quintile groups based on $CFRet^{SC}$. If the theory of information discreteness explains the cross-firm momentum effects we document using the SC weighting scheme, the return predictability of

Table 11 The results of the bivariate portfolio analysis for information discreteness

This table presents the results of bivariate portfolios formed by dependent double sorting of ID depicted in Eq. (16) and $CFRet^{SC}$ depicted in Eq. (1) during January 2010 and December 2023. We consider three ID measures with different estimation periods, specifically 3 months, 6 months, and 12 months. Each month, the individual stocks are first split into high and low ID groups according to the median of ID values. This process is further followed by an additional set of five quintile portfolios constructed by sorting on $CFRet^{SC}$. We obtain one-month-ahead equal-weighted returns of the portfolios, and then present their alphas in the table. Factor models used here are consistent with those in Table 8. Newey and West (1987) t -statistics are provided in the parentheses. The results shown in bold signify their statistical significance at the 5% level.

	3-month		6-month		12-month	
	High ID	Low ID	High ID	Low ID	High ID	Low ID
SIM alpha	0.753 (2.02)	0.636 (1.47)	0.773 (2.07)	0.525 (1.20)	0.527 (1.45)	0.732 (1.63)
CH4 alpha	1.046 (2.34)	1.023 (2.12)	1.183 (2.54)	0.761 (1.75)	1.043 (2.24)	1.098 (2.11)
q^5 alpha	0.651 (1.65)	0.750 (1.57)	0.903 (2.30)	0.185 (0.36)	0.573 (1.54)	0.587 (1.23)
FF6 alpha	0.980 (2.30)	0.824 (1.69)	1.122 (2.55)	0.634 (1.43)	0.893 (2.24)	0.948 (1.94)

$CFRet^{SC}$ is expected to be significantly stronger in the low ID group than in the high ID group.

As reported in Table 11, we do not observe more significant profits on $CFRet^{SC}$ -sorted long-short portfolios in the low ID group. Instead, the results are more significant in the high ID group, where the portfolios generate higher alphas, particularly when the estimation period equals 3 or 6 months. When the estimation period changes to 12 months, the alphas of the long-short portfolios are higher in the low ID group, but these results lack statistical significance.

To further clarify the relationship between ID and SC , we construct $CFRet$ based on the ID measure in a similar manner. Since ID ranges between -1 and 1 and we require higher weights for low- ID stocks, we therefore use $1 - ID$ to derive appropriate weights. For the three different estimation periods, we then obtain $CFRet^{ID3}$, $CFRet^{ID6}$, and $CFRet^{ID12}$.

Columns (7) to (9) of Table 10 present results from univariate FM regressions, each including a distinct ID -based $CFRet$ measure. And our results confirm that they all possess

significant return predictability. Specifically, a one-standard-deviation increase in each ID -based $CFRet$ generates at least a 0.355% monthly average return, with adjusted t -statistics above 3.7. And these results outperform even those of $CFRet^{SC}$. In contrast, columns (10) to (12) demonstrate that adding $CFRet^{SC}$ to these regressions substantially reduces the return predictability of $CFRet^{ID3}$, $CFRet^{ID6}$, and $CFRet^{ID12}$. Their associated coefficients approach zero and lose statistical significance. By comparison, $CFRet^{SC}$ retains both economic and statistical significance, as its one-standard-deviation increase generates at least a significant 0.287% monthly average return. Our findings above indicate that cross-firm momentum based on the SC weighting scheme cannot be explained by the theory of information discreteness.

6.3 The schemes based on attention metrics

According to the construction of SC , it incorporates both fundamental linkages information and measures a peer firm’s systemic importance within the information intermediary network, which differs significantly from the information contained in traditional attention metrics (e.g., firm size or the number of analysts covering the firm). However, our results show that a firm’s SC is indeed closely related to common attention metrics, and this observation raises a natural question: do weighting schemes based on these attention metrics perform better than their SC -based counterpart?

To address this issue, we construct their respective $CFRet$ measures following the approach used to build $CFRet^{SC}$, which relies on the attention metrics presented in [Table 7](#), namely *Size*, *NOA*, *CSMOM*, *AbnVolume*, *AbsCAR*, *SVIC*, and *SVIA*.

Our first specification considers the univariate FM regression model where the RHS variable includes only the $CFRet$ constructed from each attention metric as described above. Our results appear in the left panel of [Table 12](#). Clearly, all $CFRet$ based on attention-related metrics retain significantly positive return predictability. Specifically, a one-standard-deviation change in the coefficient generates an average monthly return of at least 0.277%,

Table 12 Results of the return predictability of $CFRet$ constructed using attention-related metrics

This table reports the coefficients and t -statistics derived from Fama-Macbeth (FM) regressions mainly based on $CFRet$ constructed using attention-related metrics X , as listed in the first column. The left-hand-side variable remains one-month-ahead stock excess returns (ER), while the right-hand-side (RHS) variables vary in different panels. The left panel presents the results of univariate FM regression, which is given by:

$$\text{Model 1 : } ER = \alpha + \beta_1 CFRet^X + Controls,$$

where the right-hand-side variable is each $CFRet$. Panel B showcases the results of various bivariate FM regressions, which is given by:

$$\text{Model 2 : } ER = \alpha + \beta_1 CFRet^{SC} + \beta_2 CFRet^X + Controls,$$

with the RHS variables being the combination of $CFRet^{SC}$ and attention-related $CFRet^X$. The reported coefficients represent time-series averages obtained from monthly Fama and MacBeth (1973) regressions, and the corresponding Newey and West (1987) t -statistics are presented in parentheses. For comparison, these coefficients are scaled to be interpreted as the change in a stock's monthly average excess return resulting from a one-standard-deviation increase in each right-hand-side variable. Definitions of the firm attributes are detailed in Section D of the Online Appendix. The sample for the test is January 2010 to December 2023.

X	Model 1		Model 2			
	$CFRet^X$	t -stat	$CFRet^{SC}$	t -stat	$CFRet^X$	t -stat
<i>Size</i>	0.277	(3.62)	0.393	(2.66)	-0.025	(-0.20)
<i>NOA</i>	0.341	(3.83)	0.651	(2.53)	-0.292	(-1.17)
<i>CSMOM</i>	0.843	(1.99)	0.374	(4.13)	-0.149	(-0.49)
<i>AbnVolume</i>	0.299	(3.81)	0.355	(2.70)	0.021	(0.19)
<i>AbsCAR</i>	0.278	(3.32)	0.354	(3.35)	-0.003	(-0.02)
<i>SVIC</i>	0.282	(3.43)	0.419	(3.27)	-0.056	(-0.53)
<i>SVIA</i>	0.344	(3.59)	0.369	(2.68)	0.032	(0.25)

and most adjusted t -statistics exceed three.

In the second specification, we include $CFRet^{SC}$ and each attention-related $CFRet$ as the RHS variables simultaneously to examine whether their return predictability is affected. As shown in the right panel of the table, the slope coefficients of $CFRet^{SC}$ all remain significant at the 5% level. A one-standard-deviation increase in the coefficient generates an average monthly return of at least 0.354%, and all adjusted t -statistics exceed 2.53. At the same time, the other $CFRet$ measures lose their return predictability. Our results confirm that $CFRet^{SC}$ can subsume the predictive power of these attention-related $CFRet$ measures, thus indicating the outperformance of our SC weighting scheme.

6.4 An alternative centrality

The logic underlying our construction of the SC weighting scheme is that it captures both a firm’s fundamental linkage with its focal firm and the degree of attention it receives from analysts and the market. Following this logic, any alternative weighting scheme that captures fundamental linkages and potentially describes the degree of attention the firm receives more precisely should be expected to outperform the SC weighting scheme. We test this hypothesis further.

Specifically, while the SC captures the influence of a firm’s directly connected peers, it fails to fully account for broader indirect influences that extend beyond these direct relationships or the impact of *peers of peers*. To address this limitation, we introduce *PageRank* (PR) centrality to serve this purpose (Page et al., 1999). By taking into account not only the direct connections of a firm but also its peers’ connections, thereby capturing the *indirect* influence exerted through the network.

To ensure the robustness of our findings above, we construct PR -based $CFRet$, i.e., $CFRet^{PR}$, and re-conduct the tests. The results and associated discussion appear in detail in Section E of the Online Appendix. Our results demonstrate that raw returns and alphas of $CFRet^{PR}$ -sorted long-short portfolio range from 0.82% to 1.35% per month, which surpass the performance of the $CFRet^{SC}$ -sorted long-short portfolio (see Table 2). Moreover, the results of FM regressions confirm that $CFRet^{PR}$ has superior return forecasting ability, which cannot be explained by four classic $CFRet$ versions and related firm-level characteristics.

7 Conclusions

This study introduces a network centrality-based approach that refines the detection of cross-firm momentum by explicitly accounting for a firm’s informational leadership within analyst coverage networks. We demonstrate that weighting peer returns by their strength centrality (SC), which is a measure of their overall connectedness in the network, significantly enhances

the predictability of future returns. Unlike conventional common-analyst-based measures, the *SC*-weighted strategy produces robust and persistent returns even after controlling for common risk factors and the related firm-level characteristics. Its performance also cannot be subsumed by strategies based on alternative weighting schemes or by explanations such as intra-industry cross-firm momentum and information discreteness.

The effectiveness of *SC* stems from its ability to pinpoint firms that serve as informational hubs. High-*SC* firms attract greater attention from analysts and institutional investors, operate with lower arbitrage constraints, and incorporate information more rapidly into their stock prices. As a result, returns of these central firms lead and positively predict returns of less central firms connected through shared coverage. More importantly, we document a clear directional effect: cross-firm momentum spreads from high-*SC* leaders to low-*SC* laggards, with no evidence of reverse spillovers. This leader-laggard mechanism, crystallized by the *SC* measure, is likely to be a fundamental feature of markets with concentrated information flow and heterogeneous investors.

Our findings offer a novel perspective on how information travels through financial markets and provide practical insights for constructing trading strategies and managing systematic risk in markets where information asymmetry is prominent. Our findings suggest that cross-firm momentum may be hidden (not absent) in many international markets due to methodological limitations in existing approaches. The *SC*-based method provides a generalizable way to capture leadership in networks and improves upon existing measures by emphasizing economic significance over mere connectedness. For global investors, it offers a real-time, network-informed tool to identify early movers and delayed adjusters even in data-scarce settings. For policymakers, it underscores how financial openness and institutional design shape a firm’s role in the information ecosystem.

Our future studies aim to extend this approach to other novel weighted network structures, such as those based on news co-mentions (Ge et al., 2025a) or common investment concepts (Du et al., 2022), to further explore the architecture of information diffusion in

global and cross-border markets.

References

- Albert, R. and Barabási, A.L., 2002. Statistical mechanics of complex networks. *Rev. Mod. Phys.* 74, 47. doi:[10.1103/RevModPhys.74.47](https://doi.org/10.1103/RevModPhys.74.47).
- Ali, U. and Hirshleifer, D., 2020. Shared analyst coverage: Unifying momentum spillover effects. *J. Financ. Econ.* 136, 649–675. doi:[10.1016/j.jfineco.2019.10.007](https://doi.org/10.1016/j.jfineco.2019.10.007).
- Amihud, Y., 2002. Illiquidity and stock returns: cross-section and time-series effects. *J. Financ. Mark.* 5, 31–56. doi:[10.1016/S1386-4181\(01\)00024-6](https://doi.org/10.1016/S1386-4181(01)00024-6).
- Amihud, Y. and Mendelson, H., 1986. Asset pricing and the bid-ask spread. *J. Financ. Econ.* 17, 223–249. doi:[10.1016/0304-405X\(86\)90065-6](https://doi.org/10.1016/0304-405X(86)90065-6).
- Ando, T., Greenwood-Nimmo, M. and Shin, Y., 2022. Quantile connectedness: Modeling tail behavior in the topology of financial networks. *Manag. Sci.* 68, 2401–2431. doi:[10.1287/mnsc.2021.3984](https://doi.org/10.1287/mnsc.2021.3984).
- Asparouhova, E., Bessembinder, H. and Kalcheva, I., 2013. Noisy prices and inference regarding returns. *J. Financ.* 68, 665–714. doi:[10.1111/jofi.12010](https://doi.org/10.1111/jofi.12010).
- Barrat, A., Barthélemy, M., Pastor-Satorras, R. and Vespignani, A., 2004. The architecture of complex weighted networks. *Proc. Natl. Acad. Sci. U.S.A.* 101, 3747–3752. doi:[10.1073/pnas.0400087101](https://doi.org/10.1073/pnas.0400087101).
- Bhushan, R., 1989. Firm characteristics and analyst following. *J. Accounting Econ.* 11, 255–274. doi:[10.1016/0165-4101\(89\)90008-6](https://doi.org/10.1016/0165-4101(89)90008-6).
- Billio, M., Getmansky, M., Lo, A.W. and Pelizzon, L., 2012. Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *J. Financ. Econ.* 104, 535–559. doi:[10.1016/j.jfineco.2011.12.010](https://doi.org/10.1016/j.jfineco.2011.12.010).
- Chen, X., An, L., Wang, Z. and Yu, J., 2023. Attention spillover in asset pricing. *J. Financ.* 78, 3515–3559. doi:[10.1111/jofi.13281](https://doi.org/10.1111/jofi.13281).
- Clemente, G.P., Grassi, R. and Hitaj, A., 2021. Asset allocation: new evidence through network approaches. *Ann. Oper. Res.* 299, 61–80. doi:[10.1007/s10479-019-03136-y](https://doi.org/10.1007/s10479-019-03136-y).
- Cohen, L. and Frazzini, A., 2008. Economic links and predictable returns. *J. Financ.* 63, 1977–2011. doi:[10.1111/j.1540-6261.2008.01379.x](https://doi.org/10.1111/j.1540-6261.2008.01379.x).
- Curme, C., Tumminello, M., Mantegna, R.N., Stanley, H.E. and Kenett, D.Y., 2015. Emergence of statistically validated financial intraday lead-lag relationships. *Quant. Financ.* 15, 1375–1386. doi:[10.1080/14697688.2015.1032545](https://doi.org/10.1080/14697688.2015.1032545).

- Da, Z., Engelberg, J. and Gao, P., 2011. In search of attention. *J. Financ.* 66, 1461–1499. doi:[10.1111/j.1540-6261.2011.01679.x](#).
- Da, Z., Gurun, U.G. and Warachka, M., 2014. Frog in the pan: Continuous information and momentum. *Rev. Financ. Stud.* 27, 2171–2218. doi:[10.1093/rfs/hhu003](#).
- Diebold, F.X., Liu, L. and Yilmaz, K., 2017. Commodity connectedness. Working Paper w23685. National Bureau of Economic Research. doi:[10.3386/w23685](#).
- Diebold, F.X. and Yilmaz, K., 2014. On the network topology of variance decompositions: Measuring the connectedness of financial firms. *J. Econometrics* 182, 119–134. doi:[10.1016/j.jeconom.2014.04.012](#).
- Driskill, M., Kirk, M.P. and Tucker, J.W., 2020. Concurrent earnings announcements and analysts' information production. *Account. Rev.* 95, 165–189. doi:[10.2308/accr-52489](#).
- Du, Q., Liang, D., Chen, Z. and Tu, J., 2022. Concept links and return momentum. *J. Bank. Financ.* 134, 106329. doi:[10.1016/j.jbankfin.2021.106329](#).
- Duan, B., Wang, R. and Zhang, R., 2022. Economic links and stock returns in Chinese A-share market. *J. Financ. Res. (in Chinese)* 500, 171–188.
- Durand, R.B., Limkriangkrai, M. and Fung, L., 2019. Exogenous and endogenous attention and the convergence of analysts' forecasts. *J. Behav. Financ.* 20, 154–172. doi:[10.1080/15427560.2018.1504783](#).
- D'avolio, G., 2002. The market for borrowing stock. *J. Financ. Econ.* 66, 271–306. doi:[10.1016/S0304-405X\(02\)00206-4](#).
- El-Khatib, R., Fogel, K. and Jandik, T., 2015. CEO network centrality and merger performance. *J. Financ. Econ.* 116, 349–382. doi:[10.1016/j.jfineco.2015.01.001](#).
- Evgeniou, T., Peress, J., Vermaelen, T. and Yue, L., 2022. Network centrality and managerial market-timing ability. *J. Financ. Quant. Anal.* 57, 704–760. doi:[10.1017/S0022109021000016](#).
- Fama, E.F. and French, K.R., 2018. Choosing factors. *J. Financ. Econ.* 128, 234–252. doi:[10.1016/j.jfineco.2018.02.012](#).
- Fama, E.F. and MacBeth, J.D., 1973. Risk, return, and equilibrium: Empirical tests. *J. Polit. Econ.* 81, 607–636. doi:[10.1086/260061](#).
- Fang, L. and Peress, J., 2009. Media coverage and the cross-section of stock returns. *J. Financ.* 64, 2023–2052. doi:[10.1111/j.1540-6261.2009.01493.x](#).
- Feng, J., Huo, X., Liu, X., Mao, Y. and Xiang, H., 2025. Economic links from bonds and cross-stock return predictability. *J. Financ. Econ.* 171, 104110. doi:[10.1016/j.jfineco.2025.104110](#).
- Ge, S., Li, S. and Zheng, H., 2025a. Diamond cuts diamond: News co-mention momentum

- spillover prevails in China. *J. Bank. Financ.* 171, 107356. doi:[10.1016/j.jbankfin.2024.107356](https://doi.org/10.1016/j.jbankfin.2024.107356).
- Ge, S., Li, S. and Zheng, H., 2025b. Directional clarity in news co-mentions: Attention efficiency and cross-firm price discovery. *Available at SSRN 5130201*.
- Gomes, A., Gopalan, R., Leary, M.T. and Marcet, F., 2023. Analyst coverage networks and corporate financial policies. *Manag. Sci.* doi:[10.1287/mnsc.2023.4891](https://doi.org/10.1287/mnsc.2023.4891).
- Gu, M., Kang, W. and Xu, B., 2018. Limits of arbitrage and idiosyncratic volatility: Evidence from China stock market. *J. Bank. Financ.* 86, 240–258. doi:[10.1016/j.jbankfin.2015.08.016](https://doi.org/10.1016/j.jbankfin.2015.08.016).
- Harford, J., Jiang, F., Wang, R. and Xie, F., 2019. Analyst career concerns, effort allocation, and firms' information environment. *Rev. Financ. Stud.* 32, 2179–2224. doi:[10.1093/rfs/hhy101](https://doi.org/10.1093/rfs/hhy101).
- Hendershott, T., Livdan, D. and Schürhoff, N., 2015. Are institutions informed about news? *J. Financ. Econ.* 117, 249–287. doi:[10.1016/j.jfineco.2015.03.007](https://doi.org/10.1016/j.jfineco.2015.03.007).
- Henderson, B.J. and Pearson, N.D., 2011. The dark side of financial innovation: A case study of the pricing of a retail financial product. *J. Financ. Econ.* 100, 227–247. doi:[10.1016/j.jfineco.2010.12.006](https://doi.org/10.1016/j.jfineco.2010.12.006).
- Hirshleifer, D. and Teoh, S.H., 2003. Limited attention, information disclosure, and financial reporting. *J. Accounting Econ.* 36, 337–386. doi:[10.1016/j.jacceco.2003.10.002](https://doi.org/10.1016/j.jacceco.2003.10.002).
- Hong, H. and Stein, J.C., 1999. A unified theory of underreaction, momentum trading, and overreaction in asset markets. *J. Financ.* 54, 2143–2184. doi:[10.1111/0022-1082.00184](https://doi.org/10.1111/0022-1082.00184).
- Hou, K., 2007. Industry information diffusion and the lead-lag effect in stock returns. *Rev. Financ. Stud.* 20, 1113–1138. doi:[10.1093/revfin/hhm003](https://doi.org/10.1093/revfin/hhm003).
- Hou, K., Mo, H., Xue, C. and Zhang, L., 2021. An augmented q -factor model with expected growth. *Rev. Financ.* 25, 1–41. doi:[10.1093/rof/rfaa004](https://doi.org/10.1093/rof/rfaa004).
- Hou, K. and Moskowitz, T.J., 2005. Market frictions, price delay, and the cross-section of expected returns. *Rev. Financ. Stud.* 18, 981–1020. doi:[10.1093/rfs/hhi023](https://doi.org/10.1093/rfs/hhi023).
- Hsu, J., Viswanathan, V., Wang, M. and Wool, P., 2018. Anomalies in Chinese A-shares. *J. Portf. Manag.* 44, 108–123. doi:[10.3905/jpm.2018.44.7.108](https://doi.org/10.3905/jpm.2018.44.7.108).
- Huang, S., Lee, C.M., Song, Y. and Xiang, H., 2022. A frog in every pan: Information discreteness and the lead-lag returns puzzle. *J. Financ. Econ.* 145, 83–102. doi:[10.1016/j.jfineco.2021.10.011](https://doi.org/10.1016/j.jfineco.2021.10.011).
- Israelson, R.D., 2016. Does common analyst coverage explain excess comovement? *J. Financ. Quant. Anal.* 51, 1193–1229. doi:[10.1017/S002210901600051X](https://doi.org/10.1017/S002210901600051X).
- Jansen, M., Swinkels, L. and Zhou, W., 2021. Anomalies in the China A-share market.

- Pacific-Basin Finance J.* 68, 101607. doi:[10.1016/j.pacfin.2021.101607](https://doi.org/10.1016/j.pacfin.2021.101607).
- Jiang, D., Peng, X. and Hongquan, Z., 2023. Shared analyst coverage and connected-firm momentum spillover in China. *Systems Engineering Theory Methodology Application (in Chinese)* 43, 1891–1909. doi:[10.12011/SETP2022-1974](https://doi.org/10.12011/SETP2022-1974).
- Kahneman, D., 1973. Attention and effort. volume 1063. Citeseer.
- Kaustia, M. and Rantala, V., 2021. Common analysts: Method for defining peer firms. *J. Financ. Quant. Anal.* 56, 1505–1536. doi:[10.1017/S0022109020000514](https://doi.org/10.1017/S0022109020000514).
- Kelley, E.K. and Tetlock, P.C., 2013. How wise are crowds? Insights from retail orders and stock returns. *J. Financ.* 68, 1229–1265. doi:[10.1111/jofi.12028](https://doi.org/10.1111/jofi.12028).
- Kenett, D.Y., Huang, X., Vodenska, I., Havlin, S. and Stanley, H.E., 2015. Partial correlation analysis: Applications for financial markets. *Quant. Financ.* 15, 569–578. doi:[10.1080/14697688.2014.946660](https://doi.org/10.1080/14697688.2014.946660).
- Koester, A., Lundholm, R. and Soliman, M., 2016. Attracting attention in a limited attention world: Exploring the causes and consequences of extreme positive earnings surprises. *Manag. Sci.* 62, 2871–2896. doi:[10.1287/mnsc.2015.2286](https://doi.org/10.1287/mnsc.2015.2286).
- Lee, C.M., Shi, T.T., Sun, S.T. and Zhang, R., 2024. Production complementarity and information transmission across industries. *J. Financ. Econ.* 155, 103812. doi:[10.1016/j.jfineco.2024.103812](https://doi.org/10.1016/j.jfineco.2024.103812).
- Lee, C.M., Sun, S.T., Wang, R. and Zhang, R., 2019. Technological links and predictable returns. *J. Financ. Econ.* 132, 76–96. doi:[10.1016/j.jfineco.2018.11.008](https://doi.org/10.1016/j.jfineco.2018.11.008).
- Liu, J., Stambaugh, R.F. and Yuan, Y., 2019. Size and value in China. *J. Financ. Econ.* 134, 48–69. doi:[10.1016/j.jfineco.2019.03.008](https://doi.org/10.1016/j.jfineco.2019.03.008).
- Menzly, L. and Ozbas, O., 2010. Market segmentation and cross-predictability of returns. *J. Financ.* 65, 1555–1580. doi:[10.1111/j.1540-6261.2010.01578.x](https://doi.org/10.1111/j.1540-6261.2010.01578.x).
- Merton, R.C., 1987. A simple model of capital market equilibrium with incomplete information. *J. Financ.* 42, 483–510. doi:[10.1111/j.1540-6261.1987.tb04565.x](https://doi.org/10.1111/j.1540-6261.1987.tb04565.x).
- Moskowitz, T.J. and Grinblatt, M., 1999. Do industries explain momentum? *J. Financ.* 54, 1249–1290. doi:[10.1111/0022-1082.00146](https://doi.org/10.1111/0022-1082.00146).
- Muslu, V., Rebello, M. and Xu, Y., 2014. Sell-side analyst research and stock comovement. *J. Account. Res.* 52, 911–954. doi:[10.1111/1475-679X.12057](https://doi.org/10.1111/1475-679X.12057).
- Newey, W.K. and West, K.D., 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelationconsistent covariance matrix. *Econometrica* 55, 703–708. doi:[10.2307/1913610](https://doi.org/10.2307/1913610).
- O'Brien, P.C. and Bhushan, R., 1990. Analyst following and institutional ownership. *J. Account. Res.* 28, 55–76. doi:[10.2307/2491247](https://doi.org/10.2307/2491247).

- Ozsoylev, H.N. and Walden, J., 2011. Asset pricing in large information networks. *J. Econ. Theory* 146, 2252–2280.
- Ozsoylev, H.N., Walden, J., Yavuz, M.D. and Bildik, R., 2014. Investor networks in the stock market. *Rev. Financ. Stud.* 27, 1323–1366. doi:[10.1093/rfs/hht065](https://doi.org/10.1093/rfs/hht065).
- Page, L., Brin, S., Motwani, R. and Winograd, T., 1999. The PageRank citation ranking: Bringing order to the web. Technical Report. Stanford infolab.
- Parsons, C.A., Sabbatucci, R. and Titman, S., 2020. Geographic lead-lag effects. *Rev. Financ. Stud.* 33, 4721–4770. doi:[10.1093/rfs/hhz145](https://doi.org/10.1093/rfs/hhz145).
- Peng, L. and Xiong, W., 2006. Investor attention, overconfidence and category learning. *J. Financ. Econ.* 80, 563–602. doi:[10.1016/j.jfineco.2005.05.003](https://doi.org/10.1016/j.jfineco.2005.05.003).
- Philippas, D. and Siriopoulos, C., 2013. Putting the “C” into crisis: Contagion, correlations and copulas on EMU bond markets. *J. Int. Financ. Mark. I.* 27, 161–176. doi:[10.1016/j.intfin.2013.09.008](https://doi.org/10.1016/j.intfin.2013.09.008).
- Richmond, R.J., 2019. Trade network centrality and currency risk premia. *J. Financ.* 74, 1315–1361. doi:[10.1111/jofi.12755](https://doi.org/10.1111/jofi.12755).
- Rossi, A.G., Blake, D., Timmermann, A., Tonks, I. and Wermers, R., 2018. Network centrality and delegated investment performance. *J. Financ. Econ.* 128, 183–206. doi:[10.1016/j.jfineco.2018.02.003](https://doi.org/10.1016/j.jfineco.2018.02.003).
- Scherbina, A. and Schlusche, B., 2020. Follow the leader: using the stock market to uncover information flows between firms. *Rev. Financ.* 24, 189–225. doi:[10.1093/rof/rfy038](https://doi.org/10.1093/rof/rfy038).
- Sharpe, W.F., 1963. A simplified model for portfolio analysis. *Manag. Sci.* 9, 277–293. doi:[10.1287/mnsc.9.2.277](https://doi.org/10.1287/mnsc.9.2.277).
- Shleifer, A. and Vishny, R.W., 1997. The limits of arbitrage. *J. Financ.* 52, 35–55. doi:[10.1111/j.1540-6261.1997.tb03807.x](https://doi.org/10.1111/j.1540-6261.1997.tb03807.x).
- Sorensen, T., 1948. A method of establishing groups of equal amplitude in plant sociology based on similarity of species content and its application to analyses of the vegetation on Danish commons. *Biol. Skr.* 5, 1–34.
- Tan, L., Zhang, X. and Zhang, X., 2024. Retail and institutional investor trading behaviors: Evidence from China. *Annu. Rev. Financ. Econ.* 16, 459–483. doi:[10.1146/annurev-financial-082123-110132](https://doi.org/10.1146/annurev-financial-082123-110132).
- Wang, H., Yuan, Y. and Wang, T., 2021. The dynamics of cross-boundary fire—Financial contagion between the oil and stock markets. *J. Fut. Markets* 41, 1655–1673. doi:[10.1002/fut.22239](https://doi.org/10.1002/fut.22239).
- Wu, M., Huang, W., Liu, X. and Meng, Q., 2024. Firm connection and equity return predictability—Graph-based machine learning methods. *Br. Account. Rev.* , 101436doi:[10.1016/j.bracrev.2024.101436](https://doi.org/10.1016/j.bracrev.2024.101436).

[1016/j.bar.2024.101436](#).

Xiong, H., Li, W. and Qiu, C., 2023. Shared analyst coverage and connected-firm momentum spillover effect. *Econ. Rev. (in Chinese)* 243, 123–143. doi:[10.19361/j.er.2023.05.08](#).

Xu, N., Chan, K.C., Jiang, X. and Yi, Z., 2013. Do star analysts know more firm-specific information? Evidence from China. *J. Bank. Financ.* 37, 89–102. doi:[10.1016/j.jbankfin.2012.08.014](#).

Yi, B. and Guo, S., 2022. Common analyst links and predictable returns: Evidence from China. *North Am. J. Econ. Financ.* 63, 101832. doi:[10.1016/j.najef.2022.101832](#).

Zhang, D. and Broadstock, D.C., 2020. Global financial crisis and rising connectedness in the international commodity markets. *Int. Rev. Financ. Anal.* 68, 101239. doi:[10.1016/j.irfa.2018.08.003](#).

Online Appendix

Detecting cross-firm momentum effects via shared analyst coverage: The role of leaders

Abstract

Cross-firm momentum effects via shared analyst coverage are well-documented in developed markets, but their robustness remains unclear in emerging markets, where information diffusion is asymmetric and analyst coverage is highly concentrated. Our work revisits this effect in an environment of extreme informational frictions — the Chinese market. We reconstruct the information transmission channel within the analyst coverage network by introducing a novel weighting scheme based on strength centrality (SC). This measure identifies influential leader firms that command disproportionate attention from both analysts and the market. Our results demonstrate that SC -weighted connected-firm returns robustly predict cross-sectional stock returns, yielding significant and persistent profits even under a rigorous stock filter. This performance cannot be subsumed by strategies based on alternative weighting schemes or by explanations such as intra-industry cross-firm momentum and information discreteness. Further analysis reveals that the superiority of the SC -based approach stems from its ability to effectively identify firms with stronger cross-period fundamental linkages. In addition, high- SC stocks are characterized by higher investor attention, more efficient information processing, lower arbitrage costs, and greater international exposures. With this evidence, we further confirm a directional spillover: cross-firm momentum effects flow exclusively from these high- SC leaders to low- SC laggards, and there is no reverse spillover. Our findings suggest that cross-firm momentum may be systematically underestimated in many international markets due to methodological limitations rather than economic irrelevance. The SC -based framework therefore offers a portable tool for global investors and researchers operating in environments with asymmetric information.

Keywords: Cross-firm momentum, Analyst coverage, Strength centrality, Information diffusion, Emerging markets, International exposure

JEL Codes: G12, G14, G15, C58

Summary

This appendix provides additional results to the main paper, “Detecting cross-firm momentum effects via shared analyst coverage: The role of leaders”. Specifically, it is organized as follows. Section [A](#) presents the construction of risk factors used in the study. Section [B](#) reports the results of univariate portfolio analysis and spanning tests using weekly data. Section [C](#) presents additional results of the bivariate portfolio analysis. Section [D](#) details the firm-level characteristics used in the study and presents their summary statistics. Section [E](#) conducts a robustness check using Pagerank centrality.

A Construction of pricing factors

This section describes the construction of pricing factors employed in our study. When constructing pricing factors from various models, we use the stock samples selected based on the rigorous filtering detailed in Section 2.1.

1. Market factor

The single-index model (SIM) from Sharpe (1963) consists of the only one factor, namely the market factor, denoted as MKT, which shares the same procedure for different pricing models. Specifically, MKT is defined as the return on the value-weighted portfolio of stocks, in excess of the risk-free rate (proxied by the one-year deposit interest rate).

2. CH4 factors

We follow Liu et al. (2019) to construct the factors from the CH4 model. In addition to the MKT factor, the CH4 model also includes three other factors: (1) Size factor, *SMB*; (2) Value factor, *VMG*; (3) Turnover factor, *PMO*.

The following firm-level characteristics are employed to help construct factors above: (1) Market equity, *ME*, is defined as the closing price multiplied by the number of shares outstanding at the end of the last fiscal year. (2) Earnings-to-price, *EP*, is the ratio of earnings (defined as net profit excluding nonrecurring gains/losses) to market equity (*ME*). (3) One-month abnormal turnover. A firm's abnormal turnover is calculated as the ratio of its average daily turnover over the past 20 days to its average daily turnover over the past 250 days.

Specifically, each month, we divide the stocks into three *EP* groups: top 30% (value, V), middle 40% (middle, M), and bottom 30% (growth, G). The stocks are also divided into two size groups: small (S) and big (B), with the split determined by their median market value. We then use the intersections of these groups to form value-weighted portfolios for the six resulting *ME-EP* combinations: SV, SM, SG, BV, BM, and BG. In the formation of these value-weighted portfolios, each stock is weighted by its *ME* value. Specifically, the factor *VMG* is defined as follows:

$$VMG = \frac{1}{2}(SV + BV) - \frac{1}{2}(SG + BG). \quad (A.1)$$

Similarly, the stocks can be divided into three turnover groups: top 30% (pessimistic, P), middle 40% (medium, M), and bottom 30% (optimistic, O). Using the three *ME* groups and the three turnover groups described above, we form value-weighted portfolios for the six

resulting *ME*-turnover combinations: SP, SM, SO, BP, BM, and BO. We then construct the three factors as follows:

$$PMO = \frac{1}{2}(SP + BP) - \frac{1}{2}(BO + SO). \quad (A.2)$$

We can obtain the following two SMB factors based on VMG and PMO respectively:

$$SMB_{VMG} = \frac{1}{3}(SV + SM + SG) - \frac{1}{3}(BV + BM + BG), \quad (A.3)$$

$$SMB_{PMO} = \frac{1}{3}(SP + SM + SO) - \frac{1}{3}(BP + BM + BO), \quad (A.4)$$

Regarding the size factor, it is ultimately defined as the simple average of the EP-neutralized version of SMB and the turnover-neutralized version of SMB:

$$SMB = \frac{1}{2}(SMB_{VMG} + SMB_{PMO}). \quad (A.5)$$

It is worth noting that despite the adoption of a stricter stock filter, the CH4 factors constructed here and the factors constructed by [Liu et al. \(2019\)](#) still exhibit very strong correlations.¹ Specifically, over the sample period from January 2010 to December 2023, the Pearson correlation coefficients with the factors in parentheses are 0.9996 (*MKT*), 0.9966 (*SMB*), 0.9789 (*VMG*), and 0.9841 (*PMO*), respectively.

3. q^5 factors

[Hou et al. \(2021\)](#) propose the q^5 -factor model, which consists of five factors, including: (1) Market factor, R_{Mkt} , constructed similarly to *MKT* above. (2) Size factor, R_{Me} ; (3) Investment factor, $R_{I/A}$; (4) Return on equity factor, R_{Roe} ; (5) Expected growth factor, R_{Eg} .

The following firm-level characteristics are used to construct the factors listed above: (1) Market equity (*ME*) is defined as the closing price multiplied by the number of shares outstanding at the end of the last fiscal year. (2) Investment-to-asset ratio (*IA*) is defined as the annual change in total assets divided by one-year-lagged total assets. (3) Return-on-equity (*ROE*) is defined as the ratio of operating profit to book equity, where book equity is calculated as total equity attributable to shareholders plus deferred tax liability, minus preferred stock. (4) Expected investment-to-assets changes (*EdIA*). Except for *ROE*, the variables mentioned above are constructed using data from the most recent annual report,

¹The data of CH4 factors is available from Prof. Stambaugh's website: <https://finance.wharton.upenn.edu/~stambaug/> and also Shanghai Mingshi Private Fund Management's website: <https://en.mingshiim.com/database>.

typically released by the end of April. The calculation of ROE is based on data from the most recent quarterly report.

To construct $EdIA$, for each month, we perform cross-sectional regressions of the future one-year-ahead investment-to-assets changes (dIA) on three regressors: the log of Tobin's q ($\log(q)$), cash flows (Cop), and the change in return-on-equity ($dROE$). The cross-sectional regressions are estimated using weighted least squares, with market equity as the weight. Each variable is winsorized at the 1st and 99th percentiles each month. At the beginning of each month, we calculate the expected IA changes ($EdIA$) by combining the most recent winsorized predictors with the average cross-sectional slopes. The most recent predictors, $\log(q)$ and Cop , are derived from the latest annual report, while $dROE$ is based on the most recent quarterly report. The average slopes used to calculate $EdIA$ are obtained from a 120-month rolling window (with a minimum of 30 months), where the dependent variable, dIA , uses data from the latest annual report, and the regressors are further lagged accordingly.

With these firm-level characteristics, we can readily construct the q^5 factors for the Chinese market. At the beginning of each month, we use the median of market equity (ME) for main board stocks as the breakpoint to divide all stocks into two groups: large market-capitalization stocks (Big) and small market-capitalization stocks (Small). Next, we use the 30% and 70% quantiles of return-on-equity (ROE) for main board stocks as breakpoints to divide all stocks into three groups: the top 30% (Low), the middle 40% (Middle), and the bottom 30% (High). Similarly, we use the 30% and 70% quantiles of investment-to-asset ratio (IA) for main board stocks as breakpoints to divide all stocks into three groups: the top 30% (Low), the middle 40% (Middle), and the bottom 30% (High). We then repeat this process for the expected investment-to-assets changes ($EdIA$), using the 30% and 70% quantiles of $EdIA$ for main board stocks to divide all stocks into three groups: the top 30% (Low), the middle 40% (Middle), and the bottom 30% (High). After dividing the stocks into the appropriate groups, we take the intersection of the two ME groups, the three ROE groups, and the three IA groups, resulting in 18 portfolios. We calculate the value-weighted returns of these 18 portfolios by the end of the month. Let g_1 , g_2 , and g_3 represent the classifications of each portfolio based on ME , ROE , and IA , respectively. Specifically, g_1 takes values of S (small ME) or B (large ME), while g_2 and g_3 take values of H (high), M (medium), and L (low) for ROE and IA . We use $g_1g_2g_3$ to denote the group resulting from the intersection of the classifications for these three variables. The factors are constructed

as follows:

$$R_{Me} = \frac{SLL + SML + SHL + SLM + SMM + SHM + SLH + SMH + SHH}{9} - \frac{BLL + BML + BHL + BLM + BMM + BHM + BLH + BMH + BHH}{9}, \quad (A.6)$$

$$R_{Roe} = \frac{SHL + SHM + SHH + BHL + BHM + BHH}{6} - \frac{SLL + SLM + SLH + BLL + BLM + BLH}{6}, \quad (A.7)$$

$$R_{I/A} = \frac{SLL + SML + SHL + BLL + BML + BHL}{6} - \frac{SLH + SMH + SHH + BLH + BMH + BHH}{6}, \quad (A.8)$$

For R_{Eg} , we take the intersection of the two *ME* groups and the three *EdIA* groups to form six portfolios: Small Low, Small Middle, Small High, Big Low, Big Middle, and Big High. We then calculate the value-weighted returns of these six portfolios by the end of the month. Specifically,

$$R_{Eg} = \frac{Small\ High + Big\ High}{2} - \frac{Small\ Low + Big\ Low}{2}. \quad (A.9)$$

4. Fama-French six factors

The six-factor (FF6) model proposed by [Fama and French \(2018\)](#) consists of the following factors: (1) Market factor, *MKT*; (2) Size factor, *SMB*; (3) Value factor, *HML*; (4) Profitability factor, *RMW*; (5) Investment factor, *CMA*; (6) Momentum factor, *UMD*.

The following firm-level characteristics are used to construct the factors mentioned above: (1) Market equity (*ME*) is defined as the closing price multiplied by the number of shares outstanding at the end of the last fiscal year; (2) Book-to-market ratio (*BM*) is defined as the ratio of book equity (i.e., total equity attributable to shareholders + deferred tax liability - preferred stock) to market equity (*ME*); (3) Operating profitability (*OP*) is defined as the ratio of operating profit to book equity (i.e., total equity attributable to shareholders + deferred tax liability - preferred stock); (4) Investment (*INV*) is defined as the annual change in total assets divided by the one-year-lagged total assets; (5) Momentum (*MOM*) is defined as the cumulative excess return from $t - 12$ to $t - 2$. Except for *MOM*, the variables mentioned above are constructed using data from the most recent annual report, typically released by the end of April. For *MOM*, the calculation is based on monthly updated data.

With these firm-level characteristics, we can construct the FF6 factors for the Chinese

market. At the beginning of each month, using the median ME of main board stocks as the breakpoint, all stocks are divided into two groups: large market capitalization (Big) and small market capitalization (Small). Using the 30% and 70% quantiles of the BM of main board stocks as breakpoints, all stocks are classified into three groups: the top 30% (Low), the middle 40% (Middle), and the bottom 30% (High). Similarly, using the 30% and 70% quantiles of the OP of main board stocks, all stocks are divided into three groups: the top 30% (Weak), the middle 40% (Neutral), and the bottom 30% (Robust). Using the 30% and 70% quantiles of the INV of main board stocks, all stocks are divided into three groups: the top 30% (Conservative), the middle 40% (Neutral), and the bottom 30% (Aggressive). Finally, using the 30% and 70% quantiles of the MOM of main board stocks, all stocks are divided into three groups: the top 30% (Down), the middle 40% (Middle), and the bottom 30% (Up).

We then take the intersection of two ME groups and three BM groups, two ME groups and three OP groups, and two ME groups, three INV groups, and three MOM groups, resulting in a total of 24 portfolios. We calculate the value-weighted returns of these 24 portfolios by the end of the month. Let g_1 and g_2 represent the classification of each portfolio based on ME and the other attributes (i.e., BM , OP , INV , and MOM). Specifically, g_1 takes values of S or B for small and large ME , respectively. For g_2 , it takes the following values: H , M , and L for high, medium, and low levels of BM ; R , N_{op} , and W for high, medium, and low levels of OP ; A , N_{inv} , and C for high, medium, and low levels of INV ; U , M_{mom} , and D for high, medium, and low levels of MOM . We use g_1g_2 to denote the group obtained by the intersection of a specific classification of each attribute pair. The factors are constructed as follows:

$$HML = \frac{SH + BH}{2} - \frac{SL + BL}{2}, \quad (A.10)$$

$$RMW = \frac{SR + BR}{2} - \frac{SW + BW}{2}, \quad (A.11)$$

$$CMA = \frac{SC + BC}{2} - \frac{SA + BA}{2}, \quad (A.12)$$

and

$$UMD = \frac{SU + BU}{2} - \frac{SD + BD}{2}. \quad (A.13)$$

We also have SMB_{BM} , SMB_{ROE} , and SMB_{INV} , which are given as follows:

$$SMB_{BM} = \frac{SH + SM + SL}{3} - \frac{BH + BM + BL}{3},$$

$$SMB_{OP} = \frac{SR + SN_{op} + SW}{3} - \frac{BR + BN_{op} + BW}{3},$$

$$SMB_{INV} = \frac{SC + SN_{inv} + SA}{3} - \frac{BC + BN_{inv} + BA}{3}.$$

And SMB is built as follows:

$$SMB = \frac{SMB_{BM} + SMB_{OP} + SMB_{INV}}{3}. \quad (\text{A.14})$$

B Baseline results on a weekly basis

This section presents univariate portfolio analysis and spanning tests conducted using weekly data. Specifically, at the beginning of each week, the latest data from the previous Friday-end are used to construct the analyst coverage network. Based on the weekly updated network structure, the five types of weights considered in the paper are constructed respectively:

$$weight_{i,j}^{AH} = \frac{\omega_{i,j}^{AH}}{\sum_{j \in \mathcal{N}_i} \omega_{i,j}^{AH}} = \frac{n_{i,j}}{\sum_{j \in \mathcal{N}_i} n_{i,j}}, \quad (B.1)$$

$$weight_{i,j}^{Isr} = \frac{\omega_{i,j}^{Isr}}{\sum_{j \in \mathcal{N}_i} \omega_{i,j}^{Isr}} = \frac{\frac{n_{i,j}}{\sqrt{n_i \times n_j}}}{\sum_{j \in \mathcal{N}_i} \frac{n_{i,j}}{\sqrt{n_i \times n_j}}}, \quad (B.2)$$

$$weight_{i,j}^{Sor} = \frac{\omega_{i,j}^{Sor}}{\sum_{j \in \mathcal{N}_i} \omega_{i,j}^{Sor}} = \frac{\frac{n_{i,j}}{n_i + n_j}}{\sum_{j \in \mathcal{N}_i} \frac{n_{i,j}}{n_i + n_j}}, \quad (B.3)$$

$$weight_{i,j}^{MRX} = \frac{\omega_{i,j}^{MRX}}{\sum_{j \in \mathcal{N}_i} \omega_{i,j}^{MRX}} = \frac{\frac{n_{i,j}}{n_i + n_j - n_{i,j}}}{\sum_{j \in \mathcal{N}_i} \frac{n_{i,j}}{n_i + n_j - n_{i,j}}}, \quad (B.4)$$

and

$$weight_{i,j}^{SC} = \frac{SC_j}{\sum_{j \in \mathcal{N}_i} SC_j} = \frac{\sum_{m \in \mathcal{N}_j} n_{j,m}}{\sum_{j \in \mathcal{N}_i} \sum_{m \in \mathcal{N}_j} n_{j,m}}. \quad (B.5)$$

With the above weights, we then construct their respective connected-firm return ($CFRet$):

$$CFRet_{i,t}^{type} = \sum_{j \in \mathcal{N}_{i,t}} weight_{i,j,t}^{type} \times Ret_{j,t}, \quad (B.6)$$

where $type \in \{AH, Isr, MRX, Sor, SC\}$.

We first conduct univariate portfolio analysis. Similar to analyses based on monthly data, at the beginning of each week, individual stocks screened out by the strict filter are sorted into ten decile groups based on their latest connected-firm returns, i.e., $CFRet^{AH}$, $CFRet^{Isr}$, $CFRet^{Sor}$, $CFRet^{MRX}$, and $CFRet^{SC}$. For example, in the case of $CFRet^{AH}$, decile portfolio 1, labeled High, consists of stocks with the highest average $CFRet^{AH}$, while decile portfolio 10, labeled Low, includes stocks with the lowest values. Each portfolio is held for one week, and its performance is evaluated over the full sample period spanning January 2010 to December 2023.

Table B.1 presents the weekly average returns of CFMOM portfolios. The results show that $CFRet$ -sorted long-short portfolios based on the five weighting schemes all yield significantly positive profits. Moreover, these portfolios also generate significant alphas after

adjustment for different sets of risk factors. By comparison, the traditional four schemes lead to higher returns with stronger statistical significance and lower sensitivity to market capitalization, while $CFRet^{SC}$ exhibits weaker performance. For example, in the left panel for the equal-weighted (EW) results, we observe that the $CFRet$ -sorted long-short portfolios based on the four traditional schemes produce at least 0.24% per week, whereas the SC -based counterpart yields only 0.209%. In the right panel for the value-weighted (VW) results, the return of the $CFRet^{SC}$ -sorted long-short portfolio decreases from 0.209% to 0.203%, and the adjusted t -statistic also falls from 2.64 to 1.90. In contrast, the other four types of long-short portfolios still retain their significant raw profits and significant alphas under most models.

The results above appear to indicate that the four traditional weighting schemes perform better at the weekly frequency than SC . To further examine their relative pricing advantages, we conduct the spanning tests. Results appear in [Table B.2](#). And we observe that none of the weighting schemes delivers distinct return predictability, as all can be explained by other alternatives and common factors. This finding implies that at the weekly frequency, the SC weighting scheme contains pricing information similar to that of the other four traditional weighting schemes, even though it performs not good in the univariate portfolio analysis.

Table B.1 Weekly average returns on $CFRet$ -sorted portfolios

This table presents weekly average returns for ten decile portfolios, which are constructed by sorting on $CFRet^{AH}$, $CFRet^{Isr}$, $CFRet^{MRX}$, $CFRet^{Sor}$, and $CFRet^{SC}$, respectively. At the beginning of each week, stocks are sorted into decile portfolios based on their $CFRet$ values, constructed using the procedure in Section 2.3. Portfolio 1 (10), labeled as High (Low), contains the stocks with the average highest (lowest) $CFRet$ value. All portfolios are rebalanced at the beginning of the next month and their one-month-ahead return is obtained. For each decile portfolio, we report the equal-weighted (EW) average weekly excess return. The last rows report return spreads and their alphas between decile 1 (high $CFRet$) and decile 10 (low $CFRet$). SIM alpha is obtained from Sharpe (1963) single-index model (SIM) which consists of the market factor (MKT) only. CH4 alpha is obtained from the CH4 model of Liu et al. (2019). q^5 alpha is obtained from the q^5 -factor model of Hou et al. (2021). FF6 alpha is obtained from the six-factor model of Fama and French (2018). Pricing factors above are specifically constructed for the Chinese market, which are detailed in Section A of the Online Appendix. Panels A and B respectively present the results with lax and rigorous stock filters mentioned in Section 2.1. Newey and West (1987) t -statistics are provided in the parentheses. The results shown in bold signify their statistical significance at the 5% level. The sample period is January 2010 to December 2023.

	EW					VW				
	AH	Isr	MRX	Sor	SC	AH	Isr	MRX	Sor	SC
Low	0.022 (0.15)	0.020 (0.14)	0.002 (0.02)	0.009 (0.06)	0.075 (0.53)	-0.085 (-0.63)	-0.099 (-0.73)	-0.087 (-0.64)	-0.097 (-0.70)	-0.063 (-0.47)
2	0.049 (0.36)	0.039 (0.29)	0.026 (0.19)	0.023 (0.17)	0.041 (0.30)	-0.056 (-0.43)	-0.014 (-0.11)	-0.067 (-0.52)	-0.072 (-0.56)	-0.047 (-0.37)
3	0.014 (0.11)	0.040 (0.30)	0.050 (0.37)	0.046 (0.33)	0.052 (0.38)	-0.072 (-0.58)	-0.071 (-0.57)	-0.042 (-0.35)	-0.026 (-0.21)	-0.100 (-0.81)
4	0.095 (0.69)	0.070 (0.51)	0.083 (0.61)	0.082 (0.60)	0.076 (0.56)	-0.020 (-0.17)	-0.029 (-0.24)	-0.007 (-0.06)	-0.019 (-0.16)	-0.046 (-0.38)
5	0.146 (1.04)	0.151 (1.08)	0.126 (0.90)	0.133 (0.95)	0.131 (0.97)	-0.024 (-0.19)	0.013 (0.11)	-0.027 (-0.21)	-0.037 (-0.29)	0.066 (0.54)
6	0.189 (1.34)	0.203 (1.45)	0.195 (1.39)	0.199 (1.43)	0.171 (1.23)	0.102 (0.79)	0.104 (0.83)	0.074 (0.58)	0.102 (0.82)	0.106 (0.87)
7	0.228 (1.60)	0.211 (1.50)	0.255 (1.80)	0.238 (1.67)	0.209 (1.49)	0.121 (0.95)	0.115 (0.91)	0.164 (1.32)	0.150 (1.18)	0.093 (0.78)
8	0.235 (1.66)	0.229 (1.62)	0.217 (1.55)	0.223 (1.59)	0.222 (1.59)	0.177 (1.35)	0.175 (1.34)	0.183 (1.39)	0.172 (1.34)	0.110 (0.84)
9	0.231 (1.58)	0.248 (1.68)	0.259 (1.75)	0.251 (1.69)	0.242 (1.68)	0.064 (0.46)	0.082 (0.59)	0.104 (0.75)	0.087 (0.62)	0.117 (0.91)
High	0.295 (2.01)	0.269 (1.83)	0.264 (1.80)	0.276 (1.88)	0.284 (1.90)	0.232 (1.64)	0.199 (1.41)	0.181 (1.30)	0.203 (1.44)	0.140 (0.93)
Raw	0.273 (3.03)	0.249 (2.79)	0.261 (2.93)	0.267 (2.97)	0.209 (2.64)	0.318 (2.52)	0.298 (2.35)	0.269 (2.11)	0.300 (2.31)	0.203 (1.90)
SIM	0.273 (3.03)	0.248 (2.78)	0.261 (2.93)	0.266 (2.97)	0.208 (2.65)	0.318 (2.52)	0.297 (2.34)	0.268 (2.10)	0.299 (2.30)	0.200 (1.89)
CH4	0.289 (3.20)	0.261 (2.90)	0.277 (3.07)	0.280 (3.08)	0.223 (2.80)	0.332 (2.60)	0.309 (2.40)	0.294 (2.28)	0.317 (2.41)	0.213 (1.94)
q^5	0.254 (2.52)	0.227 (2.26)	0.245 (2.46)	0.250 (2.49)	0.186 (2.14)	0.310 (2.20)	0.277 (1.94)	0.247 (1.76)	0.278 (1.94)	0.198 (1.63)
FF6	0.272 (2.89)	0.244 (2.59)	0.264 (2.84)	0.266 (2.83)	0.212 (2.55)	0.316 (2.43)	0.287 (2.19)	0.269 (2.06)	0.292 (2.20)	0.212 (1.89)

Table B.2 Spanning tests on a weekly basis

This table presents the alphas of various CFMOM long-short portfolios (denoted as $CFMOM$) using weekly data. Panels A to E highlight the alpha of $CFMOM^{SC}$, $CFMOM^{AH}$, $CFMOM^{Isr}$, $CFMOM^{MRX}$, and $CFMOM^{Sor}$, derived from time-series regressions based on an augmented model that incorporates factors from the specified model in the corresponding column and the $CFMOM$ listed in the corresponding row. Details on the construction of these pricing factors are provided in Section A of the Online Appendix. The left and right panels correspond to results based on equal-weighted (EW) and value-weighted (VW) portfolios, respectively. Newey and West (1987) t -statistics are presented in the parentheses. Statistically significant results at the 5% level are highlighted in bold. The sample period spans from January 2010 to December 2023.

	EW				VW			
	SIM	CH4	q^5	FF6	SIM	CH4	q^5	FF6
Panel A: $CFMOM^{SC}$'s alpha								
$CFMOM^{AH}$	-0.394 (-1.29)	0.006 (0.04)	0.170 (1.65)	0.057 (0.63)	-0.064 (-0.26)	-0.037 (-0.22)	0.031 (0.11)	-0.034 (-0.15)
$CFMOM^{Isr}$	-0.204 (-0.77)	0.068 (0.38)	0.172 (0.92)	0.116 (0.62)	-0.060 (-0.16)	-0.054 (-0.22)	0.198 (0.61)	0.062 (0.20)
$CFMOM^{MRX}$	-0.317 (-1.15)	-0.011 (-0.06)	0.059 (0.33)	0.002 (0.01)	-0.182 (-0.52)	-0.167 (-0.59)	-0.014 (-0.04)	-0.150 (-0.55)
$CFMOM^{Sor}$	-0.267 (-0.99)	0.013 (0.07)	0.130 (0.76)	0.052 (0.31)	-0.175 (-0.52)	-0.214 (-0.87)	-0.016 (-0.05)	-0.158 (-0.57)
Panel B: $CFMOM^{AH}$'s alpha								
$CFMOM^{SC}$	0.238 (1.10)	-0.103 (-0.59)	-0.269 (-1.59)	-0.130 (-0.73)	-0.103 (-0.40)	-0.095 (-0.45)	-0.185 (-0.53)	-0.085 (-0.40)
$CFMOM^{Isr}$	0.146 (0.64)	0.017 (0.09)	-0.046 (-0.20)	0.027 (0.09)	0.018 (0.09)	-0.017 (-0.12)	0.221 (1.33)	0.129 (0.60)
$CFMOM^{MRX}$	0.026 (0.12)	-0.070 (-0.35)	-0.184 (-0.77)	-0.106 (-0.36)	-0.109 (-0.56)	-0.140 (-0.70)	-0.029 (-0.23)	-0.127 (-1.35)
$CFMOM^{Sor}$	0.079 (0.36)	-0.043 (-0.23)	-0.096 (-0.42)	-0.047 (-0.16)	-0.111 (-0.57)	-0.215 (-1.42)	-0.042 (-0.31)	-0.144 (-1.03)
Panel C: $CFMOM^{Isr}$'s alpha								
$CFMOM^{SC}$	0.091 (0.37)	-0.158 (-0.84)	-0.271 (-1.11)	-0.183 (-0.86)	-0.172 (-0.47)	-0.115 (-0.40)	-0.445 (-1.18)	-0.249 (-0.84)
$CFMOM^{AH}$	-0.258 (-0.88)	-0.105 (-0.49)	-0.032 (-0.16)	-0.090 (-0.37)	-0.057 (-0.29)	-0.010 (-0.07)	-0.264 (-1.63)	-0.160 (-0.75)
$CFMOM^{MRX}$	-0.141 (-1.83)	-0.106 (-1.59)	-0.165 (-1.68)	-0.152 (-1.45)	-0.146 (-0.97)	-0.138 (-0.90)	-0.278 (-1.54)	-0.282 (-1.73)
$CFMOM^{Sor}$	-0.078 (-1.85)	-0.071 (-2.07)	-0.060 (-0.94)	-0.086 (-1.39)	-0.142 (-1.30)	-0.213 (-3.29)	-0.282 (-3.36)	-0.294 (-2.70)
Panel D: $CFMOM^{MRX}$'s alpha								
$CFMOM^{SC}$	0.220 (0.80)	-0.066 (-0.32)	-0.117 (-0.47)	-0.041 (-0.20)	-0.081 (-0.23)	-0.027 (-0.08)	-0.233 (-0.64)	-0.005 (-0.02)
$CFMOM^{AH}$	-0.127 (-0.44)	-0.004 (-0.02)	0.139 (0.63)	0.061 (0.24)	0.072 (0.33)	0.116 (0.53)	-0.008 (-0.06)	0.117 (1.30)
$CFMOM^{Isr}$	0.135 (1.75)	0.101 (1.55)	0.168 (1.59)	0.152 (1.38)	0.116 (0.70)	0.118 (0.75)	0.247 (1.22)	0.268 (1.52)
$CFMOM^{Sor}$	0.061 (1.06)	0.032 (0.66)	0.108 (1.65)	0.066 (1.06)	-0.007 (-0.07)	-0.086 (-0.69)	-0.021 (-0.14)	-0.023 (-0.22)
Panel E: $CFMOM^{Sor}$'s alpha								
$CFMOM^{SC}$	0.163 (0.63)	-0.097 (-0.48)	-0.222 (-0.94)	-0.104 (-0.53)	-0.070 (-0.20)	0.065 (0.21)	-0.208 (-0.56)	0.020 (0.08)
$CFMOM^{AH}$	-0.191 (-0.65)	-0.038 (-0.18)	0.028 (0.14)	-0.004 (-0.01)	0.072 (0.33)	0.199 (1.15)	0.008 (0.06)	0.135 (0.94)
$CFMOM^{Isr}$	0.075 (1.83)	0.070 (2.03)	0.059 (0.90)	0.086 (1.36)	0.125 (0.99)	0.208 (2.93)	0.275 (2.91)	0.296 (2.46)
$CFMOM^{MRX}$	-0.066 (-1.12)	-0.036 (-0.69)	-0.109 (-1.73)	-0.068 (-1.11)	-0.005 (-0.05)	0.079 (0.62)	0.009 (0.06)	0.015 (0.14)

C Additional results for identification of laggards

In this section, we evaluate the performance of portfolios formed by dependent sorting on SC and each of $CFRet$ measures, including $CFRet^{AH}$, $CFRet^{Isr}$, $CFRet^{MRX}$, and $CFRet^{Sor}$. Specifically, each month, the individual stocks are first divided into five quintile groups according to their respective SC values. This process is further followed by an additional set of quintile portfolios constructed by sorting on each $CFRet$. One-month-ahead excess returns and alphas of these portfolios are recorded. The results are presented in [Table C.1](#) through [Table C.4](#), respectively.

Table C.1 The performance of portfolios formed by dependent sorting on SC and $CFRet^{AH}$

This table presents the results of bivariate portfolios formed by dependent double sorting of one control variable — SC depicted in Eq. (10) and $CFRet^{AH}$ depicted in Eq. (1) during January 2010 and December 2023. Each month, the individual stocks are first divided into five quintile groups according to their respective SC values. This process is further followed by an additional set of quintile portfolios constructed by sorting on $CFRet^{AH}$. One-month-ahead excess returns of the portfolios are presented. “High” (“Low”) designates portfolios with the highest (lowest) SC or $CFRet^{AH}$ values. The last rows report the $CFRet^{AH}$ -sorted long-short portfolio’s return spread (“High–Low”) and associated alphas within each SC quantile. Specifically, SIM alpha is obtained from Sharpe (1963) single-index model (SIM) which consists of the market factor (MKT) only. CH4 alpha is obtained from the CH4 model of Liu et al. (2019). q^5 alpha is obtained from the q^5 -factor model of Hou et al. (2021). FF6 alpha is obtained from the six-factor model of Fama and French (2018). Pricing factors above are specifically constructed for the Chinese market, which are detailed in Section A of the Online Appendix. Left and right panels present the equal-weighted (EW) and value-weighted (VW) results, respectively. Newey and West (1987) t -statistics are provided in the parentheses. The results shown in bold signify their statistical significance at the 5% level. The sample period is January 2010 to December 2023.

	EW					VW				
	High SC	2	3	4	Low SC	High SC	2	3	4	Low SC
High $CFRet^{AH}$	0.749 (1.28)	0.631 (1.12)	1.075 (1.82)	1.145 (1.84)	1.112 (1.76)	0.429 (0.74)	0.367 (0.65)	0.835 (1.53)	0.597 (1.01)	0.527 (0.84)
2	0.525 (1.02)	0.730 (1.36)	0.987 (1.73)	0.901 (1.67)	0.668 (1.21)	0.354 (0.72)	0.450 (0.76)	0.687 (1.07)	0.592 (1.05)	0.319 (0.55)
3	0.433 (0.78)	0.428 (0.80)	0.880 (1.61)	0.626 (1.17)	0.596 (1.10)	0.345 (0.64)	0.674 (1.33)	0.390 (0.72)	0.210 (0.40)	0.296 (0.50)
4	0.476 (0.91)	0.116 (0.23)	0.117 (0.22)	0.498 (0.92)	0.555 (1.04)	0.350 (0.68)	−0.371 (−0.79)	−0.302 (−0.67)	−0.008 (−0.02)	0.126 (0.24)
Low $CFRet^{AH}$	0.498 (0.91)	0.468 (0.88)	0.382 (0.68)	0.437 (0.81)	0.364 (0.65)	0.318 (0.63)	0.168 (0.36)	0.150 (0.25)	−0.008 (−0.02)	−0.043 (−0.08)
High–Low	0.251 (0.56)	0.163 (0.37)	0.693 (1.63)	0.708 (2.15)	0.748 (2.50)	0.111 (0.23)	0.199 (0.40)	0.685 (1.64)	0.605 (1.74)	0.570 (1.85)
SIM alpha	0.266 (0.59)	0.182 (0.42)	0.709 (1.68)	0.712 (2.15)	0.747 (2.48)	0.125 (0.26)	0.214 (0.43)	0.715 (1.76)	0.614 (1.76)	0.570 (1.84)
CH4 alpha	0.304 (0.60)	0.347 (0.76)	0.897 (2.20)	0.894 (2.70)	0.923 (3.04)	0.111 (0.20)	0.373 (0.70)	0.787 (1.83)	0.836 (2.38)	0.810 (2.62)
q^5 alpha	0.548 (0.92)	0.765 (1.52)	1.337 (2.95)	1.092 (2.91)	1.136 (2.95)	0.189 (0.31)	0.835 (1.43)	1.371 (2.89)	0.866 (2.15)	0.916 (2.41)
FF6 alpha	0.459 (0.82)	0.515 (1.04)	1.134 (2.48)	1.019 (2.78)	1.096 (3.14)	0.208 (0.35)	0.514 (0.92)	0.989 (2.08)	0.951 (2.44)	0.971 (2.74)

Table C.2 The performance of portfolios formed by dependent sorting on SC and $CFRet^{Isr}$

This table presents the results of bivariate portfolios formed by dependent double sorting of one control variable — SC depicted in Eq. (10) and $CFRet^{Isr}$ depicted in Eq. (1) during January 2010 and December 2023. Each month, the individual stocks are first divided into five quintile groups according to their respective SC values. This process is further followed by an additional set of quintile portfolios constructed by sorting on $CFRet^{Isr}$. One-month-ahead excess returns of the portfolios are presented. “High” (“Low”) designates portfolios with the highest (lowest) SC or $CFRet^{Isr}$ values. The last rows report the $CFRet^{Isr}$ -sorted long-short portfolio’s return spread (“High–Low”) and associated alphas within each SC quantile. Specifically, SIM alpha is obtained from Sharpe (1963) single-index model (SIM) which consists of the market factor (MKT) only. CH4 alpha is obtained from the CH4 model of Liu et al. (2019). q^5 alpha is obtained from the q^5 -factor model of Hou et al. (2021). FF6 alpha is obtained from the six-factor model of Fama and French (2018). Pricing factors above are specifically constructed for the Chinese market, which are detailed in Section A of the Online Appendix. Left and right panels present the equal-weighted (EW) and value-weighted (VW) results, respectively. Newey and West (1987) t -statistics are provided in the parentheses. The results shown in bold signify their statistical significance at the 5% level. The sample period is January 2010 to December 2023.

	EW					VW				
	High SC	2	3	4	Low SC	High SC	2	3	4	Low SC
High $CFRet^{Isr}$	0.774 (1.33)	0.596 (1.06)	0.953 (1.62)	1.081 (1.83)	0.909 (1.46)	0.466 (0.83)	0.407 (0.71)	0.654 (1.12)	0.443 (0.76)	0.393 (0.63)
2	0.445 (0.86)	0.523 (1.00)	0.921 (1.63)	0.729 (1.36)	0.722 (1.34)	0.365 (0.71)	0.331 (0.59)	0.846 (1.38)	0.446 (0.83)	0.362 (0.64)
3	0.426 (0.79)	0.418 (0.79)	0.742 (1.37)	0.702 (1.27)	0.745 (1.33)	0.258 (0.52)	0.194 (0.39)	0.096 (0.18)	0.293 (0.53)	0.369 (0.62)
4	0.415 (0.76)	0.247 (0.46)	0.332 (0.60)	0.595 (1.08)	0.481 (0.90)	0.328 (0.60)	0.063 (0.12)	0.169 (0.31)	0.207 (0.38)	0.094 (0.17)
Low $CFRet^{Isr}$	0.611 (1.14)	0.508 (0.99)	0.436 (0.79)	0.547 (1.03)	0.390 (0.71)	0.435 (0.88)	0.144 (0.33)	0.165 (0.31)	0.085 (0.17)	−0.110 (−0.20)
High–Low	0.164 (0.39)	0.088 (0.21)	0.518 (1.24)	0.534 (1.70)	0.519 (1.87)	0.031 (0.07)	0.263 (0.53)	0.490 (1.10)	0.358 (1.02)	0.503 (1.61)
SIM alpha	0.177 (0.42)	0.105 (0.25)	0.527 (1.27)	0.534 (1.69)	0.520 (1.86)	0.050 (0.12)	0.274 (0.55)	0.509 (1.15)	0.361 (1.02)	0.508 (1.61)
CH4 alpha	0.202 (0.45)	0.260 (0.59)	0.684 (1.72)	0.748 (2.43)	0.681 (2.51)	0.063 (0.14)	0.384 (0.74)	0.615 (1.47)	0.657 (1.85)	0.697 (2.31)
q^5 alpha	0.414 (0.76)	0.833 (1.63)	1.191 (2.62)	1.086 (3.08)	0.899 (2.65)	0.128 (0.24)	0.957 (1.58)	1.213 (2.54)	0.904 (2.21)	0.770 (2.17)
FF6 alpha	0.341 (0.69)	0.476 (1.01)	0.961 (2.14)	0.905 (2.71)	0.879 (2.90)	0.097 (0.21)	0.565 (1.01)	0.798 (1.69)	0.795 (2.02)	0.886 (2.61)

Table C.3 The performance of portfolios formed by dependent sorting on SC and $CFRet^{MRX}$

This table presents the results of bivariate portfolios formed by dependent double sorting of one control variable — SC depicted in Eq. (10) and $CFRet^{MRX}$ depicted in Eq. (1) during January 2010 and December 2023. Each month, the individual stocks are first divided into five quintile groups according to their respective SC values. This process is further followed by an additional set of quintile portfolios constructed by sorting on $CFRet^{MRX}$. One-month-ahead excess returns of the portfolios are presented. “High” (“Low”) designates portfolios with the highest (lowest) SC or $CFRet^{MRX}$ values. The last rows report the $CFRet^{MRX}$ -sorted long-short portfolio’s return spread (“High–Low”) and associated alphas within each SC quantile. Specifically, SIM alpha is obtained from Sharpe (1963) single-index model (SIM) which consists of the market factor (MKT) only. CH4 alpha is obtained from the CH4 model of Liu et al. (2019). q^5 alpha is obtained from the q^5 -factor model of Hou et al. (2021). FF6 alpha is obtained from the six-factor model of Fama and French (2018). Pricing factors above are specifically constructed for the Chinese market, which are detailed in Section A of the Online Appendix. Left and right panels present the equal-weighted (EW) and value-weighted (VW) results, respectively. Newey and West (1987) t -statistics are provided in the parentheses. The results shown in bold signify their statistical significance at the 5% level. The sample period is January 2010 to December 2023.

	EW					VW				
	High SC	2	3	4	Low SC	High SC	2	3	4	Low SC
High $CFRet^{MRX}$	0.766 (1.30)	0.634 (1.12)	1.063 (1.80)	1.173 (1.94)	0.869 (1.39)	0.423 (0.73)	0.357 (0.61)	0.915 (1.54)	0.509 (0.87)	0.352 (0.56)
2	0.499 (0.98)	0.615 (1.15)	0.828 (1.50)	0.746 (1.42)	0.755 (1.42)	0.343 (0.69)	0.322 (0.56)	0.324 (0.57)	0.596 (1.10)	0.427 (0.78)
3	0.403 (0.75)	0.464 (0.89)	0.890 (1.59)	0.695 (1.23)	0.704 (1.24)	0.336 (0.64)	0.773 (1.56)	0.375 (0.69)	0.210 (0.38)	0.270 (0.44)
4	0.580 (1.07)	0.067 (0.13)	0.199 (0.38)	0.535 (0.97)	0.538 (1.00)	0.439 (0.80)	−0.491 (−0.97)	−0.392 (−0.82)	0.129 (0.23)	0.201 (0.37)
Low $CFRet^{MRX}$	0.427 (0.79)	0.515 (0.99)	0.402 (0.72)	0.510 (0.96)	0.384 (0.69)	0.271 (0.55)	0.206 (0.45)	0.340 (0.58)	0.006 (0.01)	−0.128 (−0.23)
High–Low	0.339 (0.75)	0.119 (0.27)	0.661 (1.59)	0.663 (1.95)	0.485 (1.67)	0.152 (0.32)	0.151 (0.29)	0.575 (1.41)	0.504 (1.42)	0.480 (1.46)
SIM alpha	0.357 (0.78)	0.137 (0.31)	0.673 (1.63)	0.669 (1.95)	0.485 (1.66)	0.165 (0.34)	0.158 (0.30)	0.597 (1.49)	0.512 (1.43)	0.484 (1.46)
CH4 alpha	0.420 (0.83)	0.294 (0.63)	0.851 (2.07)	0.842 (2.44)	0.679 (2.35)	0.177 (0.32)	0.312 (0.55)	0.656 (1.57)	0.762 (2.08)	0.714 (2.27)
q^5 alpha	0.645 (1.09)	0.711 (1.37)	1.335 (2.91)	1.194 (3.18)	0.885 (2.48)	0.236 (0.38)	0.698 (1.11)	1.263 (2.68)	1.002 (2.43)	0.823 (2.23)
FF6 alpha	0.581 (1.03)	0.469 (0.93)	1.090 (2.38)	1.004 (2.65)	0.877 (2.71)	0.284 (0.47)	0.441 (0.73)	0.835 (1.78)	0.895 (2.21)	0.901 (2.54)

Table C.4 The performance of portfolios formed by dependent sorting on SC and $CFRet^{Sor}$

This table presents the results of bivariate portfolios formed by dependent double sorting of one control variable — SC depicted in Eq. (10) and $CFRet^{Sor}$ depicted in Eq. (1) during January 2010 and December 2023. Each month, the individual stocks are first divided into five quintile groups according to their respective SC values. This process is further followed by an additional set of quintile portfolios constructed by sorting on $CFRet^{Sor}$. One-month-ahead excess returns of the portfolios are presented. “High” (“Low”) designates portfolios with the highest (lowest) SC or $CFRet^{Sor}$ values. The last rows report the $CFRet^{Sor}$ -sorted long-short portfolio’s return spread (“High–Low”) and associated alphas within each SC quantile. Specifically, SIM alpha is obtained from Sharpe (1963) single-index model (SIM) which consists of the market factor (MKT) only. CH4 alpha is obtained from the CH4 model of Liu et al. (2019). q^5 alpha is obtained from the q^5 -factor model of Hou et al. (2021). FF6 alpha is obtained from the six-factor model of Fama and French (2018). Pricing factors above are specifically constructed for the Chinese market, which are detailed in Section A of the Online Appendix. Left and right panels present the equal-weighted (EW) and value-weighted (VW) results, respectively. Newey and West (1987) t -statistics are provided in the parentheses. The results shown in bold signify their statistical significance at the 5% level. The sample period is January 2010 to December 2023.

	EW					VW				
	High SC	2	3	4	Low SC	High SC	2	3	4	Low SC
High $CFRet^{Sor}$	0.821 (1.38)	0.626 (1.11)	1.053 (1.77)	1.231 (2.00)	0.915 (1.45)	0.493 (0.85)	0.370 (0.63)	0.856 (1.45)	0.561 (0.94)	0.346 (0.56)
2	0.452 (0.89)	0.593 (1.09)	0.857 (1.54)	0.728 (1.42)	0.750 (1.41)	0.310 (0.63)	0.463 (0.81)	0.545 (0.96)	0.541 (1.02)	0.463 (0.84)
3	0.443 (0.82)	0.482 (0.92)	0.831 (1.51)	0.654 (1.14)	0.666 (1.18)	0.311 (0.59)	0.652 (1.32)	0.180 (0.34)	0.263 (0.47)	0.285 (0.46)
4	0.461 (0.84)	0.107 (0.21)	0.228 (0.42)	0.554 (1.03)	0.541 (0.98)	0.315 (0.57)	−0.324 (−0.65)	−0.141 (−0.29)	0.077 (0.14)	0.122 (0.22)
Low $CFRet^{Sor}$	0.498 (0.93)	0.486 (0.94)	0.412 (0.74)	0.492 (0.92)	0.379 (0.70)	0.349 (0.71)	0.141 (0.31)	0.250 (0.43)	0.017 (0.03)	−0.088 (−0.16)
High–Low	0.323 (0.72)	0.140 (0.32)	0.641 (1.52)	0.739 (2.11)	0.536 (1.79)	0.144 (0.30)	0.229 (0.43)	0.606 (1.44)	0.545 (1.51)	0.435 (1.31)
SIM alpha	0.337 (0.75)	0.160 (0.36)	0.653 (1.56)	0.744 (2.11)	0.530 (1.77)	0.158 (0.33)	0.236 (0.44)	0.628 (1.51)	0.553 (1.52)	0.433 (1.29)
CH4 alpha	0.367 (0.73)	0.322 (0.69)	0.873 (2.13)	0.937 (2.65)	0.726 (2.46)	0.177 (0.32)	0.404 (0.71)	0.777 (1.85)	0.827 (2.27)	0.669 (2.15)
q^5 alpha	0.594 (1.02)	0.779 (1.52)	1.365 (2.97)	1.253 (3.26)	0.959 (2.62)	0.251 (0.41)	0.872 (1.39)	1.425 (3.02)	1.059 (2.58)	0.822 (2.19)
FF6 alpha	0.537 (0.96)	0.516 (1.03)	1.111 (2.43)	1.090 (2.81)	0.924 (2.78)	0.284 (0.48)	0.564 (0.94)	0.966 (2.03)	0.943 (2.34)	0.872 (2.49)

D Definitions of firm-level characteristics

This section primarily introduces the firm-level characteristics relevant to the study. Unless stated otherwise, we obtain the data from the China Stock Market & Accounting Research (CSMAR) database.

IA

Investment-to-asset, denoted as *IA*, is defined as the annual change in total assets divided by one-year-lagged total assets. The calculation is based on data from the most recent annual report.

ROE

Return-to-equity, denoted as *ROE*, is defined as the ratio of operating profit to book equity (i.e., total equity attributable to shareholders + deferred tax liability - preferred stock). The calculation is performed using data from the most recent quarterly report.

LnSize

LnSize is defined as the natural logarithm of the product of a stock's last month-end closing price and its total shares outstanding.

Turn

A firm's abnormal turnover, denoted as *Turn*, is defined as the ratio of its average daily turnover over the past 20 days to its average daily turnover over the past 250 days.

CSMOM

Cross-sectional momentum (*CSMOM*) is defined as the cumulative excess return over the past 12 months, excluding the most recent month.

Illiq

Following [Amihud \(2002\)](#), the illiquidity of a stock, denoted as *Illiq*, for day t is calculated as $Illiq = |r_t|/volume_t$, where $|r_t|$ is the absolute return of the stock on day t , and $volume_t$ is the trading volume in Chinese yuan on day t . The monthly illiquidity measure is obtained by averaging the daily values over the available trading days within the month.

AbnVolume

A firm's abnormal trading volume, denoted as *AbnVolume*, is defined as the ratio of its monthly trading volume to the average trading volume of its connected firms based on shared analyst coverage.

NOA

NOA represents the number of analysts covering a specific firm. At the end of each month t , a firm or stock is considered covered by an analyst if the analyst provides at least one earnings forecast or recommendation for the stock during the period from $t - 11$ to t .

AbsCAR

AbsCAR represents the absolute excess return of an individual firm within the time window surrounding the release of its financial report. Following the approach of [Daniel et al. \(2020\)](#), we define the disclosure date of the latest financial report of a listed firm as the zero moment. *CAR* is then defined as the cumulative abnormal return of the firm's stock, calculated from two trading days before the zero moment to one trading day after:

$$CAR_i = \sum_{d=-2}^1 (R_{i,d} - R_{m,d}), \quad (\text{D.1})$$

where $R_{i,d}$ denotes the return of firm i on the d th day within the most recent disclosure period window, and $R_{m,t}$ represents the market return for the same period. We then obtain *AbsCAR* by taking the absolute value of the equation above:

$$AbsCAR_i = |CAR_i|. \quad (\text{D.2})$$

SVIC & SVIA

SVI refers to a composite search index calculated using Baidu platform data that incorporates various types of internet search data and news sentiment information. It aggregates the number of search queries related to each listed firm and reflects the intensity of online searches for that firm. Specifically, we employ two *SVI*-related metrics: the search volume with stock codes as keywords (*SVIC*) and the sum of search volumes with stock codes, company abbreviations, and full company names as keywords (*SVIA*). The data is derived from the Chinese Research Data Services Platform (CNRDS).

InstProp

InstProp represents the ratio of the total number of shares held by institutional investors to the total number of shares outstanding for listed firms. The data is obtained from the most recent financial reports.

LoA

We construct the limit-of-arbitrage (*LoA*) index following the approach of [Gu et al. \(2018\)](#). In their study, seven binary indicators are used to compose the *LoA* index. Building on their methodology, we incorporate up to ten proxies that reflect the degree of limit of arbitrage in the Chinese stock market. Specifically, we include additional factors, such as the stock connect program and newly issued financial derivatives based on stock indices, in constructing the *LoA* index. The proxies are as follows:

(1) Price limit. In the Chinese stock market, a daily price limit of 10% has been in place for regular stocks, and a 5% limit applies to stocks under special treatment since December 1996. As of August 24, 2020, the daily price limit was relaxed from 10% to 20% for stocks listed on the China Growth Enterprise Market (GEM). Stocks that hit the price limit are more likely to experience higher limits to arbitrage. Therefore, we denote $Limit_{i,t}$ as 1 if stock i reaches the daily price limit at least once during month t .

(2) Margin trading. The Chinese market introduced the margin trading and short selling (MTSS) policy in March 2010, allowing investors to borrow funds for securities trading or engage in short selling through margin transactions. Intuitively, stocks not covered by the MTSS are subject to a higher degree of arbitrage limits. Therefore, we assign a value of 1 to $MTSS_{i,t}$ if stock i is not included in the MTSS in month t .

(3) Index futures and options. Following the approach of [Gu et al. \(2018\)](#), stocks not included in widely traded index futures and options are likely to face higher arbitrage limits. Notably, China has witnessed a growing number of financial derivatives linked to various indices. Among these, futures and options based on four key indices have gained significant popularity in the market: the Shanghai and Shenzhen 300 Index (CSI 300), the Shanghai Stock Exchange 50 Index (SSE 50), the China Securities 500 Index (CSI 500), and the China Securities 1000 Index (CSI 1000).

Futures based on the CSI 300 index have been listed since April 16, 2010, and the associated options have been traded since December 23, 2019. Futures based on the SSE 50 began trading on April 16, 2015. ETF options for the SSE 50 commenced trading on February 9, 2015, while index options started trading on December 19, 2022. Futures based on the CSI 500 began trading on April 16, 2015, and ETF options for the CSI 500 were

introduced on September 19, 2022. Futures and options based on the CSI 1000 have been traded since July 22, 2022.

Therefore, a stock faces a low degree of arbitrage limits if it is a member of any of the aforementioned indices. Accordingly, we denote $CSI300_{i,t}$, $SSE50_{i,t}$, $CSI500_{i,t}$, and $CSI1000_{i,t}$ as 1 if stock i does not belong to the CSI 300, SSE 50, CSI 500, and CSI 1000, respectively.

(4) Stock connect. China launched the Shanghai-Hong Kong and Shenzhen-Hong Kong stock connect programs in November 2014 and December 2016, respectively. The two programs allow qualified mainland Chinese investors to access eligible Hong Kong shares (South-bound) and permits Hong Kong and overseas investors to trade eligible A-shares (North-bound), subject to a specific daily quota. Importantly, A-share stocks included in the stock connect can be short sold. Therefore, we define $Connect_{i,t}$, which is assigned a value of 1 if stock i is not included in the Shanghai-Hong Kong or Shenzhen-Hong Kong stock connects.

(5) Liquidity. Stocks with lower liquidity (or higher illiquidity) are more likely to experience a higher degree of arbitrage limits. Following Gu et al. (2018), we consider the illiquidity measure proposed by Amihud (2002). Specifically, $Illi_{i,t}$ is assigned a value of 1 if stock i has a value greater than the cross-sectional median in month t .

(6) Trading volume. Stocks with lower trading volumes are more likely to be associated with higher limits of arbitrage. In Gu et al. (2018), the authors use the raw trading volume. However, since trading volume is highly dependent on the size of a stock's outstanding shares, we use a refined version of trading volume: the monthly turnover rate. This rate is calculated by dividing the total number of shares traded in the past month by the total number of outstanding shares. Specifically, $Turnover_{i,t}$ is assigned a value of 1 if the monthly average volume over the past 12 months for stock i is less than or equal to the cross-sectional median in month t .

(7) Analyst coverage. Stocks with less analyst coverage tend to have higher limits of arbitrage. $Cover_{i,t}$ is assigned a value of 1 if the number of analysts following stock i is smaller than or equal to the cross-sectional median in month t .

The LoA index for stock i in month t is calculated as follows:

$$LoA_{i,t} = Limit_{i,t} + MTSS_{i,t} + CSI300_{i,t} + SSE50_{i,t} + CSI500_{i,t} + CSI1000_{i,t} \\ + Connect_{i,t} + Illi_{i,t} + Turnover_{i,t} + Cover_{i,t}. \quad (D.3)$$

Consequently, a high LoA is associated with greater limitations on arbitrage. It is important to note that stocks in the SSE 50 also belong to the CSI 300, while there is no overlap among the members of the CSI-related indices.

QFIIProp

QFIIProp represents the ratio of the total number of shares held by QFII investors to the total number of shares outstanding for listed firms. The data is obtained from the most recent financial reports.

DFIEs, DSOEs & DDPEs

We categorize listed firms in China into three broad groups: foreign-invested enterprises (FIEs), state-owned enterprises (SOEs), and domestic private enterprises (DPEs), and these three groups together account for more than 99% of all listed firms by count.

DFIE denotes a dummy variable that identifies whether a firm is a foreign-invested enterprise; specifically, a value of unity indicates the firm is an FIE, while a value of zero indicates it is a non-FIE. Note that we classify both Sino-foreign joint venture and foreign-owned sole proprietorship listed firms in the CSMAR database as FIEs. Similarly, *DSOE* denotes a dummy variable that identifies whether a firm is a state-owned enterprise, with a value of unity indicating the firm is an SOE and a value of zero indicating it is a non-SOE, and *DDPE* denotes a dummy variable that identifies whether a firm is a domestic private enterprise, where a value of unity indicates the firm is a DPE and a value of zero indicates it is a non-DPE.

We present the correlation matrix for different *CFRet* measures in [Table D.1](#). We also report the average *CFRet* values for the ten decile *CFRet*-sorted portfolios in [Table D.2](#). The associated summary statistics for the characteristics mentioned above and different *CFRet* measures are presented in [Table D.3](#). Note that all results from these three tables are based on the rigorous filter introduced in [Section 2.1](#).

Table D.1 Correlation matrix

This table presents the time-series averages of the monthly cross-sectional Pearson (in the lower triangular) and Spearman (in the upper triangular) correlations between various $CFRet$ measures.

	$CFRet^{AH}$	$CFRet^{Isr}$	$CFRet^{MRX}$	$CFRet^{Sor}$	$CFRet^{SC}$	$CFRet^{PR}$
$CFRet^{AH}$		0.852	0.922	0.946	0.872	0.890
$CFRet^{Isr}$	0.846		0.930	0.937	0.687	0.714
$CFRet^{MRX}$	0.898	0.931		0.987	0.735	0.759
$CFRet^{Sor}$	0.938	0.944	0.975		0.769	0.794
$CFRet^{SC}$	0.877	0.670	0.700	0.753		0.995
$CFRet^{PR}$	0.899	0.703	0.729	0.782	0.995	

Table D.2 Average $CFRet$ results

This table presents the monthly average connected-firm returns ($CFRet$) for ten decile portfolios, constructed by sorting on $CFRet$ according to the given weighting scheme, which is listed in the first column of the table. At the beginning of each month, stocks are sorted into decile portfolios based on its one given $CFRet$, constructed using the procedure explained in Section 2.3. Portfolio 1 (10), labeled as High (Low), contains the stocks with the highest (lowest) $CFRet$ value. For each decile portfolio, we report the equal-weighted $CFRet$ value. The sample period spans from January 2010 to December 2023.

	High	2	3	4	5	6	7	8	9	Low	High–Low
$CFRet^{SC}(\%)$	7.10	3.95	2.70	1.84	1.13	0.49	−0.17	−0.90	−1.88	−4.13	11.23
$CFRet^{AH}(\%)$	8.15	4.60	3.10	2.05	1.14	0.33	−0.47	−1.36	−2.46	−4.65	12.80
$CFRet^{Isr}(\%)$	8.51	4.54	3.07	2.04	1.17	0.37	−0.42	−1.27	−2.36	−4.66	13.17
$CFRet^{MRX}(\%)$	9.12	4.70	3.16	2.08	1.16	0.33	−0.50	−1.40	−2.55	−5.85	14.97
$CFRet^{Sor}(\%)$	8.29	4.57	3.09	2.04	1.16	0.35	−0.44	−1.31	−2.40	−4.64	12.93
$CFRet^{PR}(\%)$	7.06	3.94	2.70	1.85	1.14	0.50	−0.14	−0.87	−1.84	−4.05	11.11

Table D.3 Summary statistics

This table presents the time-series averages of cross-sectional summary statistics for connected-firm returns ($CFRet$) and other firm-level characteristics. Five versions of $CFRet$ that are listed the first five rows in the first column are constructed according to Eq. (1). The reported statistics in columns 2 to 10 include the mean value (Mean), standard deviation (St Dev), 25th percentile (25th Per), median value (Median), 75th percentile (75th Per), minimum (Min), maximum (Max), skewness (Skew), and kurtosis (Kurt). The data spans from January 2010 to December 2023.

	Mean	St Dev	25th Per	Median	75th Per	Min	Max	Skew	Kurt
$CFRet^{SC}(\%)$	1.004	3.045	-0.903	0.809	2.690	-6.064	10.044	0.388	4.018
$CFRet^{AH}(\%)$	1.032	3.553	-1.351	0.730	3.084	-6.292	10.979	0.447	3.546
$CFRet^{Isr}(\%)$	1.074	3.562	-1.268	0.763	3.057	-6.526	11.971	0.547	3.981
$CFRet^{MRX}(\%)$	1.038	3.821	-1.394	0.743	3.145	-7.899	12.343	0.442	4.047
$CFRet^{Sor}(\%)$	1.049	3.537	-1.303	0.752	3.068	-6.372	11.252	0.472	3.699
$CFRet^{PR}(\%)$	1.018	3.016	-0.870	0.824	2.689	-5.967	9.990	0.391	4.019
IA	0.224	0.464	0.032	0.118	0.253	-0.229	3.437	4.267	27.361
ROE	0.110	0.137	0.036	0.095	0.169	-0.314	0.628	0.646	6.273
$LnSize$	16.07	0.88	15.40	15.83	16.50	15.06	19.13	1.30	4.52
EP	0.033	0.048	0.010	0.026	0.050	-0.120	0.223	0.953	7.726
$Turn$	1.015	0.542	0.654	0.881	1.212	0.302	3.267	1.737	6.981
$CSMOM(\%)$	16.709	43.277	-12.053	6.673	33.968	-45.676	192.405	1.540	6.241
$Illiq$	0.032	0.030	0.013	0.025	0.042	0.002	0.173	1.919	9.064
NOA	15.29	13.27	4.67	11.28	22.62	1.00	56.67	1.11	3.65
$AbnVolume$	0.866	1.002	0.294	0.529	1.005	0.084	6.074	2.903	13.142
$AbsCAR(\%)$	3.851	3.514	1.307	2.854	5.284	0.050	17.492	1.627	5.964
$SVIC$	12258	7811	7026	10049	14911	3642	46696	2	7
$SVIA$	37212	44504	16455	25375	40399	6312	335121	4	27
$InstProp$	44.31	24.52	24.05	45.92	63.85	0.70	91.73	-0.09	2.03
LoA	6.370	1.501	5.244	6.363	7.512	3.515	9.304	0.024	2.176
$QFIIProp$	0.786	0.917	0.262	0.515	0.946	0.043	6.321	3.383	18.369
$DFIE$	0.078	—	—	—	—	—	—	—	—
$DSOE$	0.442	—	—	—	—	—	—	—	—
$DDPE$	0.475	—	—	—	—	—	—	—	—

E An alternative weighting scheme

In the analysis of our paper, we primarily use strength centrality (SC) to evaluate the influence of stocks based on their direct connections within the analyst coverage network. By computing a SC -weighted average of peer returns for each focal firm, we detect cross-firm momentum effects — namely, focal firms with higher peer returns tend to be more profitable in the future. However, while the SC -based measure captures the influence of directly connected firms, it may not fully account for the broader, indirect influence that extends beyond these direct relationships or the impact of *peers of peers*.

Suppose there is a focal firm C with directly connected firms A and B , as depicted in [Figure E.1](#). Both firm A and firm B have two connections. According to [Eq. \(10\)](#), they are assigned the same weights when calculating $CFRet^{SC}$. However, the SC measure overlooks the properties of the peers connected to firms A and B , which also influence the importance of these firms within the network. For example, B has a peer, H , which has a strength of one and is peripheral in the network. In contrast, A 's peer, D , has a strength of four and is highly central or influential in the network. Therefore, cross-firm momentum effects are more likely to occur from D to A (because D receives more attention and incorporates information faster into the price), while spillovers from H to B are unlikely. In this context, a higher (lower) weight should be assigned to firm A (B), as it is (not) more likely to experience return spillovers from its peer at an earlier stage. Thus, the relevant information would be incorporated more promptly into A 's asset price, further contributing to its return spillovers to the focal firm C .

To address the issue outlined above, we introduce *PageRank* (PR , hereafter) centrality to serve our purpose ([Page et al., 1999](#)).² Unlike SC , PR takes into account not only the direct connections of a firm but also its peers' connections, thereby capturing the *indirect* influence exerted through the network. By incorporating PR centrality into our analysis, we aim to capture both direct and indirect spillover effects, providing a more comprehensive understanding of how return spillovers occur within the analyst coverage network. Specifically,

²In addition to SC and PR , we also examine other common centrality measures, including eigenvector, betweenness, and closeness centrality, within the analyst coverage network. Our unreported results of univariate portfolio analyses confirm that these measures also yield significant cross-firm momentum returns, highlighting the broader relevance of network centrality in capturing return predictability. However, SC and PR offer a more intuitive and economically interpretable weighting scheme in our context, as it directly extends the classical common-analyst measure by incorporating each firm's total attention-weighted connectivity. In contrast, measures such as betweenness and closeness centrality, while statistically effective, are less directly tied to the economic mechanism of analyst-driven information diffusion. They reflect topological roles that do not necessarily correlate with information production or absorption speed. Thus, while multiple centrality measures prove useful, SC and PR remain preferable for its conceptual clarity and continuity with existing literature.

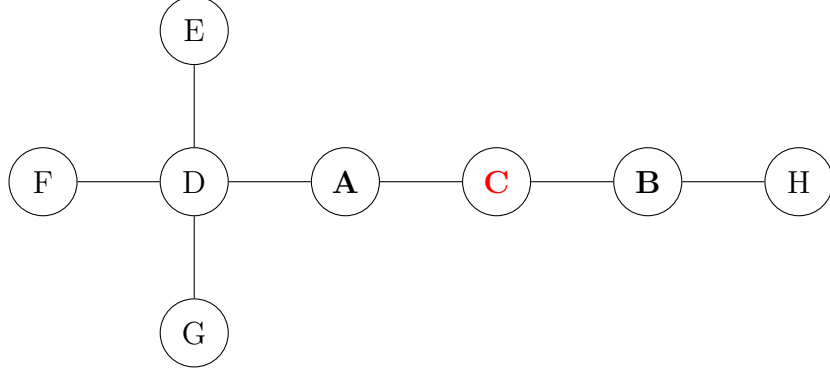


Figure E.1 A schematic of the structure of focal firm and its peers

PR centrality is defined as follows:

$$PR_j = \frac{1-d}{N} + d \sum_{m \in \mathcal{N}_j} \frac{n_{m,j}}{\sum_{k \in \mathcal{N}_m} n_{m,k}} PR_m, \quad (\text{E.1})$$

where d is the damping factor, typically set to 0.85, and N is the total number of nodes in the network. We thus build our PR -based $CFRet$:

$$weight_{i,j}^{PR} = \frac{PR_j}{\sum_{j \in \mathcal{N}_i} PR_j}. \quad (\text{E.2})$$

Note that the PR algorithm is iterative and typically converges within a few iterations. Using $weight_{i,j}^{PR}$, we rely on [Eq. \(1\)](#) to construct the connected-firm return based on the PR weighting scheme, i.e., $CFRet^{PR}$.

1. PageRank centrality calculation

We first present the iterative calculation processes of PageRank applied in our analyst coverage network, which is essentially an undirected weighted network.

The adjacency matrix $\mathbf{A}$ of the network is defined as $A_{ij} = A_{ji} = n_{ij} = n_{ji}$, where n_{ij} is the number of common analysts covering both firms i and j in their respective portfolios.

The PR vector $\mathbf{R}$ is updated iteratively using the following:

$$\mathbf{R}^{t+1} = d\mathbf{A}\mathbf{R}^t + \frac{1-d}{N}\mathbf{1}, \quad (\text{E.3})$$

where d is a damping factor set as 0.85, N is the number of nodes in the network, and $\mathbf{1}$ is a vector of all ones.

The convergence condition for PR can be defined as follows. Let ϵ be a small positive

number representing the convergence threshold. The iteration stops when:

$$\|\mathbf{R}^{t+1} - \mathbf{R}^t\|_2 < \epsilon, \quad (\text{E.4})$$

where $\|\cdot\|_2$ denotes the L_2 norm. In our work, $\epsilon = 0.0001$.

The iterative steps are as follows:

- (1) Initialize the PR vector $\mathbf{R}^0$ with equal values for all nodes, usually $\mathbf{R}^0 = \frac{1}{N}\mathbf{1}$.
- (2) At iteration t , calculate $\mathbf{R}^{t+1}$ using Eq. (E.3).
- (3) Check if the convergence condition is satisfied; if not, repeat Step (2).

2. Return predictability in terms of $CFRet^{PR}$

We then evaluate the cross-firm momentum effects based on $CFRet^{PR}$ through univariate portfolio analysis. The results are presented in Table E.1. In the equal-weighted results from Panel A, we observe a high degree of monotonicity in the raw and risk-adjusted returns from the first decile group (High) to the tenth decile group (Low), when skipping portfolio 7. Moreover, the long-short portfolios produce statistically significant return and alphas at the 5% level. Specifically, as indicated in the last column, the raw return spread amounts to 0.827% per month. Meanwhile, alphas from different models range from 0.83% to 1.35% per month, corresponding to a range of 9.96% to 16.13% per annum. Notably, this performance surpasses the adjusted return results of the long-short portfolio based on $CFRet^{SC}$ in the final column of Table 2, indicating that the cross-sectional return forecasting using $CFRet^{PR}$ exhibits superior performance.

In Panel B for the value-weighted results, we find that the long-short portfolios produce statistically significant alphas under different pricing models, except for the insignificant raw return spread and its SIM alpha. However, different from findings in the equal-weighted results above, we observe that the return spread and associated alphas are lower than those based on $CFRet^{SC}$ in the final column of Table 3. This suggests that, compared to $CFRet^{SC}$, the return predictability based on $CFRet^{PR}$ is more sensitive to firm size. Despite this, the $CFRet^{PR}$ -sorted long-short portfolios significantly outperform those based on other versions of $CFRet$, as shown in the right panels of Table 2 and Table 3.

In addition to the univariate portfolio analysis, we also rely on the FM regressions to evaluate the performance of $CFRet^{PR}$ in predicting stock returns while controlling for additional firm-level characteristics. In model (1), which incorporates $CFRet^{PR}$ as the only right-hand side (RHS) variable, the associated slope is 0.123, with a t -statistic of 3.96, indicating its

Table E.1 Excess return alphas of univariate portfolios formed by sorting on $CFRet^{PR}$

This table presents the equally-weighted (EW) and value-weighted (VW) monthly average returns for ten decile portfolios, which are constructed by sorting on $CFRet^{PR}$ in Panels A and B, respectively. Each month, the individual stocks are divided into ten decile groups according to their respective $CFRet^{PR}$ values. Portfolio 1 (10), labeled as High (Low), contains the stocks with the highest (lowest) $CFRet^{PR}$ value. One-month-ahead raw excess return (Raw) and alphas of the portfolios are recorded. Specifically, SIM alpha is obtained from [Sharpe \(1963\)](#) single-index model (SIM) which consists of the market factor (MKT) only. CH4 alpha is obtained from the CH4 model of [Liu et al. \(2019\)](#). q^5 alpha is obtained from the q^5 -factor model of [Hou et al. \(2021\)](#). FF6 alpha is obtained from the six-factor model of [Fama and French \(2018\)](#). Pricing factors above are specifically constructed for the Chinese market, which are detailed in Section A of the Online Appendix. [Newey and West \(1987\)](#) t -statistics are provided in the parentheses. The results shown in bold signify their statistical significance at the 5% level. The sample period is January 2010 to December 2023.

	High	2	3	4	5	6	7	8	9	Low	High–Low
Panel A: EW											
Raw	1.243 (1.97)	0.771 (1.42)	0.650 (1.26)	0.655 (1.30)	0.606 (1.18)	0.554 (1.09)	0.320 (0.61)	0.481 (0.93)	0.458 (0.87)	0.416 (0.78)	0.827 (2.25)
SIM	1.203 (2.08)	0.739 (1.47)	0.619 (1.32)	0.623 (1.36)	0.571 (1.24)	0.522 (1.13)	0.281 (0.60)	0.438 (0.96)	0.414 (0.90)	0.373 (0.79)	0.830 (2.25)
CH4	1.648 (2.63)	1.022 (1.91)	0.943 (1.84)	0.978 (1.91)	0.921 (1.81)	0.919 (1.81)	0.587 (1.13)	0.725 (1.43)	0.676 (1.27)	0.547 (1.04)	1.101 (3.09)
q^5	2.169 (3.34)	1.453 (2.55)	1.312 (2.42)	1.149 (2.16)	1.147 (2.18)	1.043 (1.98)	0.794 (1.48)	0.919 (1.76)	0.878 (1.66)	0.825 (1.53)	1.344 (2.99)
FF6	1.997 (2.94)	1.305 (2.26)	1.192 (2.14)	1.186 (2.17)	1.128 (2.07)	1.106 (2.02)	0.772 (1.38)	0.890 (1.65)	0.849 (1.49)	0.722 (1.29)	1.275 (3.16)
Panel B: VW											
Raw	0.831 (1.30)	0.468 (0.91)	0.494 (0.97)	0.427 (0.90)	0.356 (0.75)	−0.110 (−0.24)	0.177 (0.36)	0.257 (0.49)	0.210 (0.41)	−0.026 (−0.06)	0.857 (1.91)
SIM	0.801 (1.34)	0.443 (0.92)	0.478 (1.00)	0.415 (0.91)	0.338 (0.76)	−0.124 (−0.28)	0.150 (0.33)	0.226 (0.47)	0.171 (0.38)	−0.061 (−0.15)	0.862 (1.92)
CH4	1.183 (2.00)	0.729 (1.53)	0.774 (1.62)	0.708 (1.51)	0.645 (1.45)	0.117 (0.27)	0.393 (0.85)	0.505 (1.02)	0.391 (0.77)	0.006 (0.01)	1.177 (2.70)
q^5	1.508 (2.36)	0.950 (1.77)	0.951 (1.74)	0.659 (1.29)	0.733 (1.53)	0.140 (0.30)	0.490 (0.99)	0.545 (1.04)	0.518 (1.04)	0.214 (0.46)	1.294 (2.45)
FF6	1.442 (2.27)	0.953 (1.88)	0.924 (1.83)	0.751 (1.51)	0.755 (1.59)	0.229 (0.50)	0.476 (0.94)	0.617 (1.19)	0.548 (1.00)	0.160 (0.35)	1.282 (2.63)

robust explanatory power for cross-sectional stock returns. According to [Table D.2](#), this corresponds to a monthly average return spread between the decile portfolios with the highest and lowest values of $CFRet^{PR}$ of 1.37% ($= 0.123 \times 11.11$), resulting in an annual return of 16.44%; this profit is even greater than that based on $CFRet^{SC}$, as shown in [Table 5](#).

In models (2) to (5), we examine the FM regressions incorporating combination of $CFRet^{SC}$ and each version of $CFRet$ based on conventional weighting schemes as their RHS variables. Although the slope coefficients of $CFRet^{PR}$ in all four models show varying degrees of reduction compared to model (1), they remain statistically significant at the 5% level, indicated by the associated t -statistics exceeding 2.5. According to [Table D.2](#), the

Table E.2 The results of Fama-Macbeth regressions

This table presents results from the cross-sectional regressions of future equity returns on the $CFRet$ -sorted long-short portfolios and various control variables between 2010 and 2023. All right-hand-side variables are winsorized at the first and 99th percentiles. The related control variables are introduced in Section D. The coefficients reported represent time-series averages obtained from monthly Fama and MacBeth (1973) regressions, and the corresponding Newey and West (1987) t -statistics are presented in parentheses. The results shown in bold signify their statistical significance at the 5% level.

	(1)	(2)	(3)	(4)	(5)	(6)
$CFRet^{PR}$	0.123 (3.96)	0.065 (2.55)	0.097 (3.63)	0.092 (4.01)	0.083 (3.61)	0.063 (2.22)
$CFRet^{AH}$		0.060 (1.69)				-0.021 (-0.31)
$CFRet^{Isr}$			0.032 (1.48)			-0.049 (-0.93)
$CFRet^{MRX}$				0.037 (1.58)		-0.193 (-1.20)
$CFRet^{Sor}$					0.046 (1.75)	0.327 (1.75)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes
Adj- R^2	0.100	0.104	0.104	0.104	0.104	0.109

monthly average return spreads between the decile portfolios with the highest and lowest values of $CFRet^{PR}$ are approximately 0.72% ($= 0.065 \times 11.11$), 1.08% ($= 0.097 \times 11.11$), 1.02% ($= 0.092 \times 11.11$), and 0.92% ($= 0.083 \times 11.11$), respectively. Correspondingly, the long-short portfolios yield annual returns of 8.64%, 12.96%, 12.24%, and 11.04% for the four models, respectively. Overall, $CFRet^{PR}$ still leads to considerable profits.

In model (6), which incorporates all five $CFRet$ versions along with the related control variables, the slope of $CFRet^{SC}$ remains statistically significant at the 5% level, as indicated by a t -statistic of 2.22. The return spread between the decile portfolios with the highest and lowest values of $CFRet^{PR}$ is around 0.70% ($= 0.063 \times 11.11$) per month, or 8.40% per annum. Meanwhile, we observe that the slopes of the other $CFRet$ versions remain insignificant at the 5% level. Similar to the results in Table 5, the slopes for $CFRet^{AH}$, $CFRet^{Isr}$, and $CFRet^{MRX}$ even decrease to negative values. These findings collectively suggest $CFRet^{PR}$ has superior return forecasting ability, which cannot be explained by the other four $CFRet$ versions or related firm-level characteristics.

References

- Amihud, Y., 2002. Illiquidity and stock returns: cross-section and time-series effects. *J. Financ. Mark.* 5, 31–56. doi:[10.1016/S1386-4181\(01\)00024-6](https://doi.org/10.1016/S1386-4181(01)00024-6).
- Daniel, K., Hirshleifer, D. and Sun, L., 2020. Short- and long-horizon behavioral factors. *Rev. Financ. Stud.* 33, 1673–1736. doi:[10.1093/rfs/hhz069](https://doi.org/10.1093/rfs/hhz069).
- Fama, E.F. and French, K.R., 2018. Choosing factors. *J. Financ. Econ.* 128, 234–252. doi:[10.1016/j.jfineco.2018.02.012](https://doi.org/10.1016/j.jfineco.2018.02.012).
- Fama, E.F. and MacBeth, J.D., 1973. Risk, return, and equilibrium: Empirical tests. *J. Polit. Econ.* 81, 607–636. doi:[10.1086/260061](https://doi.org/10.1086/260061).
- Gu, M., Kang, W. and Xu, B., 2018. Limits of arbitrage and idiosyncratic volatility: Evidence from China stock market. *J. Bank. Financ.* 86, 240–258. doi:[10.1016/j.jbankfin.2015.08.016](https://doi.org/10.1016/j.jbankfin.2015.08.016).
- Hou, K., Mo, H., Xue, C. and Zhang, L., 2021. An augmented q -factor model with expected growth. *Rev. Financ.* 25, 1–41. doi:[10.1093/rof/rfaa004](https://doi.org/10.1093/rof/rfaa004).
- Liu, J., Stambaugh, R.F. and Yuan, Y., 2019. Size and value in China. *J. Financ. Econ.* 134, 48–69. doi:[10.1016/j.jfineco.2019.03.008](https://doi.org/10.1016/j.jfineco.2019.03.008).
- Newey, W.K. and West, K.D., 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelationconsistent covariance matrix. *Econometrica* 55, 703–708. doi:[10.2307/1913610](https://doi.org/10.2307/1913610).
- Page, L., Brin, S., Motwani, R. and Winograd, T., 1999. The PageRank citation ranking: Bringing order to the web. Technical Report. Stanford infolab.
- Sharpe, W.F., 1963. A simplified model for portfolio analysis. *Manag. Sci.* 9, 277–293. doi:[10.1287/mnsc.9.2.277](https://doi.org/10.1287/mnsc.9.2.277).